Chemistry - Class 11 Solved exercises - Punjab Curriculum and Textbook Board (PCTB), English medium (cover 2016-17, 1st edition, 20th impression, May 2016)

Chapter 1 - Basic Concepts

Exercise 1 - Multiple Choice Questions

1(i) Isotopes differ in

✓ A. properties which depend upon mass

B. arrangement of electrons in orbitals

C. chemical properties

D. the extent to which they may be affected in electromagnetic field

1(ii) Select the most suitable answer: isotopes are comparatively abundant when they have

A. even atomic masses

B. odd atomic masses

✓ C. even atomic masses and even atomic numbers

D. even atomic masses and odd atomic numbers

1(iii) Many elements have fractional atomic masses. This is because

A. the mass of the atom is itself fractional

B. atomic masses are average masses of isobars

C. atomic masses are average masses of isotopes

✓ D. atomic masses are average masses of isotopes proportional to their relative abundance

1(iv) The mass of one mole of electrons is

A. 1.008 mg

✓ B. 0.55 mg

C. 0.184 mg

D. 1.673 mg

1(v) 27 g of Al will react completely with how much mass of O2 to produce Al2O3?

A. 8 g of oxygen

B. 16 g of oxygen

C. 32 g of oxygen

✓ D. 24 g of oxygen

1(vi) The number of moles of CO2 which contain 8.0 g of oxygen is

✓ A. 0.25

B. 0.50

C. 1.0

D. 1.50

1(vii) The largest number of molecules are present in

✓ A. 3.6 g of H2O

B. 4.8 g of C2H5OH

C. 2.8 g of CO

D. 5.4 g of N2O5

1(viii) One mole of SO2 contains

A. 6.02 x 10^23 atoms of oxygen

B. 18.1 x 10^23 molecules of SO2

✓ C. 6.02 x 10^23 atoms of sulphur

D. 4 gram atoms of SO2

1(ix) The volume occupied by 1.4 g of N2 at S.T.P. is

A. 2.24 dm3

B. 22.4 dm3

✓ C. 1.12 dm3

D. 112 cm3

1(x) A limiting reactant is the one which

A. is taken in lesser quantity in grams as compared to other reactants

B. is taken in lesser quantity in volume as compared to the other reactants

C. gives the maximum amount of the product which is required

✓ D. gives the minimum amount of the product under consideration

Exercise 1 - Fill in the Blanks

2(i) The unit of relative atomic mass is none (it is a pure ratio, dimensionless) .

2(ii) The exact masses of isotopes can be determined by mass spectrograph .

2(iii) The phenomenon of isotopy was first discovered by Soddy .

2(iv) Empirical formula can be determined by combustion analysis for those compounds which have carbon and hydrogen in them.

2(v) A limiting reagent is that which controls the quantities of _____.

A limiting reagent is that which controls the quantities of the products formed. It is consumed first, and no more product can form once it is used up.

2(vi) 1 mole of glucose (C6H12O6) has _____ atoms of carbon, _____ of oxygen and _____ of hydrogen.

1 mole of glucose has 6 x 6.02 x 10^23 atoms of carbon, 6 x 6.02 x 10^23 atoms of oxygen and 12 x 6.02 x 10^23 atoms of hydrogen, because one molecule C6H12O6 contains 6 C, 6 O and 12 H atoms.

2(vii) 4 g of CH4 at 0 degrees C and 1 atm pressure has _____ molecules of CH4.

Step 1 - Moles of CH4 = 4 g / 16 g mol-1 = 0.25 mol. Step 2 - Molecules = 0.25 mol x 6.02 x 10^23 mol-1 = 1.505 x 10^23 molecules of CH4.

2(viii) Stoichiometric calculations can be performed only when the law of conservation of mass (a balanced chemical equation) is obeyed.

Exercise 1 - True or False

3(i) Neon has three isotopes and the fourth one with atomic mass 20.18 amu.

False

3(ii) Empirical formula gives the information about the total number of atoms present in the molecule.

False

3(iii) During combustion analysis Mg(ClO4)2 is employed to absorb water vapours.

True

3(iv) Molecular formula is the integral multiple of empirical formula and the integral multiple can never be unity.

False

3(v) The number of atoms in 1.79 g of gold and 0.023 g of sodium are equal.

False

3(vi) The number of electrons in the molecules of CO and N2 are 14 each, so 1 g of each gas will have same number of electrons.

True

3(vii) Avogadro's hypothesis is applicable to all types of gases i.e. ideal and non-ideal.

False

3(viii) Actual yield of a chemical reaction may be greater than the theoretical yield.

False

Exercise 1 - Short and Numerical Questions

4. What are ions? Under what conditions are they produced?

An ion is an atom or a group of atoms that carries a net electric charge because it has gained or lost electrons: loss of electrons gives a positive ion (cation) and gain of electrons gives a negative ion (anion). Ions are produced when an atom is

ionised, for example by heating to a high temperature, by an electric discharge, by radiation, by bombarding atoms with fast particles, or when an ionic compound dissolves in water and its ions separate.

5(a) What are isotopes? How do you deduce the fractional atomic masses of elements from the relative isotopic abundance? Give two examples in support of your answer.

Isotopes are atoms of the same element with the same atomic number but different mass numbers, that is, the same number of protons but different numbers of neutrons. The atomic mass of an element is the weighted average of the masses of its isotopes: average atomic mass = sum over isotopes of (isotopic mass x fractional abundance). Example 1 - Chlorine has Cl-35 (about 75%) and Cl-37 (about 25%), giving (35 x 0.75) + (37 x 0.25) = 35.5 amu. Example 2 - Boron has B-10 and B-11, giving an average of about 10.81 amu, between the two isotopic masses.

5(b) How does a mass spectrograph show the relative abundance of an element?

In a mass spectrograph the vaporised atoms are ionised, accelerated and deflected by a magnetic field, so that ions of different mass-to-charge ratio strike the detector at different positions. Each isotope gives a separate peak at its own mass, and the height (intensity) of each peak is proportional to the number of ions of that isotope, that is, to its relative abundance. The relative peak heights therefore read off the relative abundances directly.

5(c) What is the justification of two strong peaks in the mass spectrum for bromine, while for iodine only one peak at 127 amu is indicated?

Bromine occurs naturally as two isotopes of almost equal abundance, Br-79 and Br-81, so its mass spectrum shows two strong peaks of nearly equal height at 79 and 81 amu. Iodine, by contrast, is monoisotopic in nature: practically all iodine atoms are I-127, so only one peak, at 127 amu, appears.

6. Silver has atomic number 47 and has 16 known isotopes but two occur naturally

i.e. Ag-107 and Ag-109. Given the mass spectrometric data (Ag-107: 106.90509 amu, 51.84%; Ag-109: 108.90476 amu, 48.16%), calculate the average atomic mass of silver.

Step 1 - Convert the percentage abundances to fractions: Ag-107 = 0.5184, Ag-109 = 0.4816.

Step 2 - Contribution of Ag-107 = 106.90509 amu x 0.5184 = 55.4196 amu.

Step 3 - Contribution of Ag-109 = 108.90476 amu x 0.4816 = 52.4485 amu.

Step 4 - Average atomic mass = 55.4196 + 52.4485 = 107.868 amu.

Answer: 107.87 amu (book prints 107.88 amu; the difference is rounding of the isotopic masses).

7. Boron with atomic number 5 has two naturally occurring isotopes. Calculate the percentage abundance of 10B and 11B from: average atomic mass of boron = 10.81 amu; isotopic mass of 10B = 10.0129 amu; isotopic mass of 11B = 11.0093 amu.

Step 1 - Let the percentage of 10B be x. Then the percentage of 11B is (100 - x).

Step 2 - Average mass equation: [10.0129 x + 11.0093 (100 - x)] / 100 = 10.81.

Step 3 - Multiply out: 10.0129x + 1100.93 - 11.0093x = 1081.

Step 4 - Collect terms: -0.9964x = 1081 - 1100.93 = -19.93.

Step 5 - x = 19.93 / 0.9964 = 20.00.

Answer: 10B = 20.00%, 11B = 80.00% (book prints 20.002% and 79.998%; the difference is rounding).

8. Define the following terms and give three examples of each: (i) Gram atom (ii) Gram molecular mass (iii) Gram formula (iv) Gram ion (v) Molar volume (vi) Avogadro's number (vii) Stoichiometry (viii) Percentage yield.

(i) Gram atom: the mass in grams of one mole of atoms, numerically equal to the atomic mass; e.g. 12 g of carbon, 16 g of oxygen, 23 g of sodium. (ii) Gram molecular mass: the mass in grams of one mole of molecules, numerically equal to the molecular mass; e.g. 18 g of H2O, 32 g of O2, 44 g of CO2. (iii) Gram formula: the mass in grams of one mole of formula units of an ionic compound, numerically equal to its formula mass; e.g. 58.5 g of NaCl, 100 g of CaCO3, 98 g of H2SO4 (as formula mass). (iv) Gram ion: the mass in grams of one mole of ions; e.g. 23 g of Na+, 35.5 g of Cl-, 16 g of O2-. (v) Molar volume: the volume occupied by one mole of a gas at a stated temperature and pressure; e.g. 22.414 dm3 at S.T.P., 24 dm3 at room temperature and pressure. (vi) Avogadro's number: 6.02 x 10^23, the number of particles in one mole of any substance. (vii) Stoichiometry: the study of the quantitative (mole and mass) relationships between reactants and products in a balanced chemical equation. (viii) Percentage yield: (actual yield / theoretical yield) x 100, the efficiency of a reaction.

9. Justify: (a) 23 g of sodium and 238 g of uranium have equal number of atoms in them. (b) Mg atom is twice heavier than that of carbon atom. (c) 180 g of glucose and 342 g of sucrose have the same number of molecules but different number of atoms present in them. (d) 4.9 g of H2SO4 when completely ionized in water, have equal number of positive and negative charges but the number of positively charged ions are twice the number of negatively charged ions. (e) One mg of K2CrO4 has thrice the number of ions than the number of formula units when

ionized in water. (f) Two grams of H2, 16 g of CH4 and 44 g of CO2 occupy separately the volumes of 22.414 dm3, although the sizes and masses of molecules of three gases are very different from each other.

(a) 23 g Na = 23/23 = 1 mol and 238 g U = 238/238 = 1 mol, so both contain 6.02 x 10^23 atoms. (b) Atomic mass of Mg is 24 and of C is 12, so one Mg atom is 24/12 = 2 times heavier than one C atom. (c) 180 g glucose (molar mass 180) = 1 mol and 342 g sucrose (molar mass 342) = 1 mol, so both have the same number of molecules; but a glucose molecule has 24 atoms while a sucrose molecule has 45 atoms, so the total atom counts differ. (d) 4.9 g H2SO4 = 4.9/98 = 0.05 mol, giving 0.10 mol H+ ions and 0.05 mol SO4(2-) ions: the positive ions are twice as many in number, but their total charge (+0.10) equals the total negative charge (-0.10). (e) One formula unit K2CrO4 gives three ions (2 K+ and 1 CrO4(2-)) on ionisation, so the number of ions is three times the number of formula units. (f) Each sample is 1 mol of gas (2/2, 16/16, 44/44), and by Avogadro's hypothesis equal moles of gases occupy equal volumes at the same temperature and pressure, 22.414 dm3 at S.T.P., whatever the size or mass of the molecules.

10. Calculate each of the following quantities. (a) Mass in grams of 2.74 moles of KMnO4. (b) Moles of O atoms in 9.00 g of Mg(NO3)2. (c) Number of O atoms in 10.037 g of CuSO4.5H2O. (d) Mass in kilograms of 2.6 x 10^20 molecules of SO2. (e) Moles of Cl atoms in 0.822 g of C2H2Cl2. (f) Mass in grams of 5.136 moles of Ag2CO3. (g) Mass in grams of 2.78 x 10^21 molecules of CrO2Cl2. (h) Number of moles and formula units in 100 g of KClO3. (i) Number of K+ ions, ClO3- ions, Cl atoms and O atoms in (h).

Step 1 (a) - Molar mass of KMnO4 = 158 g mol-1. Mass = 2.74 mol x 158 g mol-1 = 432.92 g.

Step 2 (b) - Molar mass of Mg(NO3)2 = 148 g mol-1. Moles of compound = 9.00/148

= 0.0608 mol. Each formula unit has 6 O atoms, so moles of O atoms = 0.0608 x 6

= 0.36 mol.

Step 3 (c) - Molar mass of CuSO4.5H2O = 249.68 g mol-1. Moles = 10.037/249.68 = 0.0402 mol. Each formula unit has 9 O atoms (4 + 5), so O atoms = 0.0402 x 9 x 6.02 x 10^23 = 2.18 x 10^23 atoms.

Step 4 (d) - Moles of SO2 = 2.6 x 10^20 / 6.02 x 10^23 = 4.32 x 10^-4 mol. Mass = 4.32 x 10^-4 mol x 64 g mol-1 = 0.0276 g = 2.76 x 10^-5 kg. (Book prints 2.70 x 10^-5 kg; recomputation gives 2.76 x 10^-5 kg - flagged.)

Step 5 (e) - Molar mass of C2H2Cl2 = 97 g mol-1. Moles of compound = 0.822/97 = 0.00847 mol. Each molecule has 2 Cl atoms, so moles of Cl atoms = 0.0169 mol. (Book prints 0.0178 mol. RE-CHECKED by Worker B5 at 300 dpi, book p. 27: the printed mass reads 0.822 g, not 0.862 g, so the book's printed answer does not follow from its own question; 0.0169 mol is the correct value for the printed mass.)

Step 6 (f) - Molar mass of Ag2CO3 = 275.75 g mol-1. Mass = 5.136 mol x 275.75 g mol-1 = 1416.2 g.

Step 7 (g) - Moles of CrO2Cl2 = 2.78 x 10^21 / 6.02 x 10^23 = 0.00462 mol. Mass = 0.00462 mol x 155 g mol-1 = 0.7158 g.

Step 8 (h) - Molar mass of KClO3 = 122.55 g mol-1. Moles = 100/122.55 = 0.816 mol. Formula units = 0.816 x 6.02 x 10^23 = 4.91 x 10^23.

Step 9 (i) - Each formula unit gives 1 K+, 1 ClO3-, 1 Cl atom and 3 O atoms. Hence

K+ ions = 4.91 x 10^23, ClO3- ions = 4.91 x 10^23, Cl atoms = 4.91 x 10^23, O atoms = 3 x 4.91 x 10^23 = 1.47 x 10^24.

Answer: (a) 432.92 g (b) 0.36 mol (c) 2.18 x 10^23 atoms (d) 2.76 x 10^-5 kg (e) 0.0169 mol (f) 1416.2 g (g) 0.7158 g (h) 0.816 mol, 4.91 x 10^23 formula units (i) as in Step 9.

11. Aspartame, the artificial sweetner, has molecular formula C14H18N2O5. (a) What is the mass of one mole of aspartame? (b) How many moles are present in 52 g of aspartame? (c) What is the mass in grams of 10.122 moles of aspartame? (d) How many hydrogen atoms are present in 2.43 g of aspartame?

Step 1 (a) - Molar mass = (14 x 12) + (18 x 1) + (2 x 14) + (5 x 16) = 168 + 18 + 28 + 80 = 294 g mol-1.

Step 2 (b) - Moles in 52 g = 52/294 = 0.177 mol.

Step 3 (c) - Mass of 10.122 mol = 10.122 mol x 294 g mol-1 = 2975.87 g.

Step 4 (d) - Moles in 2.43 g = 2.43/294 = 0.00827 mol. Each molecule has 18 H atoms, so moles of H atoms = 0.00827 x 18 = 0.1488 mol. Number of H atoms = 0.1488 x 6.02 x 10^23 = 8.96 x 10^22.

Answer: (a) 294 g mol-1 (b) 0.177 mol (c) 2975.87 g (d) 8.96 x 10^22 atoms of H.

12. A sample of 0.600 moles of a metal M reacts completely with excess of fluorine to form 46.8 g of MF2. (a) How many moles of F are present in the sample of MF2 that forms? (b) Which element is represented by the symbol M?

Step 1 - The reaction is M + F2 -> MF2, so 0.600 mol of M gives 0.600 mol of MF2.

Step 2 (a) - Each MF2 has 2 F atoms, so moles of F = 0.600 x 2 = 1.2 mol.

Step 3 (b) - Molar mass of MF2 = 46.8 g / 0.600 mol = 78 g mol-1.

Step 4 - Mass of M = 78 - (2 x 19) = 78 - 38 = 40 g mol-1.

Answer: (a) 1.2 moles of F (b) Calcium (atomic mass 40).

13. In each pair, choose the larger of the indicated quantity, or state if the samples are equal. (a) Individual particles: 0.4 mole of oxygen molecules or 0.4 mole of oxygen atoms. (b) Mass: 0.4 mole of ozone molecules or 0.4 mole of oxygen atoms. (c) Mass: 0.6 mole of C2H2 or 0.6 mole of I2. (d) Individual particles: 4.0 g N2O4 or 3.3 g SO2. (e) Total ions: 2.3 moles of NaClO3 or 2.0 moles of MgCl2. (f) Molecules: 11.0 g H2O or 11.0 g H2O2. (g) Na+ ion: 0.500 moles of NaBr or 0.0145 kg of NaCl. (h) Mass: 6.02 x 10^23 atoms of 235U or 6.02 x 10^23 atoms of 238U.

Step 1 (a) - Both are 0.4 mol of particles, so both contain 0.4 x 6.02 x 10^23 particles. Answer: Both are equal.

Step 2 (b) - Mass of ozone = 0.4 x 48 = 19.2 g; mass of oxygen atoms = 0.4 x 16 = 6.4 g. Answer: Ozone.

Step 3 (c) - Mass of C2H2 = 0.6 x 26 = 15.6 g; mass of I2 = 0.6 x 254 = 152.4 g. Answer: I2.

Step 4 (d) - Moles of N2O4 = 4.0/92 = 0.0435 mol; moles of SO2 = 3.3/64 = 0.0516

mol. Answer: SO2.

Step 5 (e) - NaClO3 gives 2 ions per unit: 2.3 x 2 = 4.6 mol of ions. MgCl2 gives 3 ions per unit: 2.0 x 3 = 6.0 mol of ions. Answer: MgCl2.

Step 6 (f) - Moles of H2O = 11.0/18 = 0.611 mol; moles of H2O2 = 11.0/34 = 0.324 mol. Answer: H2O.

Step 7 (g) - NaBr gives 0.500 mol of Na+ ions. NaCl mass = 0.0145 kg = 14.5 g, giving 14.5/58.5 = 0.248 mol of Na+ ions. Answer: NaBr.

Step 8 (h) - 6.02 x 10^23 atoms is 1 mol in each case: mass of 235U = 235 g, mass of 238U = 238 g. Answer: 238U.

14. (a) Calculate the percentage of nitrogen in the four important fertilizers i.e. (i) NH3 (ii) NH2CONH2 (urea) (iii) (NH4)2SO4 (iv) NH4NO3. (b) Calculate the percentage of nitrogen and phosphorus in each of the following: (i) NH4H2PO4 (ii) (NH4)2HPO4 (iii) (NH4)3PO4.

Step 1 (a)(i) - NH3 molar mass = 17. N% = 14/17 x 100 = 82.35%.

Step 2 (a)(ii) - Urea NH2CONH2 molar mass = 60. N% = 28/60 x 100 = 46.67%.

Step 3 (a)(iii) - (NH4)2SO4 molar mass = 132. N% = 28/132 x 100 = 21.21%.

Step 4 (a)(iv) - NH4NO3 molar mass = 80. N% = 28/80 x 100 = 35%.

Step 5 (b)(i) - NH4H2PO4 molar mass = 115. N% = 14/115 x 100 = 12.17%. P% =

31/115 x 100 = 26.96%.

Step 6 (b)(ii) - (NH4)2HPO4 molar mass = 132. N% = 28/132 x 100 = 21.21%. P% = 31/132 x 100 = 23.48%.

Step 7 (b)(iii) - (NH4)3PO4 molar mass = 149. N% = 42/149 x 100 = 28.18%. P% = 31/149 x 100 = 20.81%.

Answer: (a) 82.35%, 46.67%, 21.21%, 35% (b) (i) N = 12.17%, P = 26.96% (ii) N = 21.21%, P = 23.48% (iii) N = 28.18%, P = 20.81%. All match the book's printed answers.

15. Glucose C6H12O6 is the most important nutrient in the cell for generating chemical potential energy. Calculate the mass % of each element in glucose and determine the number of C, H and O atoms in 10.5 g of the sample.

Step 1 - Molar mass of glucose = (6 x 12) + (12 x 1) + (6 x 16) = 72 + 12 + 96 = 180 g mol-1.

Step 2 - C% = 72/180 x 100 = 40%. H% = 12/180 x 100 = 6.66%. O% = 96/180 x 100 = 53.33%.

Step 3 - Moles in 10.5 g = 10.5/180 = 0.05833 mol.

Step 4 - C atoms = 0.05833 x 6 x 6.02 x 10^23 = 2.107 x 10^23.

Step 5 - H atoms = 0.05833 x 12 x 6.02 x 10^23 = 4.214 x 10^23.

Step 6 - O atoms = 0.05833 x 6 x 6.02 x 10^23 = 2.107 x 10^23.

Answer: C = 40%, H = 6.66%, O = 53.33%; C = 2.107 x 10^23, H = 4.214 x 10^23, O = 2.107 x 10^23 atoms. Matches the book's printed answers.

16. Ethylene glycol is used as automobile antifreeze. It has 38.7% carbon, 9.7% hydrogen and 51.6% oxygen. Its molar mass is 62.1 grams mol-1. Determine its empirical formula.

Step 1 - Take 100 g of the compound: it contains 38.7 g C, 9.7 g H and 51.6 g O.

Step 2 - Moles: C = 38.7/12 = 3.225 mol; H = 9.7/1 = 9.7 mol; O = 51.6/16 = 3.225 mol.

Step 3 - Divide by the smallest (3.225): C = 1, H = 3.01 (about 3), O = 1.

Answer: Empirical formula CH3O (book prints C H3 O).

17. Serotonin (molar mass = 176 g mol-1) is a compound that conducts nerve impulses in brain and muscles. It contains 68.2% C, 6.86% H, 15.09% N and 9.08% O. What is its molecular formula?

Step 1 - Take 100 g: moles C = 68.2/12 = 5.683; H = 6.86/1 = 6.86; N = 15.09/14 = 1.078; O = 9.08/16 = 0.5675.

Step 2 - Divide by the smallest (0.5675): C = 10.01 (about 10), H = 12.09 (about 12), N = 1.90 (about 2), O = 1.

Step 3 - Empirical formula = C10H12N2O. Empirical formula mass = (10 x 12) + (12

x 1) + (2 x 14) + 16 = 176 g mol-1.

Step 4 - n = molar mass / empirical formula mass = 176/176 = 1.

Answer: Molecular formula C10H12N2O (matches the book's printed answer).

18. An unknown metal M reacts with S to form a compound with a formula M2S3. If 3.12 g of M reacts with exactly 2.88 g of sulphur, what are the names of metal M and the compound M2S3?

Step 1 - Moles of sulphur = 2.88 g / 32 g mol-1 = 0.09 mol of S atoms.

Step 2 - In M2S3 the ratio M:S is 2:3, so moles of M = 0.09 x (2/3) = 0.06 mol.

Step 3 - Atomic mass of M = 3.12 g / 0.06 mol = 52 g mol-1.

Step 4 - The element with atomic mass 52 is chromium.

Answer: Metal M is chromium (Cr); the compound is chromium(III) sulphide, Cr2S3 (matches the book's printed answer).

19. The octane present in gasoline burns according to: 2C8H18(l) + 25O2(g) -> 16CO2(g) + 18H2O(l). (a) How many moles of O2 are needed to react fully with 4 moles of octane? (b) How many moles of CO2 can be produced from one mole of octane? (c) How many moles of water are produced by the combustion of 6 moles of octane? (d) If this reaction is to be used to synthesize 8 moles of CO2, how many grams of oxygen are needed? How many grams of octane will be used?

Step 1 (a) - From the equation, 2 mol octane need 25 mol O2. For 4 mol octane: 4 x

(25/2) = 50 mol of O2.

Step 2 (b) - 2 mol octane give 16 mol CO2, so 1 mol octane gives 16/2 = 8 mol of CO2.

Step 3 (c) - 2 mol octane give 18 mol H2O, so 6 mol octane give 6 x (18/2) = 54 mol of H2O.

Step 4 (d) - 16 mol CO2 come from 2 mol octane, so 8 mol CO2 come from 1 mol octane = 114 g of octane. Oxygen needed = 1 x (25/2) = 12.5 mol = 12.5 x 32 = 400 g.

Answer: (a) 50 moles (b) 8 moles (c) 54 moles (d) 400 g of oxygen, 114 g of octane. All match the book's printed answers.

20. Calculate the number of grams of Al2S3 which can be prepared by the reaction of 20 g of Al and 30 g of sulphur. How much the non-limiting reactant is in excess?

Step 1 - Balanced equation: 2Al + 3S -> Al2S3.

Step 2 - Moles of Al = 20/27 = 0.741 mol. Moles of S = 30/32 = 0.9375 mol.

Step 3 - Sulphur needed for all the aluminium = 0.741 x (3/2) = 1.111 mol, which is more than the 0.9375 mol available. Sulphur is therefore the limiting reactant and aluminium is in excess.

Step 4 - Moles of Al2S3 formed = 0.9375/3 = 0.3125 mol. Mass = 0.3125 x 150 = 46.87 g.

Step 5 - Aluminium used = 0.9375 x (2/3) = 0.625 mol = 0.625 x 27 = 16.875 g. Aluminium in excess = 20 - 16.875 = 3.125 g.

Answer: 46.87 g of Al2S3; 3.125 g of aluminium in excess (matches the book's printed answers).

21. A mixture of two liquids, hydrazine N2H4 and N2O4, are used in rockets. They produce N2 and water vapours. How many grams of N2 gas will be formed by reacting 100 g of N2H4 and 200 g of N2O4?

Step 1 - Balanced equation: 2N2H4 + N2O4 -> 3N2 + 4H2O.

Step 2 - Moles of N2H4 = 100/32 = 3.125 mol. Moles of N2O4 = 200/92 = 2.174 mol.

Step 3 - N2O4 needed for all the hydrazine = 3.125/2 = 1.5625 mol, which is less than the 2.174 mol available. Hydrazine is therefore the limiting reactant.

Step 4 - Moles of N2 formed = 3.125 x (3/2) = 4.6875 mol. Mass = 4.6875 x 28 = 131.25 g.

Answer: About 131 g of N2 (book prints 131.04 g; the small difference is the rounding of atomic masses).

22. Silicon carbide (SiC) is an important ceramic material. It is produced by allowing sand (SiO2) to react with carbon at high temperature: SiO2 + 3C -> SiC + 2CO. When 100 kg sand is reacted with excess of carbon, 51.4 kg of SiC is produced. What is the percentage yield of SiC?

Step 1 - Moles of SiO2 = 100 kg / 60 kg kmol-1 = 1.667 kmol.

Step 2 - From the equation, 1 mol SiO2 gives 1 mol SiC, so theoretical moles of SiC = 1.667 kmol.

Step 3 - Theoretical mass of SiC = 1.667 kmol x 40 kg kmol-1 = 66.67 kg.

Step 4 - Percentage yield = (actual yield / theoretical yield) x 100 = (51.4/66.67) x 100 = 77.1%.

Answer: 77% (matches the book's printed answer).

23(a) What is stoichiometry? Give its assumptions? Mention two important laws, which help to perform the stoichiometric calculations?

Stoichiometry is the study of the quantitative relationships between reactants and products in a chemical reaction, worked out from the balanced chemical equation. Its assumptions are: (1) the reaction goes to completion as written, (2) all reactants and products are pure, and (3) the equation is balanced so that atoms are conserved. The two laws that make stoichiometric calculations possible are the law of conservation of mass and the law of definite proportions (together with Avogadro's hypothesis for gases).

23(b) What is a limiting reactant? How does it control the quantity of the product formed? Explain with three examples?

A limiting reactant is the reactant that is used up first in a reaction and therefore limits how much product can form; once it is finished, the reaction stops even if other reactants remain. It controls the product quantity because the product moles are fixed by the limiting reactant's moles through the mole ratio of the balanced

equation. Examples: (1) In 2H2 + O2 -> 2H2O, hydrogen in short supply limits the water formed. (2) In Q.20 above, sulphur limits the Al2S3 formed from aluminium and sulphur. (3) In Q.21 above, hydrazine limits the nitrogen gas formed from hydrazine and N2O4.

24(a) Define yield. How do we calculate the percentage yield of a chemical reaction?

Yield is the amount of product actually obtained from a chemical reaction. The theoretical yield is the amount predicted by the balanced equation if the reaction went to completion. Percentage yield = (actual yield / theoretical yield) x 100. It is always 100% or less in practice, because of incomplete reactions, side reactions and losses during separation.

24(b) What are the factors which are mostly responsible for the low yield of the products in chemical reactions?

Low yields are mostly caused by: (1) the reaction not going to completion (reversible reactions reach equilibrium); (2) side reactions that consume reactants to give unwanted products; (3) loss of product during separation, filtration, washing and transfer; (4) impure reactants; and (5) unfavourable conditions of temperature, pressure or catalyst.

25. Explain the following with reasons: (i) Law of conservation of mass has to be obeyed during stoichiometric calculations. (ii) Many chemical reactions taking place in our surrounding involve the limiting reactants. (iii) No individual neon atom in the sample of the element has a mass of 20.18 amu. (iv) One mole of H2SO4 should completely react with two moles of NaOH. How does Avogadro's number help to explain it? (v) One mole of H2O has two moles of bonds, three moles of atoms, ten moles of electrons and twenty eight moles of the total fundamental particles present in it. (vi) N2 and CO have the same number of electrons, protons and neutrons.

(i) Atoms are neither created nor destroyed in a chemical reaction, so the total mass of reactants must equal the total mass of products; stoichiometric calculations use this to balance equations and to relate masses. (ii) In everyday reactions the reactants are rarely mixed in exact stoichiometric proportions, and one of them (for example oxygen from the air, or one ingredient) runs out first, so that reactant limits the reaction. (iii) 20.18 amu is the weighted average atomic mass of neon's isotopes (Ne-20, Ne-21, Ne-22); no single neon atom has that mass, each atom has mass 20, 21 or 22 amu. (iv) H2SO4 + 2NaOH -> Na2SO4 + 2H2O: one mole of acid supplies 2 moles of H+ ions, which need 2 moles of OH- ions; Avogadro's number converts these mole ratios into equal numbers of particles (2 x 6.02 x 10^23 of each). (v) One H2O molecule has 2 O-H bonds, 3 atoms (2 H + 1 O), 10 electrons (2 + 8) and 28 fundamental particles (10 protons + 8 neutrons + 10 electrons, counting H: 1 proton each and O: 8 protons + 8 neutrons); one mole multiplies each count by Avogadro's number. (vi) N2 has 14 protons, 14 electrons and 14 neutrons (7 + 7 each); CO has 6 + 8 = 14 protons, 14 electrons and 6 + 8 = 14 neutrons, so both molecules have the same counts.

Chapter 2 - Experimental Techniques in Chemistry

Exercise 2 - Multiple Choice Questions

1(i) A filtration process could be very time consuming if it were not aided by a gentle suction which is developed

A. if the paper covers the funnel upto its circumference

B. if the paper has got small sized pores in it

C. if the stem of the funnel is large so that it dips into the filtrate

✓ D. if the paper fits tightly

1(ii) During the process of crystallization, the hot saturated solution

✓ A. is cooled very slowly to get large sized crystals

B. is cooled at a moderate rate to get medium sized crystals

C. is evaporated to get the crystals of the product

D. is mixed with an immiscible liquid to get the pure crystals of the product

1(iii) Solvent extraction is an equilibrium process and it is controlled by

A. law of mass action

B. the amount of solvent used

✓ C. distribution law

D. the amount of solute

1(iv) Solvent extraction method is a particularly useful technique for separation when the product to be separated is

✓ A. non-volatile or thermally unstable

B. volatile or thermally stable

C. non-volatile or thermally stable

D. volatile or thermally unstable

1(v) The comparative rates at which the solutes move in paper chromatography, depend on

A. the size of paper

✓ B. Rf values of solutes

C. temperature of the experiment

D. size of the chromatographic tank used

Exercise 2 - Fill in the Blanks

2(i) A complete chemical characterization of a compound must include both qualitative and quantitative analyses .

2(ii) During filtration the tip of the stem of the funnel should touch the side of the

beaker to avoid splashing.

2(iii) A fluted filter paper is used to speed up the process of filtration.

2(iv) A solvent used for crystallization is required to dissolve _____ of the substance at its boiling point and _____ at the room temperature.

A solvent used for crystallization is required to dissolve a large amount of the substance at its boiling point and only a small amount at room temperature, so that crystals separate on cooling.

2(v) Repeated solvent extractions using small portions of solvent are more efficient than using a single extraction with larger volume of the solvent.

Exercise 2 - True or False (Tick the Correct Sentences)

3(i) Tick the correct sentences. If the sentence is incorrect, write the correct statements. (i) A qualitative analysis involves the identification of elements present in a compound.

Correct. Qualitative analysis identifies which elements are present in a compound.

3(ii) If the process of filtration is to run smoothly, the stem of the funnel should remain empty.

Incorrect. Correct statement: If the process of filtration is to run smoothly, the stem of the funnel should remain continuously full of liquid (not empty), so that a gentle suction is developed.

3(iii) If none of the solvents is found suitable for crystallization a combination of two or more immiscible solvents may be used.

Incorrect. Correct statement: If none of the solvents is found suitable for crystallization, a mixture of two or more miscible solvents (not immiscible solvents) may be used.

3(iv) A solute distributes itself between two immiscible liquids in a constant ratio of concentrations depending upon the amount of solvent added.

Incorrect. Correct statement: A solute distributes itself between two immiscible liquids in a constant ratio of concentrations at a given temperature, independent of the amount of solvent added (the distribution law).

3(v) Paper chromatography is a technique of partition chromatography.

Correct. In paper chromatography the stationary phase is water adsorbed on the paper and the solute partitions between this water and the mobile organic solvent, so it is a partition chromatography.

Exercise 2 - Short Questions

4. Why is there a need to crystallize the crude product?

A crude product contains impurities: soluble impurities, insoluble impurities and coloured matter, together with unchanged starting material. Crystallization purifies it because, on cooling a hot saturated solution, the pure substance separates as crystals while the impurities stay dissolved in the mother liquor (or are removed by filtration), giving a pure, well-defined product of a definite melting point.

5. A water insoluble organic compound aspirin is prepared by the reaction of salicylic acid with a mixture of acetic acid and acetic anhydride. How will you separate the product from the reaction mixture?

Aspirin is insoluble in water, whereas acetic acid and the excess acetic anhydride (which hydrolyses to acetic acid) are water-soluble. Pour the reaction mixture into cold water: aspirin separates as a solid. Filter it off, wash it with cold water to remove the acid, dry it, and recrystallize it from a suitable solvent (for example aqueous ethanol) to obtain pure aspirin.

6. A solid organic compound is soluble in water as well as in chloroform. During its preparation, it remains in aqueous layer. Describe a method to obtain it from this layer.

Use solvent extraction in a separating funnel. Shake the aqueous layer repeatedly

with small portions of chloroform: the compound distributes itself between water and chloroform and passes into the chloroform layer. Separate the lower chloroform layer each time, combine the extracts, and evaporate the chloroform (the solvent is volatile) to leave the solid compound behind.

7. The figure shows a developed chromatogram on paper with five spots: (i) Unknown mixture X (ii) Sample A (iii) Sample B (iv) Sample C (v) Sample D. Find out (i) the composition of unknown mixture X (ii) which sample is impure and what is its composition.

Spots are identified by their height above the base line, that is, by their Rf value: a component of the mixture gives a spot at exactly the same height as the same pure sample. (i) The unknown mixture X shows two spots, one at the height of sample A and one at the height of sample C. Composition of X: A + C. (ii) Sample D shows two spots, one at the height of sample B and one at the height of sample C, although D was supposed to be a single substance. Sample D is therefore impure; its composition is B + C. Samples A, B and C each show a single spot and are pure.

8. In solvent extraction technique, why repeated extraction using small portions of solvent are more efficient than using a single extraction but larger volume of solvent?

By the distribution law, each extraction removes a fixed fraction of the solute remaining in the aqueous layer. With several small portions, the solute is partitioned afresh each time, so the total fraction removed is larger than with one large portion of the same total volume, where after the first (and only) equilibrium a large fraction can remain behind in the water layer. Repeated small extractions therefore recover more solute.

9. Write down the main characteristics of a solvent selected for crystallization of a compound.

A good crystallization solvent must: (1) dissolve a large amount of the substance at

its boiling point but only a small amount at room temperature; (2) not react chemically with the substance; (3) dissolve the impurities either very well (so they stay in the mother liquor) or very poorly (so they can be filtered off hot); (4) give well-defined crystals on cooling; (5) be easily removed from the crystals (volatile) and be cheap and safe to use.

10. You have been provided with a mixture containing three inks with different colours. Write down the procedure to separate the mixture with the help of paper chromatography.

Procedure: (1) Draw a base line near the bottom of a strip of chromatography paper and put a small spot of the ink mixture on it, keeping the spot above the solvent level. (2) Hang the strip in a closed tank so that its lower end dips in a suitable solvent (the mobile phase), with the spot above the solvent. (3) The solvent rises by capillary action; each ink component partitions between the water on the paper and the solvent and moves at its own rate (its own Rf value). (4) When the solvent front has risen far enough, remove the strip, mark the solvent front at once, and let it dry. (5) Three separate coloured spots appear at different heights, one for each ink, and each can be identified by comparing its Rf value with that of a known ink.

Chapter 3 - Gases

Exercise 3 - Multiple Choice Questions

1(i) Pressure remaining constant, at which temperature the volume of a gas will become twice of what it is at 0 degrees C?

A. 546 degrees C

B. 200 degrees C

✓ C. 546 K

D. 273 K

1(ii) Number of molecules in one dm3 of water is close to

A. (6.02/22.4) x 10^23

B. (12.04/22.4) x 10^23

C. (18/22.4) x 10^23

✓ D. 55.6 x 6.02 x 10^23

1(iii) Which of the following will have the same number of molecules at STP?

✓ A. 280 cm3 of CO2 and 280 cm3 of N2O

B. 11.2 dm3 of O2 and 32 g of O2

C. 44 g of CO2 and 11.2 dm3 of CO

D. 28 g of N2 and 5.6 dm3 of oxygen

1(iv) If absolute temperature of a gas is doubled and the pressure is reduced to one half, the volume of the gas will

A. remain unchanged

✓ B. increase four times

C. reduce to 1/4

D. be doubled

1(v) How should the conditions be changed to prevent the volume of a given gas from expanding when its mass is increased?

✓ A. Temperature is lowered and pressure is increased

B. Temperature is increased and pressure is lowered

C. Temperature and pressure both are lowered

D. Temperature and pressure both are increased

1(vi) The molar volume of CO2 is maximum at

A. STP

✓ B. 127 degrees C and 1 atm

C. 0 degrees C and 2 atm

D. 273 degrees C and 2 atm

1(vii) The order of the rate of diffusion of gases NH3, SO2, Cl2 and CO2 is

A. NH3 > SO2 > Cl2 > CO2

✓ B. NH3 > CO2 > SO2 > Cl2

C. Cl2 > SO2 > CO2 > NH3

D. NH3 > CO2 > Cl2 > SO2

1(viii) Equal masses of methane and oxygen are mixed in an empty container at 25 degrees C. The fraction of total pressure exerted by oxygen is

✓ A. 1/3

B. 8/9

C. 1/9

D. 16/17

1(ix) Gases deviate from ideal behaviour at high pressure. Which of the following is correct for non-ideality?

A. At high pressure, the gas molecules move in one direction only

B. At high pressure, the collisions between the gas molecules are increased manifold

C. At high pressure, the volume of the gas becomes insignificant

✓ D. At high pressure, the intermolecular attractions become significant

1(x) The deviation of a gas from ideal behaviour is maximum at

✓ A. -10 degrees C and 5.0 atm

B. -10 degrees C and 2.0 atm

C. 100 degrees C and 2.0 atm

D. 0 degrees C and 2.0 atm

1(xi) A real gas obeying van der Waals equation will resemble ideal gas if

A. both 'a' and 'b' are large

✓ B. both 'a' and 'b' are small

C. 'a' is small and 'b' is large

D. 'a' is large and 'b' is small

Exercise 3 - Fill in the Blanks

2(i) The product PV has the S.I. unit of joule (energy) .

2(ii) Eight grams each of O2 and H2 at 27 degrees C will have total K.E. in the ratio of _____.

Total kinetic energy = (3/2) nRT. Moles: O2 = 8/32 = 0.25 mol; H2 = 8/2 = 4 mol. At the same temperature, total K.E. ratio = 0.25 : 4 = 1 : 16.

2(iii) Smell of the cooking gas during leakage from a gas cylinder is due to the property of diffusion of gases.

2(iv) Equal _____ of ideal gases at the same temperature and pressure contain _____ number of molecules.

Equal volumes of ideal gases at the same temperature and pressure contain equal number of molecules (Avogadro's law).

2(v) The temperature above which a substance exists only as a gas is called critical temperature .

Exercise 3 - True or False

3(i) Kinetic energy of molecules of a gas is zero at 0 degrees C.

False

3(ii) A gas in a closed container will exert much higher pressure at the bottom due to gravity than at the top.

False

3(iii) Real gases show ideal gas behaviour at low pressure and high temperature.

True

3(iv) Liquefaction of gases involves decrease in intermolecular spaces.

True

3(v) An ideal gas on expansion will show Joule-Thomson effect.

False

Exercise 3 - Short and Numerical Questions

4(a) What is Boyle's law of gases? Give its experimental verification.

Boyle's law: at constant temperature, the volume of a fixed mass of a gas is inversely proportional to its pressure, PV = constant. Experimental verification: a fixed mass of gas is trapped in a graduated tube over mercury (or in a syringe with a pressure gauge); as weights are added to increase the pressure, the volume decreases, and the product PV stays constant at constant temperature. A plot of P against 1/V is a straight line through the origin, and a plot of PV against P is a horizontal line.

4(b) What are isotherms? What happens to the positions of isotherms when they are plotted at high temperature for a particular gas.

Isotherms are the curves of P against V for a fixed mass of gas at one constant temperature. At higher temperatures the isotherms move away from the axes (upwards and to the right): at a higher temperature the same volume corresponds to a higher pressure, and the whole PV = constant curve is shifted outward, because PV = nRT is larger.

4(c) Why do we get a straight line when pressures exerted on a gas are plotted against inverse of volumes? This straight line changes its position in the graph by varying the temperature. Justify it.

Because PV = constant at constant temperature, P = constant/V, so P is directly proportional to 1/V and the graph is a straight line through the origin. Raising the temperature increases the constant (PV = nRT), so the line becomes steeper: for the same 1/V the pressure is higher. The slope of the line is therefore nRT and it

increases with temperature.

4(d) How will you explain that the value of the constant k in the equation PV = k depends upon (i) the temperature of a gas (ii) the quantity of a gas.

In PV = k, the constant k = nRT, so it depends on both (i) the temperature of the gas and (ii) the quantity (number of moles) of the gas. It is not a universal constant: doubling the moles or doubling the absolute temperature doubles k.

5(a) What is the Charles's law? Which scale of temperature is used to verify that V/T = k (pressure and number of moles are constant)?

Charles's law: at constant pressure, the volume of a fixed mass of a gas is directly proportional to its absolute temperature, V/T = constant. The Kelvin (absolute) scale is used, because only on that scale does V become zero at -273.16 degrees C and the proportionality V proportional to T hold with a straight line through the origin.

5(b) A sample of carbon monoxide gas occupies 150.0 mL at 25.0 degrees C. It is then cooled at constant pressure until it occupies 100.0 mL. What is the new temperature?

Step 1 - Convert to Kelvin: T1 = 25.0 + 273 = 298 K. V1 = 150.0 mL, V2 = 100.0 mL.

Step 2 - Charles's law: V1/T1 = V2/T2, so T2 = T1 x (V2/V1) = 298 x (100.0/150.0).

Step 3 - T2 = 198.7 K, which is 198.7 - 273 = -74.3 degrees C.

Answer: 198.8 K, about -74.4 degrees C (matches the book's printed answer; the 0.1 K difference is rounding of 273 vs 273.16).

5(c) Do you think that the volume of any quantity of a gas becomes zero at -273.16 degrees C? Is it not against the law of conservation of mass? How do you deduce

the idea of absolute zero from this information?

No, the volume does not really become zero: the gas liquefies (and then solidifies) long before -273.16 degrees C, so Charles's law stops applying. This does not break the law of conservation of mass, because mass is not volume: the same mass occupies a much smaller volume as a liquid. Extrapolating the straight V-T graph for a gas to zero volume gives the intercept -273.16 degrees C, the absolute zero, the temperature at which the volume of an ideal gas would be zero and molecular motion would cease.

6(a) What is Kelvin scale of temperature? Plot a graph for one mole of a real gas to prove that a gas becomes liquid, earlier than -273.16 degrees C.

The Kelvin scale starts at absolute zero, -273.16 degrees C = 0 K, and has the same degree size as the Celsius scale: T(K) = t(degrees C) + 273.16. A real gas does not reach -273.16 degrees C as a gas: on cooling it liquefies first (its V-T curve bends away from the ideal straight line and drops to the small liquid volume), so the straight-line extrapolation, not an actual measurement, gives absolute zero.

6(b) Throw some light on the factor 1/273 in Charles's law.

Charles found that the volume of a gas at constant pressure increases by 1/273 of its volume at 0 degrees C for every 1 degree C rise. That is, Vt = V0 (1 + t/273) = V0 (273 + t)/273. The factor 1/273 is therefore the coefficient of thermal expansion of an ideal gas: it is the fraction by which the volume grows per degree, and it is the same for all gases because it comes from the absolute temperature scale (273 K at 0 degrees C).

7(a) What is the general gas equation? Derive it in various forms.

The general gas equation is PV = nRT, obtained by combining Boyle's law (V proportional to 1/P), Charles's law (V proportional to T) and Avogadro's law (V proportional to n). Its forms include: PV = nRT; P = (m/M) RT/V; PM = dRT (density form, since d = m/V); P1V1/T1 = P2V2/T2 for a fixed mass; and V = nRT/P.

7(b) Can we determine the molecular mass of an unknown gas if we know the pressure, temperature and volume along with the mass of that gas?

Yes. From PV = nRT, n = PV/RT, and the mass m is known, so the molar mass M = m/n = mRT/PV. The pressure, temperature, volume and mass are all measurable, so M follows directly.

7(c) How do you justify from general gas equation that increase in temperature or decrease of pressure decreases the density of the gas?

From PM = dRT, density d = PM/RT. At constant pressure, increasing T increases the denominator, so d decreases: the same mass occupies a larger volume. Decreasing P decreases the numerator, so d also decreases. In words, heating a gas or lowering its pressure spreads the same mass over a larger volume.

7(d) Why do we feel comfortable in expressing the densities of gases in the units of g dm-3 rather than g cm-3, a unit which is used to express the densities of liquids and solids?

Gas densities are very small: for example oxygen is about 1.43 g per dm3, whereas liquids and solids are of the order of grams per cm3, a thousand times larger. Expressing a gas density in g per cm3 would give awkwardly small numbers (0.00143 g cm-3), while g per dm3 gives convenient numbers (1.43 g dm-3).

8. Derive the units for gas constant R in general gas equation: (a) when the pressure is in atmosphere and volume in dm3. (b) when the pressure is in N m-2 and volume in m3. (c) when energy is expressed in ergs.

R = PV/nT. (a) P in atm, V in dm3: R = 0.0821 atm dm3 mol-1 K-1 (from 1 mol at STP: 1 x 22.414 / 273). (b) P in N m-2 (Pa), V in m3: R = 8.314 N m mol-1 K-1 = 8.314 J mol-1 K-1. (c) Energy in ergs: 1 J = 10^7 erg, so R = 8.314 x 10^7 erg mol-1 K-1.

9(a) What is Avogadro's law of gases?

Avogadro's law: at the same temperature and pressure, equal volumes of all gases

contain equal numbers of molecules (equal moles). One mole of any gas occupies 22.414 dm3 at S.T.P.

9(b) Do you think that 1 mole of H2 and 1 mole of NH3 at 0 degrees C and 1 atm pressure will have Avogadro's number of particles?

Yes, approximately. Both are 1 mol of gas at the same temperature and pressure, so by Avogadro's law both contain Avogadro's number of particles (6.02 x 10^23 molecules). Real-gas deviations are small at 0 degrees C and 1 atm, and the count is the same for both gases.

9(c) Justify that 1 cm3 of H2 and 1 cm3 of CH4 at STP will have same number of molecules, when one molecule of CH4 is 8 times heavier than that of hydrogen.

Both samples have the same volume (1 cm3) at the same temperature and pressure, so by Avogadro's law they contain the same number of molecules. The mass of a molecule does not enter Avogadro's law: the number of molecules depends only on volume, temperature and pressure, not on how heavy each molecule is. (That the CH4 molecule is 8 times heavier only means the CH4 sample is 8 times heavier in mass, not in molecule count.)

10(a) Dalton's law of partial pressures is only obeyed by those gases which don't have attractive forces among their molecules. Explain it.

Dalton's law assumes the gases in a mixture behave independently, each exerting the pressure it would exert alone in the same volume. If the molecules attract each other, they no longer act independently: the total pressure is changed by the attractions, and the partial pressures do not simply add. The law is therefore exact only for gases with negligible intermolecular attractions, that is, for ideal gases (or real gases at low pressure and high temperature).

10(b) Derive an equation to find out the partial pressure of a gas knowing the individual moles of component gases and the total pressure of the mixture.

For a mixture, the partial pressure of each gas is pi = (ni / n total) x P total, where ni

is its moles and P total is the total pressure. Derivation: each gas obeys PV = ni RT in the same volume V at the same temperature, so pi = ni RT/V; adding all partial pressures gives P total = (sum ni) RT/V, and dividing gives pi = (ni / n total) x P total, that is, partial pressure = mole fraction x total pressure.

10(c) Explain that the process of respiration obeys the Dalton's law of partial pressures.

Respiration obeys Dalton's law because the air in the lungs is a mixture of gases, each contributing its own partial pressure. Inhaled air has a high partial pressure of oxygen (about 159 mmHg) and a low partial pressure of carbon dioxide, while venous blood has a lower oxygen and higher carbon dioxide partial pressure. Oxygen therefore diffuses from the alveoli into the blood and carbon dioxide diffuses the other way, each gas moving down its own partial-pressure gradient, exactly as Dalton's law describes for a mixture.

10(d) How do you differentiate between diffusion and effusion? Explain Graham's law of diffusion.

Diffusion is the spontaneous spreading of a gas through another gas (or through space) by random molecular motion, without a hole or barrier. Effusion is the escape of a gas through a tiny hole (pinhole) into a vacuum or a region of lower pressure, molecule by molecule. Graham's law of diffusion: the rate of diffusion (or effusion) of a gas is inversely proportional to the square root of its density (or molar mass) at constant temperature and pressure: r1/r2 = sqrt(M2/M1). Lighter gases diffuse faster.

11(a) What is critical temperature of a gas? What is its importance for liquefaction of gases? Discuss Linde's method of liquefaction of gases.

The critical temperature is the highest temperature at which a gas can be liquefied by pressure alone; above it, no amount of pressure can liquefy the gas. It matters because a gas must be cooled to or below its critical temperature before

compression can liquefy it. Linde's method: air is compressed, cooled, and then allowed to expand through a nozzle (Joule-Thomson expansion); the expansion cools the gas further, and the cold gas cools the incoming compressed gas in a heat exchanger. Repeated cycles lower the temperature until the gas liquefies, and the liquid is collected.

11(b) What is Joule-Thomson effect? Explain its importance in Linde's method of liquefaction of gases.

The Joule-Thomson effect is the cooling (for most gases below their inversion temperature) that occurs when a gas expands from a high pressure to a low pressure through a nozzle or porous plug without exchanging heat with the surroundings. In Linde's method this expansion cooling is the working step: each expansion makes the gas colder, the cold outgoing gas pre-cools the incoming compressed gas, and after many cycles the temperature falls below the critical temperature and the gas liquefies.

12(a) What is kinetic molecular theory of gases? Give its postulates.

Kinetic molecular theory of gases assumes: (1) a gas consists of a large number of tiny particles in continuous random motion; (2) the particles are negligibly small compared with the distances between them, so their own volume is negligible; (3) there are no attractive or repulsive forces between the particles; (4) collisions between particles and with the walls are perfectly elastic, so no kinetic energy is lost; (5) the average kinetic energy of the particles is directly proportional to the absolute temperature; (6) the pressure of a gas is due to the collisions of its particles with the walls of the container.

12(b) How does kinetic molecular theory of gases explain: (i) Boyle's law (ii) Charles's law (iii) Avogadro's law (iv) Graham's law of diffusion.

(i) Boyle's law: at constant temperature the kinetic energy (and hence the average speed) is fixed; compressing the gas into a smaller volume makes the particles hit

the walls more often, so the pressure rises in inverse proportion to the volume. (ii) Charles's law: heating a gas increases the kinetic energy and speed of its particles, so at constant pressure they push the walls further out and the volume grows in proportion to the absolute temperature. (iii) Avogadro's law: at the same temperature and pressure, equal volumes contain equal numbers of particles, because the pressure depends only on the number of particles and their kinetic energy, not on their chemical identity. (iv) Graham's law: at the same temperature all gases have the same average kinetic energy (1/2 m v^2), so lighter particles move faster: v is proportional to 1/sqrt(m), and the rate of diffusion is proportional to the molecular speed.

13(a) Gases show non-ideal behaviour at low temperature and high pressure. Explain this with the help of a graph.

At low temperature the kinetic energy of the molecules is small, so intermolecular attractions can hold molecules together and reduce the pressure below the ideal value; at high pressure the molecules are crowded together, so their own finite volume becomes a significant part of the container volume and the free volume is less than V. On a PV-versus-P plot an ideal gas gives a horizontal line (PV = nRT constant), but a real gas curve dips below it at moderate pressures (attractions dominate) and rises above it at high pressures (molecular volume dominates).

13(b) Do you think that some of the postulates of kinetic molecular theory of gases are faulty? Point out these postulates.

Yes, two postulates are faulty for real gases: (1) that the volume of the gas molecules themselves is negligible - at high pressure it is a significant fraction of the container volume; and (2) that there are no intermolecular attractions - at low temperature attractions are significant and reduce the pressure. The other postulates (random motion, elastic collisions, pressure from wall collisions, kinetic energy proportional to temperature) remain good approximations.

13(c) Hydrogen and helium are ideal at room temperature, but SO2 and Cl2 are non-ideal. How will you explain this?

Hydrogen and helium are very small, light, non-polar molecules with extremely weak intermolecular attractions, so both faulty postulates are nearly true for them even at room temperature: their molecular volume is tiny and their attractions are negligible. SO2 and Cl2 are larger molecules with stronger attractions (SO2 is polar, Cl2 is large and easily polarised), so their attractions and molecular volumes are significant and they deviate noticeably from ideal behaviour.

14(a) Derive van der Waals equation for real gases.

The ideal equation PV = nRT is corrected in two ways. (1) The molecules themselves occupy volume nb, so the free volume is (V - nb). (2) Attractions between molecules reduce the pressure by a term proportional to the square of the concentration, a n^2/V^2, so the observed pressure P must be increased by that amount. Hence (P + a n^2/V^2)(V - nb) = nRT, the van der Waals equation for n moles (for 1 mol: (P + a/V^2)(V - b) = RT).

14(b) What is the physical significance of van der Waals constants, 'a' and 'b'? Give their units.

'a' measures the strength of the intermolecular attractions: larger 'a' means stronger attractions and a larger pressure reduction. Its units are atm dm6 mol-2 (pressure x volume^2 / moles^2). 'b' measures the volume actually occupied by one mole of molecules (the excluded volume): larger 'b' means larger molecules. Its units are dm3 mol-1.

15. Explain: (i) The plot of PV versus P is a straight line at constant temperature and with a fixed number of moles of an ideal gas. (ii) The straight line in (a) is parallel to pressure-axis and goes away from the pressure axis at higher pressures for many gases. (iii) Pressure of NH3 gas at given conditions (say 20 atm pressure and room temperature) is less as calculated by van der Waals equation than that calculated

by general gas equation. (iv) Water vapours do not behave ideally at 273 K. (v) SO2 is comparatively non-ideal at 273 K but behaves ideally at 327 degrees C.

(i) For an ideal gas at constant temperature, PV = nRT is constant, so a plot of PV against P is a horizontal straight line. (ii) For many real gases the line is horizontal only at low pressure; at higher pressures it moves away from the pressure axis (rises), because the finite molecular volume makes PV larger than nRT. (iii) NH3 has strong intermolecular attractions (hydrogen bonding, polar), so the attraction term lowers its pressure below the ideal value, as van der Waals equation predicts. (iv) At 273 K water vapour is close to its condensation point, so attractions between molecules are strong and it behaves non-ideally. (v) SO2 is fairly easily liquefied and its attractions are significant at 273 K, but at 327 degrees C (600 K) the kinetic energy is so large that attractions become negligible and it behaves almost ideally.

16. Helium gas in a 100 cm3 container at a pressure of 500 torr is transferred to a container with a volume of 250 cm3. What will be the new pressure (a) if no change in temperature occurs (b) if its temperature changes from 20 degrees C to 15 degrees C?

Step 1 (a) - Boyle's law at constant temperature: P1V1 = P2V2, so P2 = P1 x (V1/V2) = 500 torr x (100/250).

Step 2 (a) - P2 = 200 torr.

Step 3 (b) - Combined gas law: P2 = P1 x (V1/V2) x (T2/T1). T1 = 20 + 273 = 293 K, T2 = 15 + 273 = 288 K.

Step 4 (b) - P2 = 500 x (100/250) x (288/293) = 500 x 0.4 x 0.9829 = 196.6 torr.

Answer: (a) 200 torr (b) 196.56 torr (matches the book's printed answers).

17. (a) What are the densities in kg/m3 of the following gases at STP (P = 101325 N m-2, T = 273 K, molecular masses are in kg mol-1): (i) methane, (ii) oxygen, (iii) hydrogen. (b) Compare the values of densities in proportion to their mole masses. (c) How do you justify that increase of volume up to 100 dm3 at 27 degrees C of 2 moles of NH3 will allow the gas behave ideally, as compared to S.T.P. conditions?

Step 1 (a) - At STP one mole occupies 22.414 dm3 = 0.022414 m3. Density = molar mass / molar volume.

Step 2 - Methane: 0.016 kg mol-1 / 0.022414 m3 mol-1 = 0.714 kg m-3. Oxygen: 0.032/0.022414 = 1.428 kg m-3. Hydrogen: 0.002/0.022414 = 0.089 kg m-3.

Step 3 - These are the same numbers as 0.714, 1.428 and 0.089 g dm-3; the book's printed values (0.714, 1.428, 0.089) are in kg m-3, i.e. g dm-3, not kg dm-3 as the question's wording states - flagged.

Step 4 (b) - The densities are in the ratio 0.714 : 1.428 : 0.089, which is the same as the molar-mass ratio 16 : 32 : 2, because at the same temperature and pressure all gases have the same molar volume.

Step 5 (c) - At STP, 2 mol of NH3 occupy only 44.8 dm3, so the molecules are close together and attractions (and the molecular volume) matter. Expanded to 100 dm3 at 27 degrees C, the pressure falls and the molecules are much further apart, so attractions become negligible and the free volume is essentially the whole volume: the gas behaves ideally.

Answer: (a) CH4 = 0.714 kg m-3, O2 = 1.428 kg m-3, H2 = 0.089 kg m-3 (matches the book's printed answers, with the unit note in Step 3).

18. A sample of krypton with a volume of 6.25 dm3, a pressure of 765 torr and a temperature of 20 degrees C is expanded to a volume of 9.55 dm3 and a pressure of 375 torr. What will be its final temperature in degrees C?

Step 1 - Combined gas law: T2 = T1 x (P2/P1) x (V2/V1). T1 = 20 + 273 = 293 K.

Step 2 - P2/P1 = 375/765 = 0.4902. V2/V1 = 9.55/6.25 = 1.528.

Step 3 - T2 = 293 x 0.4902 x 1.528 = 219.5 K.

Step 4 - In Celsius: 219.5 - 273 = -53.5 degrees C.

Answer: T = -53.6 degrees C (matches the book's printed answer; the 0.1 degree difference is rounding).

19. Working at a vacuum line, a chemist isolated a gas in a weighing bulb with a volume of 255 cm3, at a temperature of 25 degrees C and under a pressure in the bulb of 10.0 torr. The gas weighed 12.1 mg. What is the molecular mass of this gas?

Step 1 - Convert: V = 255 cm3 = 0.255 dm3; T = 25 + 273 = 298 K; P = 10.0/760 = 0.01316 atm.

Step 2 - Moles from PV = nRT: n = PV/RT = (0.01316 x 0.255) / (0.0821 x 298) = 0.003356 / 24.466 = 1.372 x 10^-4 mol.

Step 3 - Molar mass = mass / moles = 0.0121 g / 1.372 x 10^-4 mol = 88.2 g mol-1.

Answer: About 88 g mol-1 (book prints 87.93 g mol-1; the small difference is rounding of R and the pressure conversion).

20. What pressure is exerted by a mixture of 2.00 g of H2 and 8.00 g of N2 at 273 K in a 10 dm3 vessel?

Step 1 - Moles of H2 = 2.00/2 = 1.00 mol. Moles of N2 = 8.00/28 = 0.2857 mol.

Step 2 - Total moles = 1.00 + 0.2857 = 1.2857 mol.

Step 3 - From PV = nRT: P = nRT/V = (1.2857 x 0.0821 x 273) / 10.

Step 4 - P = 28.82 / 10 = 2.88 atm.

Answer: P = 2.88 atm (matches the book's printed answer).

21. (a) The relative densities of two gases A and B are 1:1.5. Find out the volume of B which will diffuse in the same time in which 150 dm3 of A will diffuse. (b) Hydrogen (H2) diffuses through a porous plate at a rate of 500 cm3 per minute at 0 degrees C. What is the rate of diffusion of oxygen through the same porous plate at 0 degrees C? (c) The rate of effusion of an unknown gas A through a pinhole is found to be 0.279 times the rate of effusion of H2 gas through the same pinhole. Calculate the molecular mass of the unknown gas at STP.

Step 1 (a) - By Graham's law, volumes diffused in the same time are in the ratio of the rates: VB/VA = sqrt(dA/dB) = sqrt(1/1.5) = 0.8165.

Step 2 (a) - VB = 150 x 0.8165 = 122.47 dm3.

Step 3 (b) - Rate of O2 / rate of H2 = sqrt(M_H2 / M_O2) = sqrt(2/32) = sqrt(1/16) = 0.25. Rate of O2 = 500 x 0.25 = 125 cm3 per minute.

Step 4 (c) - Rate ratio = sqrt(M_H2 / M_A) = 0.279. Squaring: M_H2 / M_A = 0.279^2 = 0.07784.

Step 5 (c) - M_A = 2 / 0.07784 = 25.7 g mol-1.

Answer: (a) 122.47 dm3 (b) 125 cm3 per minute (c) 25.7 g mol-1. All match the book's printed answers.

22. Calculate the number of molecules and the number of atoms in the given amounts of each gas: (a) 20 cm3 of CH4 at 0 degrees C and pressure of 700 mm of mercury. (b) 1 cm3 of NH3 at 100 degrees C and pressure of 1.5 atm.

Step 1 (a) - Convert: P = 700/760 = 0.921 atm; V = 20 cm3 = 0.020 dm3; T = 273 K.

Step 2 (a) - Moles = PV/RT = (0.921 x 0.020) / (0.0821 x 273) = 0.01842 / 22.413 = 8.22 x 10^-4 mol.

Step 3 (a) - Molecules = 8.22 x 10^-4 x 6.02 x 10^23 = 4.95 x 10^20. Each CH4 molecule has 5 atoms, so atoms = 5 x 4.95 x 10^20 = 24.7 x 10^20.

Step 4 (b) - Convert: V = 1 cm3 = 0.001 dm3; T = 100 + 273 = 373 K; P = 1.5 atm.

Step 5 (b) - Moles = (1.5 x 0.001) / (0.0821 x 373) = 0.0015 / 30.62 = 4.90 x 10^-5 mol.

Step 6 (b) - Molecules = 4.90 x 10^-5 x 6.02 x 10^23 = 2.94 x 10^19. Each NH3 molecule has 4 atoms, so atoms = 4 x 2.94 x 10^19 = 1.177 x 10^20.

Answer: (a) 4.936 x 10^20 molecules, 24.7 x 10^20 atoms (b) 2.94 x 10^19 molecules, 1.177 x 10^20 atoms. Matches the book's printed answers.

23. Calculate the masses of 10^20 molecules of each of H2, O2 and CO2 at STP. What will happen to the masses of these gases, when the temperature of these gases are increased by 100 degrees C and the pressure is decreased by 100 torr?

Step 1 - Moles in 10^20 molecules = 10^20 / 6.02 x 10^23 = 1.661 x 10^-4 mol.

Step 2 - Mass of H2 = 1.661 x 10^-4 x 2 = 3.3 x 10^-4 g. Mass of O2 = 1.661 x 10^-4 x 32 = 5.31 x 10^-3 g. Mass of CO2 = 1.661 x 10^-4 x 44 = 7.30 x 10^-3 g.

Step 3 - The mass of a given number of molecules is fixed by the count and the molar mass (mass = n x M). Changing the temperature or the pressure changes the volume the gas occupies, not the mass of those molecules, so the masses do not change.

Answer: H2 = 3.3 x 10^-4 g, O2 = 5.31 x 10^-3 g, CO2 = 7.30 x 10^-3 g; the masses are unchanged by the temperature rise and pressure fall (matches the book's printed answers).

24. (a) Two moles of NH3 are enclosed in a 5 dm3 flask at 27 degrees C. Calculate the pressure exerted by the gas assuming that (i) it behaves like an ideal gas (ii) it behaves like a real gas (a = 4.17 atm dm6 mol-2, b = 0.0371 dm3 mol-1). (b) Also calculate the amount of pressure lessened due to forces of attraction at these conditions of volume and temperature. (c) Do you expect the same decrease in the pressure of two moles of NH3 having a volume of 40 dm3 and at temperature of 27 degrees C?

Step 1 (a)(i) - Ideal gas: P = nRT/V = (2 x 0.0821 x 300) / 5 = 49.26 / 5 = 9.85 atm.

Step 2 (a)(ii) - Van der Waals: P = nRT/(V - nb) - a n^2/V^2. V - nb = 5 - (2 x 0.0371) = 4.9258 dm3. nRT/(V - nb) = 49.26/4.9258 = 10.00 atm. Attraction term = 4.17 x 4 / 25 = 0.667 atm. Hence P = 10.00 - 0.667 = 9.33 atm.

Step 3 (b) - The net pressure lessening against the ideal value is 9.85 - 9.33 = 0.52 atm (book prints 0.51 atm; the difference is rounding). The attraction term alone is 0.667 atm, partly offset by the volume correction, which raises the pressure.

Step 4 (c) - No. At 40 dm3 the attraction term is 4.17 x 4 / 1600 = 0.0104 atm, about sixty times smaller, and the volume correction is also negligible, so the gas is almost ideal and the pressure decrease is only about 0.01 atm, not 0.51 atm.

Answer: (a)(i) 9.85 atm (ii) 9.33 atm (b) about 0.51 atm lessened (c) No, the decrease is negligible at 40 dm3.

Chapter 4 - Liquids and Solids

Exercise 4 - Questions of Liquids: Multiple Choice

Liquids 1(i) London dispersion forces are the only forces present among the

A. molecules of water in liquid state

✓ B. atoms of helium in gaseous state at high temperature

C. molecules of solid iodine

D. molecules of hydrogen chloride gas

Liquids 1(ii) Acetone and chloroform are soluble in each other due to

✓ A. intermolecular hydrogen bonding

B. ion-dipole interaction

C. instantaneous dipole

D. all of the above

Liquids 1(iii) NH_3 shows a maximum boiling point among the hydrides of Vth group elements due to

A. very small size of nitrogen

✓ B. lone pair of electrons present on nitrogen

C. enhanced electronegative character of nitrogen

D. pyramidal structure of NH_3

Liquids 1(iv) When water freezes at 0 degrees C, its density decreases due to

A. cubic structure of ice

✓ B. empty spaces present in the structure of ice

C. change of bond lengths

D. change of bond angles

Liquids 1(v) In order to mention the boiling point of water at 110 degrees C, the external pressure should be

✓ A. between 760 torr and 1200 torr

B. between 200 torr and 760 torr

C. 765 torr

D. any value of pressure

Exercise 4 - Questions of Liquids: Fill in the Blanks

Liquids 2(i) The polarizability of noble gases increases down the group and results in the increase in their boiling points.

Liquids 2(ii) is developed in acetone and chloroform when they are mixed together.

Liquids 2(iii) Exceptionally weak acidity of HF is due to strong hydrogen bonding present in it.

Liquids 2(iv) The concept of dynamic equilibrium is the ultimate state of all reversible systems.

Liquids 2(v) Change in enthalpy of vaporisation of C6H14 should be _____ than that of C2H6.

The enthalpy of vaporisation of C6H14 should be greater than that of C2H6, because the larger molecule has stronger London dispersion forces and more surface area, so more energy is needed to separate its molecules.

Liquids 2(vi) During the formation of ice from liquid water there is a 9 percent % increase in volume.

Liquids 2(vii) The rate of increase of vapour pressure of water increases at high temperatures.

Liquids 2(viii) A layer of ice on the surface of water insulates the water underneath for further heat loss.

Liquids 2(ix) Evaporation is a cooling (endothermic) process.

Liquids 2(x) Liquid crystals are used in the display of electronic devices.

Exercise 4 - Questions of Liquids: True or False

Liquids 3(i) Dipole-dipole forces are weaker than dipole-induced dipole forces.

False

Liquids 3(ii) The ion dipole interactions are responsible for the dissolution of an ionic substance in water.

True

Liquids 3(iii) The high polarizability of iodine is responsible for its existence in solid form and its difference from other halogens.

True

Liquids 3(iv) The strong hydrogen bonding in H2S makes it different from water.

False

Liquids 3(v) Hydrocarbons are soluble in water because they are polar compounds.

False

Liquids 3(vi) The viscosities of liquids partially depend upon the extent of hydrogen bonding.

True

Liquids 3(vii) The state of equilibrium between liquid state and vapours is dynamic in nature.

True

Liquids 3(viii) Heat of vaporization of liquids depend upon the intermolecular forces of attraction present between their molecules.

True

Liquids 3(ix) Ice does not show any vapour pressure on its surface at -1 degrees C.

False

Liquids 3(x) Boiling point of a liquid is independent of external pressure.

False

Exercise 4 - Questions of Liquids

Liquids 4(a) What type of intermolecular forces will dominate in the following liquids: (i) Ammonia, NH3 (ii) Octane, C8H18 (iii) Argon, Ar (iv) Propanone, CH3COCH3 (v) Methanol, CH3OH.

(i) Ammonia, NH3: hydrogen bonding dominates (N-H with a lone pair on nitrogen). (ii) Octane, C8H18: London dispersion forces only (non-polar hydrocarbon). (iii) Argon, Ar: London dispersion forces only (single non-polar atom). (iv) Propanone, CH3COCH3: dipole-dipole forces dominate (polar C=O, no O-H hydrogen bonding). (v) Methanol, CH3OH: hydrogen bonding dominates (O-H group).

Liquids 4(b) Propanone, propanol and butane have very similar relative molecular masses. List them in the expected increasing boiling points. Explain your answer.

Increasing order of boiling points: butane < propanone < propanol. Butane is non-polar, held only by weak London forces, so it boils lowest. Propanone is polar, held by dipole-dipole forces, so it boils higher. Propanol can form hydrogen bonds through its O-H group, the strongest of the three, so it boils highest, even though all three have similar molar masses.

Liquids 5. Explain with reasons: (i) In the hydrogen bonded structure of HF, which is the stronger bond: the shorter covalent bond or the longer hydrogen bond between different molecules? (ii) In a very cold winter the fish in garden ponds owe their lives to hydrogen bonding? (iii) Water and ethanol can mix easily and in all proportions. (iv) The origin of the intermolecular forces in water.

(i) The covalent H-F bond inside one HF molecule is much stronger (and shorter) than the hydrogen bond between two HF molecules: the covalent bond is a full shared-electron bond, while the hydrogen bond is only a weak electrostatic attraction between molecules. (ii) In a very cold winter the surface water of a pond freezes to ice, which floats and insulates the water below, so the water under the ice stays liquid (near 4 degrees C) and the fish survive; this floating ice is a consequence of hydrogen bonding in water. (iii) Water and ethanol both form hydrogen bonds, and their molecules can hydrogen-bond to each other, so they mix in all proportions. (iv) The intermolecular forces in water are hydrogen bonding (dominant), dipole-dipole forces (water is polar) and London dispersion forces (present in all molecules).

Liquids 6(a) Briefly consider some of the effects on our lives if water has only a very weak hydrogen bonding present among its molecules. (b) All gases have a characteristic critical temperature. Above the critical temperature it is impossible to liquefy a gas. The critical temperatures of carbon dioxide and methane are 31.14

degrees C and -81.9 degrees C, respectively. Which gas has the stronger intermolecular forces? Briefly explain your choice.

(a) If water had only very weak hydrogen bonding, it would be a gas at room temperature (like H2S), ice would not float, the oceans would freeze solid from the bottom up, and life as we know it could not exist; the high boiling point, high heat capacity and solvent power of water all depend on its strong hydrogen bonding. (b) Carbon dioxide has the stronger intermolecular forces: its critical temperature (31.14 degrees C) is far higher than methane's (-81.9 degrees C). A higher critical temperature means stronger attractions must be overcome, and CO2 is larger and more polarizable (with a quadrupole moment) than the small, symmetrical CH4 molecule.

Liquids 7. Three liquids have the properties: Water: H2O, molar mass 18, enthalpy of vaporisation 41.1 kJ mol-1, boiling point 100 degrees C. Propanone: C3H6O, molar mass 58, enthalpy of vaporisation 31.9 kJ mol-1, boiling point 56 degrees C. Pentane: C5H12, molar mass 72, enthalpy of vaporisation 27.7 kJ mol-1, boiling point 36 degrees C. (a) What type of intermolecular force predominates in each liquid? (b) What do you deduce about the relative strength of these forces? Justify. (c) If the liquids are shaken together in pairs: (i) Which pair would be unlikely to mix? (ii) Explain this immiscibility in terms of the forces between the molecules. (iii) Choose one of the pairs that mix and say whether the enthalpy change on mixing would be positive or negative.

(a) Water: hydrogen bonding predominates. Propanone: dipole-dipole forces predominate. Pentane: London dispersion forces predominate. (b) Relative strength: water > propanone > pentane. Justification: water has the highest enthalpy of vaporisation (41.1 kJ mol-1) and the highest boiling point (100 degrees C) although it has the lowest molar mass (18), which is only possible if its intermolecular forces are the strongest. Propanone is next (31.9 kJ mol-1, 56 degrees C) and pentane is

weakest (27.7 kJ mol-1, 36 degrees C) although it has the highest molar mass (72), showing its forces are the weakest. (c) (i) The water-pentane pair is unlikely to mix: water is polar and hydrogen-bonded, pentane is non-polar, and mixing would break strong water-water hydrogen bonds without replacing them. (ii) This immiscibility is explained by the forces: like dissolves like; water-water and pentane-pentane attractions are each stronger than water-pentane attractions. (iii) For a pair that does mix, for example water-propanone, the enthalpy change on mixing is negative (heat is given out), because new hydrogen bonds and dipole attractions form between the different molecules.

Liquids 8. Describe the various forces responsible for keeping the particles together in the following elements and compounds and their effects on physical properties, from the data: Neon (Ne, molar mass 20, M.P. -248 degrees C); Argon (Ar, 40, -189 degrees C); Water (H2O, 18, 0 degrees C); Sodium fluoride (NaF, 42, 993 degrees C); Diamond (C, 12, 3350 degrees C).

Neon and argon are held together only by very weak London forces between small atoms, so their melting points are extremely low (-248 and -189 degrees C). Water, although its molar mass is only 18, melts at 0 degrees C because its molecules are held by strong hydrogen bonds. Sodium fluoride is ionic, held by strong electrostatic forces between Na+ and F- ions, so it melts at 993 degrees C. Diamond is a covalent network solid with strong covalent bonds throughout the crystal, so it melts at 3350 degrees C. The melting points therefore follow the strength of the bonding: London < hydrogen bonding < ionic < covalent network.

Liquids 9. The boiling points and molar masses of hydrides of some first row elements are: CH4 109 K, 16 g mol-1; NH3 240 K, 17 g mol-1; H2O 373 K, 18 g mol-1. Suggest reasons for the difference in their boiling points in terms of the type of molecules involved and the nature of the forces present between them.

Although the molar masses are similar (16, 17, 18), the boiling points rise sharply:

CH4 109 K, NH3 240 K, H2O 373 K. Methane is non-polar, held only by weak London forces, so it boils lowest. Ammonia is polar and forms hydrogen bonds, so it boils much higher. Water forms the most extensive hydrogen bonding (each molecule can form up to four hydrogen bonds in a network) and is the most polar, so it boils highest. The differences are therefore due to the type and strength of the intermolecular forces, not the molar masses.

Liquids 10. Explain the term saturated vapour pressure. Arrange in order of increasing vapour pressure: 1 dm3 water, 1 dm3 ethanol, 50 cm3 water, 50 cm3 ethanol and 50 cm3 of ether.

Saturated vapour pressure is the pressure exerted by the vapour of a liquid when it is in dynamic equilibrium with the liquid in a closed container at a given temperature. It depends only on the liquid and the temperature, not on the amount of liquid or the volume of the container. Increasing order of vapour pressure: the two water samples are equal and lowest, the two ethanol samples are equal and next, and ether is highest. So: 1 dm3 water = 50 cm3 water < 1 dm3 ethanol = 50 cm3 ethanol < 50 cm3 ether.

Liquids 11. While a volatile liquid standing in a beaker evaporates, the temperature of the liquid remains the same as that of its surrounding. If the same liquid is allowed to vaporize into atmosphere in an insulated vessel, its temperature falls below that of its surrounding. Explain the difference in behaviour.

In the open beaker, heat is continuously supplied from the surroundings to replace the heat used in evaporation, so the liquid stays at the surrounding temperature. In the insulated vessel no heat can enter from outside, so the heat needed for evaporation is taken from the liquid itself: the fastest molecules escape, the average kinetic energy of the remaining liquid falls, and its temperature drops below that of the surroundings.

Liquids 12. How does hydrogen bonding explain: (i) Structure of DNA (ii) Hydrogen

bonding in proteins (iii) Formation of ice and its lesser density than liquid water (iv) Solubilities of compounds.

(i) Structure of DNA: the two strands of the double helix are held together by hydrogen bonds between complementary base pairs (A-T with two hydrogen bonds, G-C with three). (ii) Hydrogen bonding in proteins: hydrogen bonds between the C=O and N-H groups of the peptide chain hold the alpha-helix and beta-sheet secondary structures. (iii) Formation of ice and its lesser density than liquid water: in ice each water molecule is hydrogen-bonded to four neighbours in an open, cage-like structure with empty spaces, so ice occupies more volume and is less dense than liquid water, and floats. (iv) Solubilities: a substance dissolves in water if it can form hydrogen bonds with water (or ion-dipole attractions), which is why alcohols, sugars and many polar compounds are soluble, while non-polar hydrocarbons are not.

Liquids 13. What are liquid crystals? Give their uses in daily life.

Liquid crystals are substances that flow like liquids but keep some of the ordered molecular arrangement of a crystal: they are an intermediate state between a solid crystal and an ordinary liquid. Uses in daily life: liquid crystal displays (LCDs) in watches, calculators, mobile phones and televisions; liquid crystal thermometers (temperature strips); and optical shutters and display panels.

Liquids 14. Explain with reasons: (i) Evaporation causes cooling. (ii) Evaporation takes place at all temperatures. (iii) Boiling needs a constant supply of heat. (iv) Earthenware vessels keep water cool. (v) One feels sense of cooling under the fan after bath. (vi) Dynamic equilibrium is established during evaporation of a liquid in a closed vessel at constant temperature. (vii) The boiling point of water is different at Murree hills and at Mount Everest. (viii) Vacuum distillation can be used to avoid decomposition of a sensitive liquid. (ix) Heat of sublimation of a substance is greater than that of heat of vaporization. (x) Heat of sublimation of iodine is very

high.

(i) Evaporation causes cooling because only the molecules with the highest kinetic energy can escape from the surface; their departure lowers the average kinetic energy (temperature) of the remaining liquid. (ii) Evaporation takes place at all temperatures because at any temperature some surface molecules have enough energy to overcome the intermolecular forces, even though the number increases rapidly with temperature. (iii) Boiling needs a constant supply of heat because the heat is used to break the intermolecular forces and separate the molecules, while the temperature stays constant at the boiling point. (iv) Earthenware vessels keep water cool because water seeps through the porous walls and evaporates from the outer surface; the evaporation takes heat from the vessel and the water inside. (v) One feels cooling under a fan after a bath because the moving air removes the vapour and speeds up the evaporation of water from the skin, taking heat from the body. (vi) Dynamic equilibrium is established during evaporation of a liquid in a closed vessel at constant temperature: the rate of evaporation becomes equal to the rate of condensation, so the vapour pressure stays constant. (vii) The boiling point of water is different at Murree hills and at Mount Everest because the atmospheric pressure is lower at higher altitude: water boils when its vapour pressure equals the external pressure, so at lower pressure it boils at a lower temperature. (viii) Vacuum distillation can be used to avoid decomposition of a sensitive liquid because lowering the pressure lowers its boiling point, so it can be distilled at a temperature below its decomposition temperature. (ix) Heat of sublimation of a substance is greater than its heat of vaporization because sublimation includes both melting (fusion) and vaporisation: solid to liquid to gas needs more energy than liquid to gas alone. (x) Heat of sublimation of iodine is very high because I2 molecules are large and highly polarizable, so the London forces between them are strong and much energy is needed to separate them.

Exercise 4 - Questions of Solids: Multiple Choice

Solids 1(i) Ionic solids are characterized by

A. low melting points

B. good conductivity in solid state

C. high vapour pressures

✓ D. solubility in polar solvents

Solids 1(ii) Amorphous solids

A. have sharp melting points

B. undergo clean cleavage when cut with knife

C. have perfect arrangement of atoms

✓ D. can possess small regions of orderly arrangement of atoms

Solids 1(iii) The molecules of CO_2 in dry ice form the

A. ionic crystals

B. covalent crystals

✓ C. molecular crystals

D. any type of crystal

Solids 1(iv) Which of the following is a pseudo solid?

A. CaF_2

✓ B. Glass

C. NaCl

D. All

Solids 1(v) Diamond is a bad conductor because

A. it has a tight structure

B. it has a high density

✓ C. there are no free electrons present in the crystal of diamond to conduct electricity

D. it is transparent to light

Exercise 4 - Questions of Solids: Fill in the Blanks

Solids 2(i) In a crystal lattice, the number of nearest neighbours to each atom is called the coordination number .

Solids 2(ii) There are 14 Bravais lattices.

Solids 2(iii) A pseudo solid is regarded as a supercooled liquid.

Solids 2(iv) Glass may begin to crystallize by a process called devitrification .

Solids 2(v) Crystalline solids which exhibit the same _____ in all directions are called _____.

Crystalline solids which exhibit the same physical properties in all directions are called isotropic (the same property, e.g. refractive index or conductivity, is the same in every direction).

Solids 2(vi) The branch of science which deals with the structure of crystals is called crystallography.

Exercise 4 - Questions of Solids: True or False (p. 116)

Solids 3(i) There are five parameters in unit cell dimensions of a crystal.

False

Solids 3(ii) Ionic crystals are very hard, low volatility and very low melting and boiling points.

False

Solids 3(iii) The value of lattice energy of the ionic substances depends upon the size of ions.

True

Solids 3(iv) Molecular orbital theory of solids is also called band theory.

True

Solids 3(v) Ionic solid is good conductor of electricity in the molten state.

True

Exercise 4 - Questions of Solids: True or False (p. 117, as printed)

Solids True/False list as printed on p. 117 (v) The vapour pressures of solids are far less than those of liquids.

True

Solids True/False list as printed on p. 117 (vi) Amorphous solid like glass is also called super cooled liquid.

True

Solids True/False list as printed on p. 117 (vii) Cleavage of the crystals is itself anisotropic behaviour.

True

Solids True/False list as printed on p. 117 (viii) The crystals showing isomorphism mostly have the same atomic ratios.

True

Solids True/False list as printed on p. 117 (ix) The transition temperature is shown by elements having allotropic forms and by compounds showing polymorphism.

True

Solids True/False list as printed on p. 117 (x) One of the unit cell angles of hexagonal crystal is 120 degrees.

True

Solids True/False list as printed on p. 117 (xi) The electrical conductivity of the metals decrease by increasing temperature.

True

Solids True/False list as printed on p. 117 (xii) In the closest packing of atoms of metals, only 74% space is occupied.

True

Solids True/False list as printed on p. 117 (xiii) Ionic crystals do not conduct electricity in the solid state.

True

Solids True/False list as printed on p. 117 (xiv) Ionic crystals are highly brittle.

True

Solids True/False list as printed on p. 117 (xv) The number of positive ions surrounding the negative ion in the ionic crystal lattice depends upon the sizes of the two ions.

True

Exercise 4 - Questions of Solids

Solids 4. What are solids? Give general properties of solids. How do you differentiate between crystalline solids and amorphous solids?

Solids are substances with a definite shape and a definite volume, whose particles are held in fixed positions by strong forces. General properties: definite shape and volume, high density, incompressibility, slow diffusion, and a sharp (crystalline) or gradual (amorphous) melting point. Crystalline solids have a regular, repeating arrangement of particles, sharp melting points, flat faces and clean cleavage; amorphous solids have no regular arrangement, soften over a range of temperature, and break with irregular fracture (for example glass, plastic, rubber).

Solids 5(a) Explain the following properties of crystalline solids, giving three examples in each case: (i) Anisotropy (ii) Cleavage (iii) Habit of a crystal (iv) Isomorphism (v) Polymorphism (vi) Transition temperature (vii) Symmetry (viii) Growing of a crystal. (b) How polymorphism and allotropy are related to each other? Give examples.

Crystalline solids show: (i) Anisotropy: properties differ in different directions (for example cleavage and conductivity). (ii) Cleavage: they split along definite planes. (iii) Habit of a crystal: the characteristic external shape of the crystal. (iv) Isomorphism: different substances with similar formulae crystallise in the same form

(for example $NaNO_3$ and $CaCO_3$). (v) Polymorphism: the same substance exists in more than one crystalline form (for example $CaCO_3$ as calcite and aragonite). (vi) Transition temperature: the temperature at which one crystalline form changes into another. (vii) Symmetry: the crystal looks the same after certain rotations or reflections. (viii) Growing of a crystal: crystals grow by adding particles layer by layer from a saturated solution or melt. (b) Polymorphism and allotropy are related: allotropy is polymorphism in elements (the same element in different forms, for example carbon as diamond and graphite, sulphur as rhombic and monoclinic); polymorphism is the general term, used for compounds as well as elements.

Solids 6(a) Define unit cell. What are unit cell dimensions? How the idea of crystal lattice is developed from the concept of unit cell? (b) Explain seven crystal systems and draw the shapes of their unit cells.

A unit cell is the smallest repeating unit of a crystal lattice which, when repeated in three dimensions, builds the whole crystal. Unit cell dimensions are its edge lengths (a, b, c) and the angles between them (alpha, beta, gamma): six parameters in all. The crystal lattice is developed by repeating the unit cell in all three directions, so that the pattern of points (lattice points) extends through the whole crystal. (b) The seven crystal systems are: cubic, tetragonal, orthorhombic, hexagonal, rhombohedral (trigonal), monoclinic and triclinic, distinguished by their edge lengths and angles; their unit-cell shapes are drawn as boxes with the corresponding equalities of edges and angles (for example cubic: $a = b = c$, all angles 90 degrees; hexagonal: $a = b$, not equal to c, gamma = 120 degrees). This is a drawing task: sketch each of the seven unit cells with its edges and angles labelled.

Solids 7(a) What are ionic solids? Give their properties. Explain the structure of NaCl. Sketch a model to justify that unit cell of NaCl has four formula units in it. (b) What are covalent solids? Give their properties. Explain the structure of diamond.

(c) What are molecular crystals? Give their properties. Justify that molecular crystals are softer than ionic crystals.

Ionic solids consist of positive and negative ions held by strong electrostatic forces. Properties: high melting and boiling points, hard and brittle, soluble in polar solvents, non-conductors in the solid state but conductors in the molten state and in solution. NaCl structure: a face-centred cubic arrangement in which each Na+ is surrounded by 6 Cl- ions and each Cl- by 6 Na+ ions; the unit cell contains 4 Na+ and 4 Cl- ions, that is, 4 formula units of NaCl (corner and face-centre counting: 8 corners x 1/8 + 6 faces x 1/2 = 4). (b) Covalent solids consist of atoms joined by covalent bonds in a network: hard, very high melting points, poor conductors (for example diamond, in which each carbon is tetrahedrally bonded to four others). (c) Molecular crystals consist of molecules held by weak intermolecular forces: soft, low melting points, poor conductors. Molecular crystals are softer than ionic crystals because their molecules are held only by weak van der Waals or hydrogen-bond forces, whereas ionic crystals are held by strong electrostatic forces between ions.

Solids 8(a) Give different theories of a metallic bond. How does electron sea theory justify the electrical conductivity, thermal conductivity and shining surfaces of metals? (b) Explain with the help of a diagram: (i) Cubic close packing in the structure of metals (ii) Hexagonal close packing in the structure of metals.

Theories of metallic bond: (1) Electron sea theory: metal atoms release their valence electrons into a common sea of mobile electrons in which positive metal ions (kernels) are immersed. (2) Band theory: the atomic orbitals overlap to form energy bands, and the partly filled conduction band allows electrons to move. Electron sea theory explains: electrical conductivity (the mobile electrons carry current), thermal conductivity (the mobile electrons carry heat energy), and shining surfaces/lustre (the mobile electrons absorb and re-emit light). (b) Cubic close packing: layers of spheres in ABCABC stacking, giving a face-centred cubic

arrangement with 74% space occupied. Hexagonal close packing: layers in ABAB stacking, giving a hexagonal arrangement, also 74% occupied. These are drawing tasks: sketch the layer stacking for each case.

Solids 9. Crystals of salts fracture easily, but metals are deformed under stress without fracturing. Explain the difference.

In an ionic crystal, if one layer of ions is shifted by stress, ions of like charge come to face each other and repel strongly, so the crystal cleaves (fractures) along the plane. In a metal, the layers of positive ions can slide over each other without changing the nature of the bonding, because the mobile electron sea adjusts to the new positions; the metal therefore deforms (is malleable and ductile) instead of fracturing.

Solids 10. What is the coordination number of an ion? What is the coordination number of the cation in (a) NaCl structure and (b) CsCl structure? Explain the reason for this difference.

Coordination number is the number of nearest neighbours (opposite ions) surrounding an ion in the crystal. (a) In NaCl structure the cation (Na+) is surrounded by 6 Cl- ions, so its coordination number is 6. (b) In CsCl structure the cation (Cs+) is surrounded by 8 Cl- ions, so its coordination number is 8. The difference is due to the radius ratio: the larger Cs+ ion can accommodate 8 large Cl- ions around it, while the smaller Na+ ion can accommodate only 6.

Solids 11. Give examples of ionic solids, molecular solids and covalent macromolecular solids. What are the factors which determine whether each of these types of solid will dissolve in water or not?

Examples: ionic solids: NaCl, KCl, CaF2; molecular solids: dry ice (CO2), ice, iodine; covalent macromolecular solids: diamond, graphite, silicon dioxide (quartz). Whether a solid dissolves in water depends on: whether water can break its bonding (ion-dipole attractions for ionic solids, hydrogen bonding for molecular solids that

can hydrogen-bond), the lattice energy (high lattice energy resists dissolution), and the hydration energy of the ions (high hydration energy favours dissolution); a solid dissolves when the hydration energy can overcome the lattice energy.

Solids 12. Explain with reasons: (i) Sodium is softer than copper, but both are very good electrical conductors. (ii) Diamond is hard and an electrical insulator. (iii) Sodium chloride and caesium chloride have different structures. (iv) Iodine dissolves readily in tetrachloromethane.

(i) Sodium is softer than copper, but both are very good electrical conductors: both have a sea of mobile electrons (hence conductivity), but sodium has only one valence electron per atom and weaker metallic bonding than copper, so its lattice is softer. (ii) Diamond is hard because every carbon atom is joined by strong covalent bonds to four neighbours in a rigid network; it is an electrical insulator because all its valence electrons are used in those bonds and none are free to move. (iii) Sodium chloride and caesium chloride have different structures because of their different radius ratios: Na+ is small enough for 6-coordination (NaCl structure), while the larger Cs+ allows 8-coordination (CsCl structure). (iv) Iodine dissolves readily in tetrachloromethane because both are non-polar: iodine molecules are held by London forces, which are similar to the forces in CCl4, so like dissolves like.

Chapter 5 - Atomic Structure

Exercise 5 - Multiple Choice Questions

1(i) The nature of the positive rays depend on

A. the nature of the electrode

B. the nature of the discharge tube

✓ C. the nature of the residual gas

D. all of the above

1(ii) The velocity of photon is

✓ A. independent of its wavelength

B. depends on its wavelength

C. equal to square of its amplitude

D. depends on its source

1(iii) The wave number of the light emitted by a certain source is 2 x 10^6 m^-1. The wavelength of this light will be

✓ A. 500 nm

B. 500 m

C. 200 nm

D. 5 x 10^7 m

1(iv) Rutherford's model of atom failed because

A. the atom did not have a nucleus and electrons

B. it did not account for the attraction between protons and neutrons

✓ C. it did not account for the stability of the atom

D. there is actually no space between the nucleus and the electrons

1(v) Bohr model of atom is contradicted by

A. Planck's quantum theory

B. dual nature of matter

✓ C. Heisenberg's uncertainty principle

D. all of the above

1(vi) Splitting of spectral lines when atoms are subjected to strong electric field is called

A. Zeeman effect

✓ B. Stark effect

C. Photoelectric effect

D. Compton effect

1(vii) In the ground state of an atom, the electron is present

A. in the nucleus

B. in the second shell

✓ C. nearest to the nucleus

D. farthest from the nucleus

1(viii) Quantum number values for 2p orbitals are

✓ A. n = 2, l = 1

B. n = 1, l = 2

C. n = 1, l = 0

D. n = 2, l = 0

1(ix) Orbitals having same energy are called

A. hybrid orbitals

B. valence orbitals

✓ C. degenerate orbitals

D. d-orbitals

1(x) When 6d orbital is complete, the entering electron goes into

A. 7f

B. 7s

✓ C. 7p

D. 7d

Exercise 5 - Fill in the Blanks

2(i) Beta-particles are nothing but electrons moving with a very high speed.

2(ii) The charge on one mole of electrons is 96500 coulombs.

2(iii) The mass of hydrogen atom is 1.67 x 10^-24 grams.

2(iv) The mass of one mole of electrons is 0.55 mg .

2(v) Energy is emitted when electron jumps from higher to a lower orbit.

2(vi) The ionization energy of hydrogen atom can be calculated from Bohr model of atom.

2(vii) For d-subshell, the azimuthal quantum number has value of 2 .

2(viii) The number of electrons in a given subshell is given by formula 2(2l + 1) .

2(ix) The electronic configuration of H is 1s^1 .

Exercise 5 - True or False

3(i) A neutron is slightly lighter particle than a proton.

False

3(ii) A photon is the massless bundle of energy but has momentum.

True

3(iii) The unit of Rydberg constant is the reciprocal of unit of length.

True

3(iv) The actual isotopic mass is a whole number.

False

3(v) Heisenberg's uncertainty principle is applicable to macroscopic bodies.

False

3(vi) The nodal plane in an orbital is the plane of zero electron density.

True

3(vii) The number of orbitals present in a sublevel is given by the formula (2l + 1).

True

3(viii) The magnetic quantum number was introduced to explain Zeeman and Stark effect.

True

3(ix) Spin quantum number tells us the direction of spin of electron around the nucleus.

False

Exercise 5 - Short Questions

Note on True/False 3(ix) - Spin quantum number tells us the direction of spin of electron around the nucleus.

Marked False. The spin quantum number (+1/2 or -1/2) gives the direction of the electron's spin about its OWN axis (clockwise or anticlockwise). Motion around the nucleus is described by the orbital quantum numbers, so the statement as printed, with 'around the nucleus', is False.

4(a) Keeping in mind the discharge tube experiment, why is it necessary to decrease the pressure in the discharge tube to get the cathode rays?

At low pressure there are few gas molecules in the tube, so the cathode rays can travel a long distance without colliding with gas molecules, and a clear beam reaches the anode. At ordinary pressure the rays are scattered by collisions and no beam is seen.

4(b) Whichever gas is used in the discharge tube, the nature of the cathode rays remains the same. Why?

Cathode rays are streams of electrons knocked out of the atoms of the cathode material itself. Whatever gas is present, the particles emitted are the same electrons, so their nature does not depend on the gas in the tube.

4(c) Why is the e/m value of the cathode rays just equal to that of electron?

The measured e/m of cathode rays is 1.7588 x 10^11 C kg^-1, exactly the e/m of an electron. Since every cathode-ray particle has this same charge-to-mass ratio, the rays consist of electrons.

4(d) How does the bending of the cathode rays in the electric and magnetic fields show that they are negatively charged?

In an electric field the rays bend towards the positive plate, and in a magnetic field

they bend in the direction expected for a negatively charged particle. Only a negative charge bends in these directions, so the rays are negatively charged.

4(e) Why are the positive rays also called canal rays?

Positive rays pass through holes (canals) drilled in the cathode and travel in the opposite direction to the cathode rays. Because they stream through these canals, they are called canal rays.

4(f) The e/m values of positive rays for different gases are different, but those for cathode rays are the same. Justify it.

Positive rays are the positive ions left when gas atoms lose electrons, so their mass is the mass of the gas atom (or ion). Different gases give ions of different mass, so e/m differs from gas to gas. Cathode rays are always electrons, whose mass is the same, so their e/m is always the same.

4(g) The e/m value for positive rays obtained from hydrogen gas is 1836 times less than that of cathode rays. Justify it.

For hydrogen the positive ray is the proton, mass 1.6726 x 10^-27 kg, while the electron mass is 9.109 x 10^-31 kg. The ratio is about 1836, so the e/m of the proton is 1836 times smaller than the e/m of the electron, the charge being the same in both cases.

5(a) Explain Millikan's oil drop experiment to determine the charge of an electron.

Millikan sprayed fine oil drops between two charged plates. By adjusting the voltage he balanced a drop so that the electric force on it exactly cancelled its weight. From the balancing voltage, the drop's radius (from its rate of fall) and the density of oil, he found the charge on the drop. Repeating for many drops, every charge was a whole-number multiple of 1.602 x 10^-19 C, the charge of one electron.

5(b) What is J.J. Thomson's experiment for determining e/m value of electron?

J.J. Thomson passed cathode rays through crossed electric and magnetic fields and adjusted them until the beam was undeflected. From the balancing fields he found

the velocity of the rays, and from the deflection in the magnetic field alone he found e/m = 1.7588 x 10^11 C kg^-1 for the electron.

5(c) Evaluate mass of electron from the above two experiments.

Mass of electron = charge / (charge/mass) = 1.602 x 10^-19 C / 1.7588 x 10^11 C kg^-1 = 9.11 x 10^-31 kg. This is the mass of the electron.

6(a) Discuss Chadwick's experiment for the discovery of neutron. Compare the properties of electron, proton and neutron.

Chadwick bombarded beryllium with alpha-particles. The radiation given out knocked protons out of paraffin wax, and from the energies of those protons he showed the radiation consisted of neutral particles of about the mass of a proton: the neutron. Electron: charge -1.602 x 10^-19 C, mass 9.109 x 10^-31 kg. Proton: charge +1.602 x 10^-19 C, mass 1.6726 x 10^-27 kg. Neutron: charge 0, mass 1.6749 x 10^-27 kg. All three are the fundamental particles of the atom.

6(b) Rutherford's atomic model is based on the scattering of alpha-particles from a thin gold foil. Discuss it and explain the conclusions.

Rutherford fired alpha-particles at a very thin gold foil. Most passed straight through, some were deflected through small angles, and a very few bounced almost straight back. Conclusions: (i) most of the atom is empty space, since most particles pass through; (ii) the positive charge and almost all the mass are concentrated in a tiny central nucleus, which repels the few particles that come close; (iii) electrons occupy the large empty space around the nucleus.

7(a) Give the postulates of Bohr's atomic model. Which postulate tells us that orbits are stationary and energy is quantized?

Postulates: (i) the electron revolves around the nucleus in certain permitted circular orbits of fixed energy, without radiating energy; (ii) the angular momentum of the electron is quantized, mvr = nh/2pi; (iii) energy is absorbed or emitted only when the electron jumps from one permitted orbit to another, as a photon of energy delta

E = h x nu. The postulate that tells us orbits are stationary and energy is quantized is the quantized-angular-momentum postulate, mvr = nh/2pi.

7(b) Derive the equation for the radius of nth orbit of hydrogen atom using Bohr's model.

Centripetal force = electrostatic attraction: mv^2/r = (1/4pi epsilon0) x (Ze^2/r^2). With mvr = nh/2pi, eliminating v gives r = (epsilon0 h^2 n^2)/(pi m Z e^2). For hydrogen Z = 1, r_n = 0.529 x n^2 angstrom.

7(c)(i) How does the radius equation tell you that radius is directly proportional to the square of the number of orbit?

r_n = 0.529 x n^2 angstrom, so r is proportional to n^2. Doubling n multiplies the radius by four, so the radius is directly proportional to the square of the number of the orbit.

7(c)(ii) How does the radius equation tell you that radius is inversely proportional to the number of protons in the nucleus?

For hydrogen r_n = 0.529 x n^2 / Z angstrom with Z = 1, and in general r_n is proportional to n^2/Z. The radius is therefore inversely proportional to the number of protons in the nucleus (the atomic number Z): a larger nuclear charge pulls the electron into a smaller orbit.

7(d) How do you come to know that the velocities of electrons in higher orbits are less than those in lower orbits of hydrogen atom?

Bohr's velocity equation is v = (Z e^2)/(2 epsilon0 h n), so v is proportional to 1/n. As n increases the velocity falls, so electrons in higher orbits move more slowly than those in lower orbits.

7(e) Justify that the distance gaps between different orbits go on increasing from the lower to the higher orbits.

r_n = 0.529 x n^2 angstrom, so successive radii are 0.529, 2.116, 4.761, 8.464 angstrom. The gaps are 1.587, 2.645, 3.703 angstrom, growing by 2 x 0.529

angstrom each time, so the distance gaps between orbits increase from lower to higher orbits.

8(a) Derive the formula for calculating the energy of an electron in nth orbit using Bohr's model.

Total energy = kinetic energy + potential energy = (1/2)mv^2 - (1/4pi epsilon0)(Ze^2/r). Substituting Bohr's r_n and v gives E_n = -(m Z^2 e^4)/(8 epsilon0^2 h^2 n^2). For hydrogen Z = 1, E_n = -2.18 x 10^-18 / n^2 joule.

8(b) The potential energy of the bounded electron is negative.

The potential energy of the electron is -(1/4pi epsilon0)(Ze^2/r), negative because the zero of potential energy is taken at infinite separation, where electron and nucleus no longer attract. Bringing them together from infinity releases energy, so at any finite distance the potential energy is below zero.

8(c) Total energy of the bounded electron is also negative.

The total energy is negative because the electron is bound: energy must be supplied to remove it to infinity (zero energy). A negative total energy is the sign of a bound system; a free electron has zero or positive energy.

8(d) Energy of an electron is inversely proportional to n^2, but energy of higher orbits are always greater than those of the lower orbits.

E_n = -2.18 x 10^-18 / n^2 joule, so E is inversely proportional to n^2 in magnitude, but the values are -2.18, -0.545, -0.242 x 10^-18 J for n = 1, 2, 3. The magnitude falls, so the energy itself (a negative number) increases towards zero: higher orbits have greater (less negative) energy than lower orbits.

9(a) Derive the equations for hydrogen atom related to (i) energy difference between two levels n1 and n2, (ii) frequency of photon emitted when an electron jumps from n2 to n1, (iii) wave number of the photon when the electron jumps from n2 to n1.

(i) Energy difference: delta E = 2.18 x 10^-18 x (1/n1^2 - 1/n2^2) joule, for a jump

from n2 down to n1. (ii) Frequency: nu = delta E / h. (iii) Wave number: wavenumber = (1/lambda) = delta E/(h c) = R x (1/n1^2 - 1/n2^2), with R = 1.097 x 10^7 m^-1.

9(b) Justify that Bohr's equation for the wave number can explain the spectral lines of Lyman, Balmer and Paschen series.

Wavenumber = R x (1/n1^2 - 1/n2^2). With n1 = 1 the lines are the Lyman series (ultraviolet), with n1 = 2 the Balmer series (visible), and with n1 = 3 the Paschen series (infrared). One formula, with n1 = 1, 2, 3, reproduces all three series, which is the success of Bohr's equation.

10(a) What is spectrum? Differentiate between continuous spectrum and line spectrum.

A spectrum is the pattern of radiation obtained when light is dispersed by wavelength, for example by a prism. A continuous spectrum shows all wavelengths without gaps, as from a hot solid. A line spectrum shows only certain sharp wavelengths, as from excited atoms, each line being one permitted electron jump.

10(b) Compare line emission and line absorption spectra.

In a line emission spectrum the excited atom emits photons and gives bright lines on a dark background. In a line absorption spectrum white light passing through a gas loses exactly those wavelengths, giving dark lines on a bright continuous background. The two sets of lines occur at the same wavelengths.

10(c) What is the origin of line spectrum?

Line spectra arise because an atom's electron can occupy only certain energy levels. When it jumps from a higher to a lower level it emits a photon of one fixed energy, hence one fixed wavelength; each permitted jump gives one line. No other wavelengths can be emitted, so the spectrum is a set of lines, not continuous.

11(a) Hydrogen atom and He+ are mono-electronic systems, but the size of He+ is much smaller than H, why?

Both are one-electron systems, but He+ has nuclear charge Z = 2 while H has Z =

1. Bohr's radius r = 0.529 x n^2 / Z angstrom, so for n = 1 the He+ radius is 0.529/2 = 0.265 angstrom, half the hydrogen radius. The stronger nuclear pull shrinks the He+ atom.

11(b) Do you think that the size of Li2+ is even smaller than He+? Justify with calculations.

For Li2+, Z = 3, so r1 = 0.529/3 = 0.176 angstrom, smaller than He+ (0.265 angstrom) and smaller than H (0.529 angstrom). Yes, Li2+ is even smaller, because the radius is inversely proportional to Z.

12(a) What are X-rays? What is their origin? How was the idea of atomic number derived from the discovery of X-rays?

X-rays are high-energy electromagnetic radiations of very short wavelength, produced when fast electrons strike a heavy metal target and knock out inner-shell electrons; outer electrons falling into the vacancies emit X-rays. Moseley found that the square root of the X-ray frequency is proportional to (Z - b), where Z is the atomic number. This regular dependence on Z, not on atomic mass, showed that the atomic number, the number of protons, is the fundamental identity of an element.

12(b) How does Bohr's model justify Moseley's equation?

Moseley's law, sqrt(nu) = a(Z - b), follows from Bohr's model because the X-ray frequency depends on the effective nuclear charge felt by the inner electron, which grows with Z. Bohr's energy levels, E proportional to Z^2, give frequencies that increase regularly with Z, exactly as Moseley observed.

13 Point out the defects of Bohr's model. How are these defects partially covered by dual nature of electron and Heisenberg's uncertainty principle?

Defects of Bohr's model: (i) it keeps electrons in fixed orbits, contradicting Heisenberg's uncertainty principle and the dual nature of the electron, which forbid an exact path; (ii) it works only for one-electron systems (H, He+, Li2+), not for multi-electron atoms; (iii) it cannot explain the fine splitting of spectral lines, nor the

Zeeman and Stark effects. The dual nature and the uncertainty principle replace fixed orbits by probability distributions (orbitals), which is the wave-mechanical model.

14(a) Briefly discuss the wave mechanical model of atom. How has it given the idea of orbital? Compare orbit and orbital.

In the wave-mechanical model the electron is not on a fixed path; Schrodinger's wave equation gives a wave function whose square is the probability of finding the electron in a region. That region of high probability is an orbital. An orbit is a fixed circular path of definite radius (Bohr); an orbital is a three-dimensional region where the electron is most likely to be found, with no definite path.

14(b) What are quantum numbers? Discuss their significance.

Quantum numbers: principal (n) gives the shell and the energy/size; azimuthal (l) gives the subshell and the shape (l = 0 s, 1 p, 2 d, 3 f); magnetic (m) gives the orientation of the orbital in space; spin (s) gives the direction of the electron's own spin (+1/2 or -1/2). Together they completely specify an electron in an atom.

14(c) When azimuthal quantum number has a value 3, then there are seven values of magnetic quantum number. Give reasons.

For l = 3, m runs from -3 to +3, giving -3, -2, -1, 0, +1, +2, +3: seven values. Each value is one orientation of the f-subshell, so there are seven f-orbitals.

15(a) Discuss rules for the distribution of electrons in energy subshells and in orbitals.

Distribution rules: (i) Aufbau: electrons fill subshells in order of increasing (n + l) energy; (ii) Pauli's exclusion principle: no two electrons in an atom have all four quantum numbers the same, so an orbital holds at most two electrons of opposite spin; (iii) Hund's rule: in degenerate orbitals electrons occupy separate orbitals singly first, with parallel spins, before pairing.

15(b) What is (n + l) rule? Arrange the orbitals according to this rule. Do you think

that this rule is applicable to degenerate orbitals?

The (n + l) rule: a subshell of lower (n + l) is filled first; if two subshells have the same (n + l), the one with lower n is filled first, e.g. 4s (4 + 0 = 4) before 3d (3 + 2 = 5). Degenerate orbitals (same n and l, e.g. the three 2p orbitals) have the same (n + l) and the same energy, so the rule orders subshells, not orbitals within a degenerate set; within the set Hund's rule applies.

15(c) Distribute electrons in orbitals of 57La, 29Cu, 79Au, 24Cr, 53I, 86Rn.

57La: [Xe] 6s^2 5d^1. 29Cu: [Ar] 3d^10 4s^1 (the 4s electron moves into 3d to complete the d-subshell). 79Au: [Xe] 4f^14 5d^10 6s^1 (same reason as copper). 24Cr: [Ar] 3d^5 4s^1 (half-filled d-subshell is extra stable). 53I: [Kr] 4d^10 5s^2 5p^5. 86Rn: [Xe] 4f^14 5d^10 6s^2 6p^6.

16 Draw the shapes of s, p and d-orbitals. Justify these by keeping in view the azimuthal and magnetic quantum numbers.

Labelled drawing instruction: draw an s-orbital as a sphere, centred on the nucleus. Draw each p-orbital as a dumb-bell along one axis: px along x, py along y, pz along z, with two lobes and a node at the nucleus. Draw d-orbitals as four-lobed clover leaves (dxy, dxz, dyz, dx^2-y^2), except dz^2, which is a dumb-bell along z with a doughnut-shaped ring around its middle. Justification: l = 0 (s) has no angular node, so it is spherical; l = 1 (p) has one angular node, giving two lobes; l = 2 (d) has two angular nodes, giving four lobes (dz^2 is the exception). The magnetic quantum number m gives the orientations: one s, three p, five d orbitals.

Exercise 5 - Numerical Questions

17(a) A photon of light with energy 10^-19 J is emitted by a source of light. Convert this energy into the wavelength, frequency and wave number of the photon in terms of meters, hertz and m^-1, respectively.

Step 1 - Write the given data: E = 10^-19 J, h = 6.626 x 10^-34 J s, c = 3 x 10^8 m

s^-1.

Step 2 - Frequency from Planck's equation E = h x nu: nu = E/h = 10^-19 / 6.626 x 10^-34 = 1.51 x 10^14 s^-1 (hertz).

Step 3 - Wavelength from c = nu x lambda: lambda = c/nu = 3 x 10^8 / 1.51 x 10^14 = 1.98 x 10^-6 m.

Step 4 - Wave number = 1/lambda = 1 / 1.98 x 10^-6 = 5.0 x 10^5 m^-1.

Answer: frequency 1.51 x 10^14 Hz; wavelength 1.98 x 10^-6 m; wave number 5 x 10^5 m^-1. (Note: the book prints the wavelength as 1.98 x 10^6 m, but the minus sign is required: 1.98 x 10^-6 m is the value consistent with the book's own part (b) answer of 1.98 x 10^4 cm.)

17(b) Convert this energy of the photon into ergs and calculate the wavelength in cm, frequency in Hz and wave number in cm^-1. [h = 6.626 x 10^-34 J s or 6.625 x 10^-27 erg s, c = 3 x 10^8 m s^-1 or 3 x 10^10 cm s^-1]

Step 1 - Convert energy to ergs: 1 J = 10^7 erg, so E = 10^-19 x 10^7 = 10^-12 erg.

Step 2 - Frequency: nu = E/h = 10^-12 / 6.625 x 10^-27 = 1.51 x 10^14 s^-1 (hertz). The frequency is unchanged by the change of units.

Step 3 - Wavelength in cm: lambda = c/nu = 3 x 10^10 / 1.51 x 10^14 = 1.98 x 10^-4 cm.

Step 4 - Wave number in cm^-1: 1/lambda = 1 / 1.98 x 10^-4 = 5 x 10^3 cm^-1.

Answer: frequency 1.51 x 10^14 Hz; wavelength 1.98 x 10^-4 cm; wave number 5 x 10^3 cm^-1 (matches the book's printed answers).

18 The formula for calculating the energy of an electron in hydrogen atom given by Bohr's model is E_n = -(m e^4)/(8 epsilon0^2 h^2 n^2). Calculate the energy of the electron in first orbit of hydrogen atom. The values of various parameters are same as provided in Q19.

Step 1 - For the first orbit n = 1, so E_1 = -(m e^4)/(8 epsilon0^2 h^2).

Step 2 - Substituting m = 9.108 x 10^-31 kg, e = 1.602 x 10^-19 C, epsilon0 = 8.85 x 10^-12 C^2 J^-1 m^-1 and h = 6.624 x 10^-34 J s gives E_1 = -2.18 x 10^-18 J.

Answer: -2.18 x 10^-18 J (matches the book's printed answer).

19(a) Bohr's equation for the radius of nth orbit of electron in hydrogen atom is r_n = (epsilon0 h^2 n^2)/(pi e^2 m). When the electron moves from n = 1 to n = 2, how much does the radius of the orbit increase?

Step 1 - Radius of first orbit: r_1 = 0.529 x 1^2 = 0.529 angstrom.

Step 2 - Radius of second orbit: r_2 = 0.529 x 2^2 = 0.529 x 4 = 2.116 angstrom.

Step 3 - Increase in radius = r_2 - r_1 = 2.116 - 0.529 = 1.587 angstrom.

Answer: 1.587 angstrom (matches the book's printed answer).

19(b) What is the distance travelled by the electron when it goes from n = 2 to n = 3 and n = 9 to n = 10? [epsilon0 = 8.85 x 10^-12 C^2 J^-1 m^-1, h = 6.624 x 10^-34 J s, pi = 3.14, m = 9.108 x 10^-31 kg, e = 1.602 x 10^-19 C]

Step 1 - Use r_n = 0.529 x n^2 angstrom. Radii: r_2 = 2.116 angstrom, r_3 = 4.761 angstrom, r_9 = 42.849 angstrom, r_10 = 52.9 angstrom.

Step 2 - Distance from n = 2 to n = 3 = r_3 - r_2 = 4.761 - 2.116 = 2.645 angstrom, about 2.65 angstrom.

Step 3 - Distance from n = 9 to n = 10 = r_10 - r_9 = 52.9 - 42.849 = 10.051 angstrom, about 10.05 angstrom.

Answer: 2.65 angstrom and 10.05 angstrom (matches the book's printed answers). Note: the 'distance travelled' here is the increase in orbital radius between the two orbits.

20(a) Calculate the energy of first five orbits of hydrogen atom and determine the energy differences between them.

Step 1 - Use E_n = -2.18 x 10^-18 / n^2 J. E_1 = -2.18 x 10^-18 J; E_2 = -2.18/4 = -0.545 x 10^-18 J; E_3 = -2.18/9 = -0.242 x 10^-18 J; E_4 = -2.18/16 = -0.136 x 10^-18 J; E_5 = -2.18/25 = -0.087 x 10^-18 J.

Step 2 - Differences: E_2 - E_1 = 1.635 x 10^-18 J; E_3 - E_2 = 0.303 x 10^-18 J; E_4 - E_3 = 0.106 x 10^-18 J; E_5 - E_4 = 0.049 x 10^-18 J.

Answer: energies -2.18, -0.545, -0.242, -0.136, -0.087 (all x 10^-18 J) for n = 1 to 5; the gaps shrink rapidly as n increases.

20(b) Justify that energy difference between second and third orbits is approximately five times smaller than that between first and second orbits.

Step 1 - Difference between first and second orbits = 1.635 x 10^-18 J (from Q20(a)).

Step 2 - Difference between second and third orbits = 0.303 x 10^-18 J.

Step 3 - Ratio = 1.635 / 0.303 = 5.4, approximately 5.

Answer: the second-to-third gap is about five times smaller than the first-to-second gap, as stated.

20(c) Calculate the energy of electron in He+ in first five orbits and justify that the energy differences are different from those of hydrogen atom.

Step 1 - For He+, Z = 2, so E_n = -2.18 x Z^2 / n^2 x 10^-18 = -8.72 / n^2 x 10^-18 J. E_1 = -8.72, E_2 = -2.18, E_3 = -0.969, E_4 = -0.545, E_5 = -0.349 (all x 10^-18 J).

Step 2 - Differences: E_2 - E_1 = 6.54 x 10^-18 J; E_3 - E_2 = 1.211 x 10^-18 J, and so on.

Step 3 - Every He+ energy and every gap is exactly Z^2 = 4 times the corresponding hydrogen value.

Answer: the He+ energies differ from hydrogen's by the factor Z^2 = 4, because the energy is proportional to the square of the nuclear charge.

20(d) Do you think that groups of the spectral lines of He+ are at different places than those for hydrogen atom? Give reasons.

Step 1 - Spectral line positions are set by energy differences, delta E = h x nu.

Step 2 - For He+ every energy difference is 4 times the hydrogen difference (Q20(c)).

Step 3 - Hence every He+ line has 4 times the frequency (one quarter the wavelength) of the corresponding hydrogen line.

Answer: yes, the He+ line groups lie at different places, shifted to higher frequency and shorter wavelength by the factor 4.

21 Calculate the value of principal quantum number if an electron in hydrogen atom revolves in an orbit of energy -0.242 x 10^-18 J.

Step 1 - Use E_n = -2.18 x 10^-18 / n^2 J, with E_n = -0.242 x 10^-18 J.

Step 2 - So 0.242 = 2.18 / n^2, giving n^2 = 2.18 / 0.242 = 9.0.

Step 3 - n = 3.

Answer: n = 3 (matches the book's printed answer).

22(a) Bohr's formula for the energy levels of hydrogen atom for any system H, He+, Li2+, etc. is E_n = -(Z^2 e^4 m)/(8 epsilon0^2 h^2 n^2) or E_n = -K [Z^2/n^2]. For hydrogen Z = 1 and for He+ Z = 2. Draw an energy level diagram for hydrogen atom and He+.

Step 1 - Labelled drawing instruction: draw a vertical energy axis with zero at the top (free electron) and negative energies below.

Step 2 - For hydrogen mark levels at -2.18 (n = 1), -0.545 (n = 2), -0.242 (n = 3), -0.136 (n = 4), -0.087 (n = 5), all x 10^-18 J, the gaps shrinking upwards.

Step 3 - For He+ mark levels at -8.72 (n = 1), -2.18 (n = 2), -0.969 (n = 3), -0.545 (n = 4), -0.349 (n = 5), all x 10^-18 J, on the same scale, so the He+ ladder is stretched four times deeper than hydrogen's.

Answer: see the drawing instruction above; the two ladders differ by the factor Z^2 = 4.

22(b) Thinking that K = 2.18 x 10^-18 J, calculate the energy needed to remove the electron from hydrogen atom and from He+.

Step 1 - Ionization energy = energy to take the electron from n = 1 to infinity (E = 0). For hydrogen: 0 - (-2.18 x 10^-18) = 2.18 x 10^-18 J.

Step 2 - For He+: $E_1 = -K \times Z^2 = -2.18 \times 4 = -8.72 \times 10^{-18}$ J, so the energy needed = 8.72 x 10^-18 J.

Answer: 2.18 x 10^-18 J for H and 8.72 x 10^-18 J for He+ (matches the book's printed answers).

22(c) How do you justify that the energies calculated in (b) are the ionization energies of H and He+?

Step 1 - Ionization energy is defined as the energy needed to remove the electron

completely from the atom, i.e. to take it from its ground state to infinity.

Step 2 - At infinity the electron is free and its energy is defined as zero.

Step 3 - The energy supplied must therefore equal 0 - E_1, which is exactly what was calculated in (b).

Answer: the calculated values are ionization energies because they are the ground-state-to-infinity energy differences.

22(d) Use Avogadro's number to convert ionization energy values in kJ mol^-1 for H and He+.

Step 1 - Multiply the per-atom energy by Avogadro's number, 6.022 x 10^23 mol^-1. For H: 2.18 x 10^-18 x 6.022 x 10^23 = 1.313 x 10^6 J mol^-1 = 1313 kJ mol^-1.

Step 2 - For He+: 8.72 x 10^-18 x 6.022 x 10^23 = 5.251 x 10^6 J mol^-1 = 5251 kJ mol^-1.

Answer: about 1313.3 kJ mol^-1 for H and 5249.4 kJ mol^-1 for He+ as printed in the book (the small differences are rounding of K and Avogadro's number).

22(e) The experimental values of ionization energy of H and He+ are 1331 kJ mol^-1 and 5250 kJ mol^-1, respectively. How do you compare your values with experimental values?

Step 1 - Calculated: about 1313 kJ mol^-1 (H) and 5249 kJ mol^-1 (He+). Experimental: 1331 and 5250 kJ mol^-1.

Step 2 - The He+ values agree closely (5249 vs 5250). The hydrogen value is about 1.3% low (1313 vs 1331), the difference coming from the rounded constants used in the book's calculation.

Answer: the calculated values agree well with experiment; the hydrogen value is slightly low because of rounding in the constants (the book prints 5249 kJ mol^-1 for He+ in this comparison).

23 Calculate the wave number of the photon when the electron jumps from (i) n = 5 to n = 2, (ii) n = 5 to n = 1. In which series of spectral lines and spectral regions these photons will appear?

Step 1 - Use wavenumber = R x (1/n1^2 - 1/n2^2) with R = 1.097 x 10^7 m^-1.

Step 2 - (i) n = 5 to n = 2: 1/4 - 1/25 = 0.21, so wavenumber = 1.097 x 10^7 x 0.21 = 2.3 x 10^6 m^-1. The lower level is n = 2, so this is the Balmer series, in the visible region.

Step 3 - (ii) n = 5 to n = 1: 1 - 1/25 = 0.96, so wavenumber = 1.097 x 10^7 x 0.96 = 1.05 x 10^7 m^-1. The lower level is n = 1, so this is the Lyman series, in the ultraviolet region.

Answer: (i) 2.3 x 10^6 m^-1, Balmer series, visible; (ii) 1.05 x 10^7 m^-1, Lyman series, ultraviolet (matches the book's printed answers).

24(a) A photon of a wave number 102.70 x 10^5 m^-1 is emitted when electron jumps from higher to n = 1. Determine the number of that orbit from where the electron falls.

Step 1 - Wavenumber = 102.70 x 10^5 = 1.027 x 10^7 m^-1. Use wavenumber = R x (1 - 1/n^2) with R = 1.097 x 10^7 m^-1, since the lower level is n = 1.

Step 2 - 1 - 1/n^2 = 1.027 x 10^7 / 1.097 x 10^7 = 0.936.

Step 3 - 1/n^2 = 0.064, so n^2 = 15.6, about 16, giving n = 4.

Answer: n = 4 (matches the book's printed answer).

24(b) Indicate the name of the series to which this photon belongs.

Step 1 - The electron falls to n = 1.

Step 2 - All lines ending at n = 1 form the Lyman series.

Answer: Lyman series (matches the book's printed answer).

24(c) If the electron will fall from higher orbit to n = 2, then calculate the wave number of the photon emitted. Why is this energy difference so small as compared to that in part (a)?

Step 1 - From part (a) the higher orbit is n = 4. Falling to n = 2: wavenumber = R x (1/4 - 1/16) = 1.097 x 10^7 x 0.1875 = 2.06 x 10^6 m^-1 = 20.6 x 10^5 m^-1, about 20.5 x 10^5 m^-1 as printed.

Step 2 - The energy difference is smaller because the energy levels get closer together as n increases (E proportional to -1/n^2): the gap between n = 4 and n = 2 is much smaller than the gap between n = 4 and n = 1.

Answer: about 20.5 x 10^5 m^-1; the gap is smaller because higher levels are crowded together near zero energy.

25(a) What is de-Broglie's wavelength of an electron in meters travelling at half a speed of light? [m = 9.109 x 10^-31 kg, c = 3 x 10^8 m s^-1]

Step 1 - Velocity v = c/2 = 3 x 10^8 / 2 = 1.5 x 10^8 m s^-1.

Step 2 - de Broglie wavelength lambda = h/(m v) = 6.626 x 10^-34 / (9.109 x 10^-31 x 1.5 x 10^8).

Step 3 - Denominator = 1.366 x 10^-22, so lambda = 4.85 x 10^-12 m.

Answer: 4.85 x 10^-12 m = 0.048 angstrom, about 0.048 angstrom as printed in the book.

25(b) Convert the mass of electron into grams and velocity of light into cm s^-1 and then calculate the wavelength of an electron in cm.

Step 1 - Mass in grams: 9.109 x 10^-31 kg = 9.109 x 10^-28 g. Velocity of light in cm s^-1: 3 x 10^10 cm s^-1, so half the speed of light = 1.5 x 10^10 cm s^-1.

Step 2 - Planck's constant in cgs: h = 6.626 x 10^-27 erg s. lambda = h/(m v) = 6.626 x 10^-27 / (9.109 x 10^-28 x 1.5 x 10^10).

Step 3 - Denominator = 1.366 x 10^-17, so lambda = 4.85 x 10^-10 cm = 0.048 x 10^-8 cm.

Answer: 0.048 x 10^-8 cm (matches the book's printed answer).

25(c) Convert the wavelength of electron from meters to (i) nm, (ii) angstrom, (iii) pm.

Step 1 - From part (a), lambda = 4.85 x 10^-12 m.

Step 2 - (i) In nm: 1 nm = 10^-9 m, so lambda = 4.85 x 10^-12 / 10^-9 = 0.00485 nm, about 0.0048 nm.

Step 3 - (ii) In angstrom: 1 angstrom = 10^-10 m, so lambda = 0.0485 angstrom, about 0.048 angstrom. (iii) In pm: 1 pm = 10^-12 m, so lambda = 4.85 pm.

Answer: 0.0048 nm; 0.048 angstrom; 4.85 pm (matches the book's printed answers).

Chapter 6 - Chemical Bonding

Exercise 6 - Multiple Choice Questions

1(i) An ionic compound A+ B- is most likely to be formed when

A. the ionization energy of A is high and electron affinity of B is low

✓ B. the ionization energy of A is low and electron affinity of B is high

C. both the ionization energy of A and electron affinity of B are high

D. both the ionization energy of A and electron affinity of B are low

1(ii) The number of bonds in nitrogen molecule is

A. one sigma and one pi

✓ B. one sigma and two pi

C. three sigma only

D. two sigma and one pi

1(iii) Which of the following statements is not correct regarding bonding molecular orbitals?

A. Bonding molecular orbitals possess less energy than atomic orbitals from which they are formed.

✓ B. Bonding molecular orbitals have low electron density between the two nuclei.

C. Every electron in the bonding molecular orbitals contributes to the attraction between atoms.

D. Bonding molecular orbitals are formed when the electron waves undergo constructive interference.

1(iv) Which of the following molecules has zero dipole moment?

A. NH3

B. CHCl3

C. H2O

✓ D. BF3

1(v) Which of the hydrogen halides has the highest percentage of ionic character?

A. HCl

B. HBr

✓ C. HF

D. HI

1(vi) Which of the following species has unpaired electrons in antibonding molecular orbitals?

A. O2^2+

✓ B. N2^2-

C. B2

D. F2

Exercise 6 - Fill in the Blanks

2(i) The tendency of atoms to attain maximum of eight electrons in the valence shell is called completion of octet.

2(ii) The geometrical shape of SiCl4 and PCl3 can be explained on the basis of sp3 and sp3 hybridizations.

2(iii) The VSEPR theory stands for Valence Shell Electron Pair Repulsion .

2(iv) For N2 molecule, the energy of sigma (2p)x orbital is greater than pi (2p)y orbital.

2(v) The paramagnetic property of O2 is well explained on the basis of MO theory in terms of the presence of two unpaired electrons in two MO orbitals.

2(vi) The values of dipole moment for CS2 is zero while for SO2 is 1.61 D.

2(vii) The bond order of N2 is three while that of Ne2 is zero.

Exercise 6 - True or False

3(i) The core of an atom is the atom minus its valence shell.

True

3(ii) The molecules of nitrogen (N triple bond N) and acetylene (HC triple bond CH) are not isoelectronic.

False

3(iii) There are four coordinate covalent bonds in NH4+ ion.

False

3(iv) A sigma-bond is stronger than a pi-bond and the electrons of sigma-bond are more diffused than pi-bond.

False

3(v) The bond energy of heteroatomic diatomic molecules increases with the decrease in the electronegativities of the bonded atoms.

False

3(vi) With increase in bond order, bond length decreases and bond strength increases.

True

3(vii) The first ionization energies of the elements rise steadily with the increasing atomic number from top to bottom in a group.

False

3(viii) A double bond is stronger than a single bond and a triple bond is weaker than a double bond.

False

3(ix) The bonds formed between the elements having electronegativity difference more than 1.7 are said to be covalent in nature.

False

3(x) The repulsive force between the two bonding pairs is less than that between the two lone pairs.

True

3(xi) The number of covalent bonds an atom can form is related to the number of unpaired electrons it has.

True

3(xii) The rules which govern the filling of electrons into the atomic orbitals also govern filling of electrons into the molecular orbitals.

True

Exercise 6 - Short and Numerical Questions

3(i) reason - The core of an atom is the atom minus its valence shell.

True. The core is the inner, non-valence part of the atom: the nucleus plus all inner shells. Removing the valence shell leaves exactly the core.

3(ii) reason - The molecules of nitrogen and acetylene are not isoelectronic.

False. N_2 has 7 + 7 = 14 electrons and acetylene C_2H_2 has 12 + 2 = 14 electrons, with the same number of atoms in the same connectivity pattern, so the two molecules ARE isoelectronic. The statement that they are not isoelectronic is false.

3(iii) reason - There are four coordinate covalent bonds in NH_4^+ ion.

False. In NH_4^+ the nitrogen forms three ordinary covalent bonds with three hydrogens and only ONE coordinate covalent bond, with the fourth hydrogen (the H^+ that accepts the lone pair). So there is one coordinate bond, not four.

3(iv) reason - A sigma-bond is stronger than a pi-bond and the electrons of sigma-bond are more diffused than pi-bond.

False, as a whole. A sigma-bond IS stronger than a pi-bond, but sigma electrons are concentrated on the line between the nuclei, while pi electrons are spread above and below it. So sigma electrons are LESS diffused than pi electrons, not more.

3(v) reason - The bond energy of heteroatomic diatomic molecules increases with the decrease in the electronegativities of the bonded atoms.

False. The bond energy of a heteroatomic molecule increases as the electronegativity difference of the bonded atoms INCREASES, because the bond gains partial ionic character, which strengthens it. Decreasing the difference weakens this effect.

3(vi) reason - With increase in bond order, bond length decreases and bond strength increases.

True. Higher bond order means more shared electron pairs pulling the nuclei together, so the bond is shorter and stronger: single < double < triple in strength, and the reverse in length.

3(vii) reason - The first ionization energies of the elements rise steadily from top to bottom in a group.

False. First ionization energy DECREASES down a group, because the outer electron is farther from the nucleus and better shielded by inner shells, so it is held less

tightly. It generally increases across a period, not down a group.

3(viii) reason - A double bond is stronger than a single bond and a triple bond is weaker than a double bond.

False. A double bond is stronger than a single bond, and a triple bond is stronger than a double bond: bond strength increases with bond order.

3(ix) reason - Bonds with electronegativity difference more than 1.7 are covalent in nature.

False. On Pauling's scale, an electronegativity difference greater than about 1.7 marks a bond as predominantly IONIC, not covalent. Smaller differences give covalent (or polar covalent) bonds.

3(x) reason - The repulsive force between the two bonding pairs is less than that between the two lone pairs.

True. The repulsion order is lone pair-lone pair > lone pair-bonding pair > bonding pair-bonding pair, so two bonding pairs repel each other less than two lone pairs do. This is the basis of VSEPR shapes.

3(xi) reason - The number of covalent bonds an atom can form is related to the number of unpaired electrons it has.

True. Each unpaired electron can pair with an electron from another atom to form one covalent bond, so the number of bonds an atom normally forms equals its number of unpaired electrons (before hybridization or promotion).

3(xii) reason - The rules for filling atomic orbitals also govern filling of molecular orbitals.

True. Molecular orbitals are filled by the Aufbau order, Pauli's exclusion principle (at most two electrons of opposite spin per orbital) and Hund's rule, exactly as for atomic orbitals.

4 What is a chemical bond? Discuss the formation of ionic and covalent bonds. How does the electronegativity difference differentiate between ionic and covalent bond?

A chemical bond is the force that holds two or more atoms or ions together in a molecule or crystal, formed because the combined system has lower energy than the separate atoms. An ionic bond forms by complete transfer of electrons from a metal (low ionization energy) to a non-metal (high electron affinity), giving oppositely charged ions held by electrostatic attraction, e.g. Na+ Cl-. A covalent bond forms by mutual sharing of electron pairs between atoms, e.g. H2, Cl2. Electronegativity difference decides the type: a difference near zero gives a non-polar covalent bond; a moderate difference gives a polar covalent bond; a difference above about 1.7 gives a predominantly ionic bond.

5(a) Define ionization energy and electron affinity. How do these quantities change in the periodic table? What factors are responsible for their variation?

Ionization energy is the minimum energy needed to remove the most loosely held electron from an isolated gaseous atom. Electron affinity is the energy released when an electron is added to an isolated gaseous atom. Both generally increase across a period (greater nuclear charge, smaller size) and decrease down a group (larger size, more shielding). Factors responsible: nuclear charge, atomic size, shielding effect of inner electrons, and the stability of half-filled or completely filled subshells.

5(b) Explain what you understand by the term electronegativity. Discuss its variations in the periodic table. How does it affect the bond strengths?

Electronegativity is the tendency of an atom in a molecule to attract the shared pair of electrons towards itself. It increases across a period and decreases down a group; fluorine (4.0) is the most electronegative element. Greater electronegativity difference makes a bond more polar and, up to a point, stronger, because partial ionic character adds electrostatic attraction to the covalent sharing; when the difference exceeds about 1.7 the bond becomes predominantly ionic.

6 Write the Lewis structures for the following compounds: (i) HCN (ii) CCl4 (iii) CS2

(iv) H3N->AlF3 (v) NH4OH (vi) H2SO4 (vii) H3PO4 (viii) K2Cr2O7 (ix) N2O5 (x) Ag(NH3)2NO3. (Note: the book's printed numbering of (vi)/(vii) is unclear under show-through; the ten compounds are given here in the printed order.)

Labelled drawing instructions (draw dots for valence electrons, lines for shared pairs, and show charges where present): (i) HCN: H-C triple bond N, with one lone pair on N. (ii) CCl4: central C singly bonded to four Cl atoms, each Cl with three lone pairs. (iii) CS2: S double bond C double bond S, each S with two lone pairs. (iv) H3N->AlF3: NH3 donating its lone pair to AlF3, shown by an arrow from N to Al; Al bonded to three F atoms, each F with three lone pairs. (v) NH4OH: [NH4]+ with N bonded to four H atoms (one bond shown as the coordinate bond) and OH- with O bonded to H and three lone pairs on O. (vi) H2SO4: central S double bonded to two O atoms and singly bonded to two OH groups, each O with lone pairs as required. (vii) H3PO4: central P double bonded to one O and singly bonded to three OH groups. (viii) K2Cr2O7: two K+ ions with the dichromate ion Cr2O7^2-, two CrO4 tetrahedra sharing one oxygen. (ix) N2O5: O2N-O-NO2, each N bonded to three O atoms (one bridging). (x) Ag(NH3)2NO3: [Ag(NH3)2]+ with Ag bonded to two NH3 molecules by coordinate bonds, and NO3- as a separate ion with N bonded to three O atoms.

7(a) Explain qualitatively the valence bond theory. How does it differ from molecular orbital theory?

Valence bond theory: a covalent bond forms when two half-filled atomic orbitals, one from each atom, overlap, and the two electrons of opposite spin pair in the overlap region. Greater overlap gives a stronger bond; head-on overlap gives a sigma bond, sideways overlap gives a pi bond. Difference from molecular orbital theory: valence bond theory keeps the electrons in localized bonds between two atoms and retains atomic-orbital character, while molecular orbital theory combines atomic orbitals into new orbitals spread over the whole molecule, with bonding and antibonding

levels, and explains bond order and paramagnetism (e.g. of O2) that valence bond theory cannot.

7(b) How can the bonding in Cl2, O2, N2, HF, H2S be explained with respect to valence bond theory?

Cl2: overlap of two half-filled 3p orbitals head-on gives one sigma bond. O2: one sigma bond from head-on 2p overlap plus one pi bond from sideways 2p overlap, a double bond. N2: one sigma plus two pi bonds from 2p overlaps, a triple bond. HF: overlap of the H 1s orbital with a half-filled F 2p orbital gives one sigma bond. H2S: overlap of two H 1s orbitals with two half-filled S 3p orbitals gives two sigma bonds, with two lone pairs left on sulphur.

8 Explain VSEPR theory. Discuss the structures of CH4, NH3, H2O, BeCl2, BF3, SO2, SO3 with reference to this theory.

VSEPR theory: electron pairs around a central atom (bonding pairs and lone pairs) repel each other and take up positions as far apart as possible, giving the molecule its shape; lone pairs repel more strongly than bonding pairs. CH4: four bonding pairs, tetrahedral, 109.5 degrees. NH3: three bonding pairs and one lone pair, pyramidal, 107.5 degrees. H2O: two bonding pairs and two lone pairs, bent, 104.5 degrees. BeCl2: two bonding pairs, linear, 180 degrees. BF3: three bonding pairs, trigonal planar, 120 degrees. SO2: two bonding pairs and one lone pair on sulphur, bent. SO3: three bonding pairs, trigonal planar, 120 degrees.

9 The molecules NF3 and BF3 both have molecular formulae of the type XF3, but they have different structural formulas. Keeping in view VSEPR theory, sketch the shape of each molecule and explain the origin of differing in shapes.

Both are XF3, but NF3 has a lone pair on nitrogen while BF3 has none. NF3: three bonding pairs plus one lone pair on N, so the electron geometry is tetrahedral but the shape is pyramidal, and the molecule is polar. BF3: three bonding pairs and no lone pair on B, so the shape is trigonal planar with 120 degree angles, and the three

equal bond dipoles cancel, so it is non-polar. The lone pair on nitrogen is the origin of the difference in shape.

10 The species NH2-, NH3, NH4+ have bond angles of 105 degrees, 107.5 degrees and 109.5 degrees respectively. Justify these values by drawing their structures.

Labelled drawing instructions: draw NH2- as a bent (angular) ion with two N-H bonds and two lone pairs on N; draw NH3 as a pyramid with three N-H bonds and one lone pair on N; draw NH4+ as a regular tetrahedron with four N-H bonds and no lone pair on N. Justification: lone pair-bonding pair repulsion is stronger than bonding pair-bonding pair repulsion, so each lone pair squeezes the H-N-H angle below the tetrahedral 109.5 degrees. NH4+ has no lone pair, so it keeps 109.5 degrees; NH3 with one lone pair is squeezed to 107.5 degrees; NH2- with two lone pairs is squeezed most, to 105 degrees.

11(a) Explain atomic orbital hybridization with reference to sp3, sp2 and sp modes of hybridization for PH3, C2H4 and C2H2. Discuss geometries of CCl4, PCl3 and H2S by hybridization of central atoms.

Hybridization is the mixing of atomic orbitals of an atom to form an equal number of new hybrid orbitals of the same energy, which then form bonds. sp3: one s and three p orbitals mix to give four sp3 orbitals pointing to the corners of a tetrahedron (109.5 degrees), e.g. PH3 (pyramidal, one orbital holding a lone pair) and CCl4 (tetrahedral). sp2: one s and two p orbitals mix to give three sp2 orbitals in a plane (120 degrees) plus one unhybridized p orbital, e.g. C2H4 (planar, with a pi bond from the p orbitals). sp: one s and one p orbital mix to give two sp orbitals in a line (180 degrees) plus two unhybridized p orbitals, e.g. C2H2 (linear, with two pi bonds). PCl3: central P is sp3 hybridized, pyramidal (one lone pair). H2S: central S is sp3 hybridized, bent (two lone pairs).

11(b) The linear geometry of BeCl2 suggests that central Be atom is sp-hybridized. What type of hybridization does a central atom undergo when the atoms bonded to

it are located at the corners of (a) an equilateral triangle (b) a regular tetrahedron

(c) a trigonal bipyramid?

(a) Atoms at the corners of an equilateral triangle: the central atom is sp2 hybridized (three hybrid orbitals in a plane, 120 degrees apart). (b) Atoms at the corners of a regular tetrahedron: the central atom is sp3 hybridized (four hybrid orbitals, 109.5 degrees apart). (c) Atoms at the corners of a trigonal bipyramid: the central atom is sp3d hybridized (five hybrid orbitals). BeCl2 is linear because Be is sp hybridized, with two hybrid orbitals at 180 degrees.

12(a) Give the basis of the molecular orbital theory and discuss the molecular orbital configurations of (i) He2 (ii) N2 (iii) O2 (iv) O2^2+ (v) O2^2-.

Molecular orbital theory: atomic orbitals combine to form molecular orbitals spread over the whole molecule; n atomic orbitals give n molecular orbitals, half bonding (lower energy) and half antibonding (higher energy). Bond order = (bonding electrons - antibonding electrons)/2. (i) He2: (sigma 1s)^2 (sigma* 1s)^2, bond order 0, so He2 does not exist. (ii) N2: (sigma 2s)^2 (sigma* 2s)^2 (pi 2p)^4 (sigma 2p)^2, bond order 3. (iii) O2: (sigma 2s)^2 (sigma* 2s)^2 (sigma 2p)^2 (pi 2p)^4 (pi* 2p)^2, bond order 2, with two unpaired electrons in the pi* orbitals. (iv) O2^2+: as O2 but with the pi* orbitals empty, bond order 3. (v) O2^2-: as O2 but with the pi* orbitals full (four electrons), bond order 1.

12(b) How does molecular orbital theory explain the paramagnetic character of O2, O2+ and O2^2- species?

O2 has two unpaired electrons in its pi* antibonding orbitals, so it is paramagnetic. O2+ has lost one pi* electron, leaving one unpaired pi* electron, so it is also paramagnetic (less strongly). O2^2- has a full pi* set (four electrons, all paired), so it has no unpaired electrons and is diamagnetic. Molecular orbital theory explains this directly from the pi* occupancy, which valence bond theory cannot.

13(a) Sketch the molecular orbital pictures of (i) pi (2p)x and pi* (2p)x (ii) O2, O2+,

O2- (iii) He2 and Ne2.

Labelled drawing instructions: (i) pi (2p)x bonding orbital: draw two p orbitals overlapping sideways, with electron density above and below the line joining the nuclei and no node between the nuclei. pi* (2p)x antibonding orbital: draw the same two p orbitals with opposite phases, with a node (zero electron density) between the nuclei and density pushed to the outer sides. (ii) O2, O2+, O2-: draw energy-level diagrams with the pi* level holding two, one and four electrons respectively (see Q12). (iii) He2 and Ne2: draw energy-level diagrams in which bonding and antibonding levels hold equal numbers of electrons, giving bond order zero, so neither molecule exists as a stable species.

13(b) Sketch the hybrid orbitals of the species PCl3, SiCl4 and NH4+.

Labelled drawing instructions: draw PCl3 with P at the apex of a pyramid, three Cl atoms at the base, and one lone pair on P (sp3, tetrahedral electron geometry, pyramidal shape). Draw SiCl4 with Si at the centre of a tetrahedron and four Cl atoms at the corners (sp3, tetrahedral). Draw NH4+ with N at the centre of a tetrahedron and four H atoms at the corners, with a + charge on the ion (sp3, tetrahedral). In each case the central atom uses four sp3 hybrid orbitals; the shapes differ only because PCl3 has one lone pair while SiCl4 and NH4+ have none.

14(a) Define bond energy. Explain the various parameters which determine its strength.

Bond energy is the average energy required to break one mole of a particular type of bond in a gaseous molecule, a measure of bond strength. Parameters that determine it: bond length (shorter bonds are stronger), bond order (higher order is stronger), electronegativity difference (greater difference adds ionic character and strength), atomic size and orbital overlap (better overlap gives a stronger bond), and lone-pair repulsions.

14(b) How do you compare the bond strengths of (i) polar and non-polar molecules

(ii) sigma- and pi-bonds?

(i) Polar vs non-polar: a polar bond, with partial ionic character from an electronegativity difference, is stronger than a comparable non-polar bond, because electrostatic attraction adds to the covalent sharing. (ii) Sigma vs pi: a sigma bond is stronger than a pi bond, because head-on overlap puts electron density directly between the nuclei, while sideways pi overlap is less effective.

14(c) Calculate the bond energy of H-Br. The bond energy of H-H is 436 kJ mol^-1 and that of Br-Br is 193 kJ mol^-1.

Step 1 - Use the mean of the two homonuclear bond energies: E(H-Br) = (E(H-H) + E(Br-Br)) / 2.

Step 2 - Substitute: E(H-Br) = (436 + 193) / 2 = 629 / 2 = 314.5 kJ mol^-1.

Answer: 314.5 kJ mol^-1 (matches the book's printed answer).

15(a) Define dipole moment. Give its various units. Find relationship between Debye and C m. How does it help to find out the shapes of molecules?

Dipole moment is the product of the charge (q) at either end of a polar bond and the distance (r) between the two oppositely charged centres: mu = q x r. It is a vector quantity, measured in Debye (D) or in coulomb-metre (C m). 1 D = 3.336 x 10^-30 C m. A molecule's shape follows from its dipole moment: if the individual bond moments cancel (zero resultant), the molecule is symmetrical (linear, trigonal planar, tetrahedral); if they do not cancel, the molecule is unsymmetrical (bent, pyramidal) and polar.

15(b) The bond length of H-Br is 1.4 x 10^-10 m. Its observed dipole moment is 0.79 D. Find the percentage ionic character of the bond. Unit positive charge = 1.6022 x 10^-19 C and 1 D = 3.336 x 10^-30 C m.

Step 1 - Dipole moment if the bond were 100% ionic: mu(ionic) = q x r = 1.6022 x 10^-19 C x 1.4 x 10^-10 m = 2.243 x 10^-29 C m.

Step 2 - Convert to Debye: mu(ionic) = 2.243 x 10^-29 / 3.336 x 10^-30 = 6.72 D.

Step 3 - Percentage ionic character = (mu(observed) / mu(ionic)) x 100 = (0.79 / 6.72) x 100 = 11.7%.

Answer: 11.7% (matches the book's printed answer).

16 PF3 is a polar molecule with dipole moment 1.02 D and thus the P-F bond is polar. Si is in the proximity of P in the periodic table. It is expected that Si-F bond would also be polar, but SiF4 has no dipole moment. Explain it.

PF3 is pyramidal (three P-F bonds plus one lone pair on P), so its three bond dipoles do not cancel and the molecule has a dipole moment of 1.02 D: the P-F bond is polar. SiF4 is tetrahedral and perfectly symmetrical, so its four equal Si-F bond dipoles cancel exactly, giving zero resultant dipole moment, even though each Si-F bond is polar. Symmetry, not bond polarity alone, decides whether a molecule is polar.

17 Which of the following molecules will be polar or non-polar? Sketch the structures and justify your answer: (i) CCl4 (ii) SO3 (iii) NF3 (iv) SO2.

Labelled drawing instructions: (i) CCl4: draw C at the centre of a tetrahedron with four Cl atoms at the corners; the four equal bond dipoles cancel, so CCl4 is non-polar. (ii) SO3: draw S at the centre of a triangle with three O atoms at the corners in one plane; the three bond dipoles cancel, so SO3 is non-polar. (iii) NF3: draw N at the apex of a pyramid with three F atoms at the base and one lone pair on N; the dipoles do not cancel, so NF3 is polar. (iv) SO2: draw S with two O atoms

at an angle (bent shape) and one lone pair on S; the dipoles do not cancel, so SO2 is polar (1.61 D).

18 Explain the following with reasons (as printed, the list runs (i), (ii), (iii), (vi), (v), (vi), (vii): the (iv) entry is missing and (vi) appears twice in print; the items are answered here in printed order).

(i) Bond distance is the distance at which the attraction between the nuclei and the shared electrons exactly balances the repulsion between the nuclei and between the electrons: the energy minimum. If the atoms come closer, repulsion rises steeply; if they move apart, attraction is lost. The observed bond length is this compromise distance. (ii) After a coordinate bond forms, the donated pair becomes indistinguishable from any other shared pair in the molecule, so in NH4+, H3O+ and CH3NH3+ no experiment can tell which bond was the coordinate one: the distinction vanishes. (iii) H2O and NH3 are sp3 hybridized like CH4, but their lone pairs repel bonding pairs more strongly than bonding pairs repel each other, squeezing the angles from 109.5 degrees to 104.5 degrees (two lone pairs) and 107.5 degrees (one lone pair). (iv) [printed as (vi)] Pi-bonds are formed by sideways overlap, so their electron density lies above and below the bond axis and is spread over a larger region: pi-bonds are more diffused than sigma-bonds, whose density is concentrated on the axis. (v) In HI the iodine atom is much larger and its orbitals more diffused than in HCl, so the H-I bond is longer and weaker and the abnormality of bond length and strength (compared with the trend) is less prominent than in HCl. (vi) CO2 and CS2 are linear and symmetrical, so their equal and opposite bond dipoles cancel and the dipole moment is zero; SO2 is bent, so its bond dipoles add to 1.61 D. (vii) Electrovalent (ionic) compounds are held in a crystal lattice by strong electrostatic forces between ions, so much more energy is needed to melt, boil, vaporize or sublime them than for covalent compounds, which are held only by weak intermolecular forces.

Chapter 7 - Thermochemistry

Exercise 7 - Multiple Choice Questions

1(i) If an endothermic reaction is allowed to take place very rapidly in the air, the temperature of the surrounding air

A. remains constant

B. increases

✓ C. decreases

D. remains unchanged

1(ii) In endothermic reactions, the heat content of the

✓ A. products is more than that of reactants

B. reactants is more than that of products

C. both (a) and (b)

D. reactants and products are equal

1(iii) Calorie is equivalent to

A. 0.4184 J

B. 41.84 J

✓ C. 4.184 J

D. 418.4 J

1(iv) The change in heat contents of a chemical reaction at constant temperature and pressure is called

✓ A. enthalpy change

B. bond energy

C. heat of sublimation

D. internal energy change

1(v) Which of the following statements is contrary to the first law of

thermodynamics?

A. Energy can neither be created nor destroyed.

B. One form of energy can be transferred into an equivalent amount of other kinds of energy.

C. In an adiabatic process, the work done is independent of its path.

✓ D. Continuous production of mechanical work without supplying an equivalent amount of heat is possible.

1(vi) For a given process, the heat changes at constant pressure (qp) and at constant volume (qv) are related to each other as

A. qp = qv

B. qp < qv

✓ C. qp > qv

D. qp = qv/2

1(vii) For the reaction: NaOH + HCl -> NaCl + H2O the change in enthalpy is called

A. heat of reaction

B. heat of formation

✓ C. heat of neutralization

D. heat of combustion

1(viii) The net heat change in a chemical reaction is same, whether it is brought about in two or more different ways in one or several steps. It is known as

A. Henry's law

B. Joule's principle

✓ C. Hess's law

D. Law of conservation of energy

1(ix) Enthalpy of neutralisation of all the strong acids and strong bases has the same value because

A. neutralisation leads to the formation of salt and water.

B. strong acids and bases are ionic substances.

C. acids always give rise to H+ ions and bases always furnish OH- ions.

✓ D. the net chemical change involves the combination of H+ and OH- ions to form water.

Exercise 7 - Fill in the Blanks

2(i) The substance undergoing a physical or a chemical change forms a chemical system .

2(ii) The change in internal energy can be measured.

2(iii) Solids which have more than one crystalline forms possess different values of heats of formation.

2(iv) A process is called spontaneous if it takes place on its own without any external assistance.

2(v) A state function is a macroscopic property of a system which is

2(v) Complete: A ___ is a macroscopic property of a system which is ___ of the path adopted to bring about that change.

A state function is a macroscopic property of a system which is independent of the path adopted to bring about that change.

Exercise 7 - True or False

3(i) It is necessary that a spontaneous reaction should be exothermic.

False

3(ii) Amount of heat absorbed at constant volume is internal energy change.

True

3(iii) The work done by the system is given the positive sign.

False

3(iv) Enthalpy is a state function but internal energy is not.

False

3(v) Total heat content of a system is called enthalpy of the system.

True

Exercise 7 - Short and Numerical Questions

3(i) reason - It is necessary that a spontaneous reaction should be exothermic.

False. Spontaneity is decided by the Gibbs energy change, delta G = delta H - T delta S. An endothermic reaction (positive delta H) can still be spontaneous if the entropy increase is large enough that T delta S outweighs delta H, e.g. melting of ice at room temperature.

3(ii) reason - Amount of heat absorbed at constant volume is internal energy change.

True. At constant volume no expansion work is done (w = 0), so by the first law the heat absorbed equals the change in internal energy: qv = delta E.

3(iii) reason - The work done by the system is given the positive sign.

False. In the chemistry sign convention, work done BY the system on the surroundings is given a NEGATIVE sign (the system loses energy), while work done ON the system is positive. So delta E = q + w with w negative for expansion work done by the system.

3(iv) reason - Enthalpy is a state function but internal energy is not.

False. Both enthalpy (H = E + PV) and internal energy (E) are state functions: their values depend only on the present state of the system, not on the path taken to reach it.

3(v) reason - Total heat content of a system is called enthalpy of the system.

True. Enthalpy is defined as H = E + PV and represents the total heat content of a system at constant pressure; the heat absorbed or evolved at constant pressure equals the enthalpy change.

4 Define the following terms and give three examples of each: (i) System (ii) Surroundings (iii) State function (iv) Units of energy (v) Exothermic reaction (vi) Endothermic reaction (vii) Internal energy of the system (viii) Enthalpy of the system.

System: the part of the universe chosen for study, e.g. the reacting mixture in a flask; a solution in a calorimeter; a gas in a cylinder. Surroundings: everything outside the system that can exchange energy or matter with it, e.g. the air around a flask; the water bath around a calorimeter; the laboratory. State function: a property whose value depends only on the present state of the system, not on its history, e.g. temperature; pressure; enthalpy. Units of energy: joule (J); kilojoule (kJ); calorie (cal, 1 cal = 4.184 J). Exothermic reaction: a reaction that releases heat to the surroundings, e.g. combustion of methane; neutralization of an acid by a base; respiration. Endothermic reaction: a reaction that absorbs heat from the surroundings, e.g. melting of ice; photosynthesis; decomposition of calcium carbonate. Internal energy of the system: the total energy (kinetic + potential) of all the particles of the system, e.g. the internal energy of a gas rises when it is heated; it falls when the gas expands and does work; it is a state function. Enthalpy of the system: H = E + PV, the total heat content of the system, e.g. the enthalpy of steam is greater than that of liquid water; the enthalpy change of a reaction at constant pressure is the heat evolved or absorbed; enthalpy is a state function.

5(a) Differentiate between: (i) Internal energy and enthalpy (ii) Internal energy change and enthalpy change (iii) Exothermic and endothermic reactions.

(i) Internal energy (E) is the total energy of the particles of a system; enthalpy (H = E + PV) is the total heat content, i.e. internal energy plus the pressure-volume energy. (ii) Internal energy change (delta E) is the heat change at constant volume (qv); enthalpy change (delta H) is the heat change at constant pressure (qp); they differ by the expansion work, delta H = delta E + delta n RT. (iii) Exothermic

reactions release heat (delta H negative, surroundings warm up); endothermic reactions absorb heat (delta H positive, surroundings cool down).

5(b) Define the following enthalpies and give two examples of each: (i) Standard enthalpy of reaction (ii) Standard enthalpy of combustion (iii) Standard enthalpy of atomization (iv) Standard enthalpy of solution.

(i) Standard enthalpy of reaction: the enthalpy change when a reaction occurs with all substances in their standard states (1 atm, 25 degrees C), e.g. H2(g) + 1/2 O2(g) -> H2O(l), delta H = -285.8 kJ mol^-1; C(s) + O2(g) -> CO2(g), delta H = -393.7 kJ mol^-1. (ii) Standard enthalpy of combustion: the enthalpy change when one mole of a substance is completely burnt in excess oxygen in its standard state, e.g. CH4(g) + 2O2(g) -> CO2(g) + 2H2O(l), delta H = -890 kJ mol^-1; C2H5OH(l) + 3O2(g) -> 2CO2(g) + 3H2O(l), delta H = -1367 kJ mol^-1. (iii) Standard enthalpy of atomization: the enthalpy change when one mole of a substance is converted into gaseous atoms, e.g. Na(s) -> Na(g), delta H = +108 kJ mol^-1; 1/2 Cl2(g) -> Cl(g), delta H = +121 kJ mol^-1. (iv) Standard enthalpy of solution: the enthalpy change when one mole of a solute dissolves in a large excess of solvent, e.g. NH4Cl(s) + aq -> NH4Cl(aq), endothermic; NaOH(s) + aq -> NaOH(aq), exothermic.

6(a) What are spontaneous and non-spontaneous processes? Give examples.

A spontaneous process is one that takes place on its own without external assistance, e.g. flow of heat from a hot body to a cold one; rusting of iron. A non-spontaneous process is one that does not take place on its own and needs a continuous supply of energy, e.g. electrolysis of water; pumping water uphill.

6(b) Explain that burning of a candle is a spontaneous process.

Burning a candle is spontaneous because once ignited, the wax burns in air on its own, releasing heat (exothermic), and the overall Gibbs energy change is negative. The reaction needs only a small initial push (the match flame) to start; after that it sustains itself without further help.

6(c) Is it true that a non-spontaneous process never happens in the universe?

Explain it.

No, that is not true. A non-spontaneous process can be made to happen by supplying energy from outside, e.g. water does not split into hydrogen and oxygen on its own, but electrolysis splits it using electrical energy; a refrigerator pumps heat from cold to hot using electrical work. Non-spontaneous only means 'not on its own', not 'never'.

7(a) What is the first law of thermodynamics? How does it explain that (i) qv = delta E (ii) qp = delta H?

The first law of thermodynamics states that energy can neither be created nor destroyed, only changed from one form to another: delta E = q + w, where q is heat absorbed by the system and w is work done on the system. It explains that (i) qv = delta E: at constant volume no work is done, so all heat goes into internal energy; (ii) qp = delta H: at constant pressure the heat equals the enthalpy change, because part of the energy appears as expansion work.

7(b) How will you differentiate between delta E and delta H? Is it true that delta H and delta E have the same values for the reactions taking place in the solution state?

Delta E and delta H differ by the expansion work: delta H = delta E + delta n RT, where delta n is the change in moles of gas. In the solution state there is almost no volume change (delta n RT is negligible, since liquids and solids hardly expand), so delta H and delta E have practically the same values for reactions in solution.

8(a) What is the difference between heat and temperature? Write a mathematical relationship between these two parameters.

Heat is energy in transit due to a temperature difference, measured in joules; temperature is the measure of the average kinetic energy of the particles, measured in degrees. They are related by q = m x c x delta T, where m is the mass,

c the specific heat capacity, and delta T the temperature change: the same heat produces a larger temperature rise in a smaller mass or a substance of lower specific heat.

8(b) How do you measure the heat of combustion of a substance by bomb calorimeter?

A weighed sample is placed in a steel bomb, filled with oxygen at high pressure, and ignited electrically while the bomb sits in a known mass of water in an insulated calorimeter. The temperature rise of the water (and calorimeter) is measured. Heat evolved = (mass of water x specific heat of water + heat capacity of calorimeter) x temperature rise. Dividing by the moles of sample burnt gives the heat of combustion (at constant volume, delta E; corrected to delta H if needed).

9 Define heat of neutralization. When a dilute solution of a strong acid is neutralized by a dilute solution of a strong base, the heat of neutralization is found to be nearly the same in all cases. How do you account for this?

Heat of neutralization is the enthalpy change when one mole of H+ ions from an acid reacts with one mole of OH- ions from a base to form one mole of water. For a strong acid and a strong base in dilute solution, the only net reaction is H+ + OH- -> H2O, whatever the acid and base are, because both are fully ionized and the salt formed stays in solution. Hence the heat evolved is nearly the same in all cases (about -57.3 kJ mol^-1).

10(a) State the laws of thermochemistry and show how they are based on the first law of thermodynamics.

Laws of thermochemistry: (i) Lavoisier-Laplace law: the heat change of a reaction in one direction is equal in magnitude and opposite in sign to that in the reverse direction. (ii) Hess's law of constant heat summation: the total heat change of a reaction is the same whether it takes place in one step or in several steps. Both are based on the first law of thermodynamics: since energy is conserved and enthalpy

is a state function, the total energy change depends only on the initial and final states, not on the path.

10(b) What is a thermochemical equation? Give three examples. What information do they convey?

A thermochemical equation is a balanced chemical equation that also states the enthalpy change and the physical states of all reactants and products, e.g. H2(g) + 1/2 O2(g) -> H2O(l), delta H = -285.8 kJ mol^-1; C(s) + O2(g) -> CO2(g), delta H = -393.7 kJ mol^-1; N2(g) + 3H2(g) -> 2NH3(g), delta H = -92.4 kJ mol^-1. It conveys the reaction, the amounts reacting, the physical states, and whether heat is evolved (negative delta H) or absorbed (positive delta H).

10(c) Why is it necessary to mention the physical states of reactants and products in a thermochemical reaction? Apply Hess's law to justify your answer.

Physical states must be stated because the enthalpy of a substance depends on its state: e.g. H2O(l) and H2O(g) have very different enthalpies (the difference is the heat of vaporization), so a reaction producing liquid water releases more heat than one producing steam. Hess's law justifies this: the enthalpy change is fixed by the initial and final states, so unless the states are specified, the initial and final states are not fully specified and the enthalpy change is not defined.

11(a) Define and explain Hess's law of constant heat summation. Explain it with examples and give its application.

Hess's law of constant heat summation: the enthalpy change of a reaction is the same whether the reaction takes place in one step or in a series of steps, because enthalpy is a state function. Example: carbon can burn directly to CO2 (delta H = -393.7 kJ mol^-1), or first to CO and then to CO2; the sum of the two steps equals -393.7 kJ mol^-1. Application: it lets us calculate enthalpy changes for reactions that cannot be carried out directly (e.g. formation of CO from C and O2, heats of formation, lattice energies) from steps that can be measured.

11(b) Hess's law helps us to calculate the heats of those reactions, which cannot be normally carried out in a laboratory. Explain it.

Many reactions cannot be carried out cleanly in a laboratory, or their heat changes are too small or too slow to measure, e.g. the formation of methane from its elements, or the conversion of graphite to diamond. Hess's law lets us build the impossible reaction out of measurable steps (combustion reactions, solution reactions) whose enthalpy changes are known, and add them to get the unknown enthalpy change, since the total depends only on the start and end states.

12(a) What is lattice energy? How does Born-Haber cycle help to calculate the lattice energy of NaCl?

Lattice energy is the enthalpy change when one mole of an ionic crystal is formed from its gaseous ions, e.g. Na+(g) + Cl-(g) -> NaCl(s), delta H = -787 kJ mol^-1 (energy is released). The Born-Haber cycle applies Hess's law to a cycle: elements -> gaseous atoms (atomization) -> gaseous ions (ionization energy, electron affinity) -> crystal (lattice energy), with the direct formation enthalpy measured by calorimetry. For NaCl: formation enthalpy -411 kJ mol^-1 = atomization of Na (+108) + ionization of Na (+496) + atomization of Cl (+121) + electron affinity of Cl (-349) + lattice energy, giving lattice energy = -411 - 376 = -787 kJ mol^-1.

12(b) Justify that heat of formation of a compound is the sum of all the other enthalpies.

By Hess's law, the heat of formation of a compound is the sum of the enthalpy changes of all the steps that build it from its elements: atomization of the elements, ionization, electron affinity, and lattice formation. Since enthalpy is a state function, the sum of the step enthalpies must equal the directly measured formation enthalpy, whatever route is taken.

13 50 cm^3 of 1.0 M HCl is mixed with 50 cm^3 of 1.00 M NaOH in a glass calorimeter. The temperature of the resultant mixture increases from 21.0 degrees

C to 27.5 degrees C. Assume that calorimeter losses of heat are negligible.

Calculate the enthalpy change per mole for the reaction. The density of solution is 1 g cm^-3 and specific heat is 4.18 J g^-1 K^-1.

Step 1 - Total volume = 50 + 50 = 100 cm^3, so mass of solution = 100 cm^3 x 1 g cm^-3 = 100 g.

Step 2 - Temperature rise = 27.5 - 21.0 = 6.5 degrees C (= 6.5 K).

Step 3 - Heat evolved = m x c x delta T = 100 g x 4.18 J g^-1 K^-1 x 6.5 K = 2717 J = 2.717 kJ.

Step 4 - Moles of acid (and base) reacting = 50/1000 dm^3 x 1.0 mol dm^-3 = 0.05 mol.

Step 5 - Enthalpy change per mole = -2.717 kJ / 0.05 mol = -54.3 kJ mol^-1 (heat is evolved, so the sign is negative).

Answer: about -54 kJ mol^-1 (matches the book's printed answer).

14 Hydrazine (N2H4) is a rocket fuel. It burns in O2 to give N2 and H2O: N2H4(l) + O2(g) -> N2(g) + 2H2O(g). 1.00 g of N2H4 is burned in a bomb calorimeter. An increase of temperature 3.51 degrees C is recorded. The heat capacity of the calorimeter is 5.5 kJ K^-1. Calculate the quantity of heat evolved. Also, calculate the heat of combustion of 1 mole of N2H4.

Step 1 - Heat evolved = heat capacity x temperature rise = 5.5 kJ K^-1 x 3.51 K = 19.3 kJ.

Step 2 - Molar mass of N2H4 = 2 x 14 + 4 x 1 = 32 g mol^-1, so 1.00 g = 1.00/32 = 0.03125 mol.

Step 3 - Heat of combustion per mole = -19.3 kJ / 0.03125 mol = -618 kJ mol^-1 (negative because heat is evolved).

Answer: heat evolved 19.3 kJ; heat of combustion -618 kJ mol^-1 (matches the book's printed answers).

15 Octane (C8H18) is a motor fuel. 1.80 g of a sample of octane is burned in a bomb calorimeter having heat capacity 11.66 kJ K^-1. The temperature of the calorimeter increases from 21.36 degrees C to 28.78 degrees C. Calculate the heat of combustion for 1.8 g of octane. Also, calculate the heat for 1 mole of octane.

Step 1 - Temperature rise = 28.78 - 21.36 = 7.42 degrees C.

Step 2 - Heat evolved for 1.80 g = 11.66 kJ K^-1 x 7.42 K = 86.51 kJ.

Step 3 - Molar mass of octane C8H18 = 8 x 12 + 18 x 1 = 114 g mol^-1, so 1.80 g = 1.80/114 = 0.01579 mol.

Step 4 - Heat of combustion per mole = -86.51 kJ / 0.01579 mol = -5478.8 kJ mol^-1.

Answer: 86.51 kJ for 1.8 g; -5478.84 kJ mol^-1 for one mole (matches the book's printed answers).

16 By applying Hess's law calculate the enthalpy change for the formation of an

aqueous solution of NH4Cl from NH3 gas and HCl gas. The results for the various reactions are: (i) NH3(g) + aq -> NH3(aq), delta H = -35.16 kJ mol^-1; (ii) HCl(g) + aq -> HCl(aq), delta H = -72.41 kJ mol^-1; (iii) NH3(aq) + HCl(aq) -> NH4Cl(aq), delta H = -51.48 kJ mol^-1.

Step 1 - Add the three steps: NH3(g) + aq -> NH3(aq); HCl(g) + aq -> HCl(aq); NH3(aq) + HCl(aq) -> NH4Cl(aq).

Step 2 - The aqueous NH3 and HCl cancel, leaving the overall reaction NH3(g) + HCl(g) + aq -> NH4Cl(aq).

Step 3 - By Hess's law, delta H = (-35.16) + (-72.41) + (-51.48) = -159.05 kJ mol^-1.

Answer: about -159.08 kJ mol^-1 as printed in the book (the 0.03 difference is rounding).

17 Calculate the heat of formation of ethyl alcohol from the following information: (i) Heat of combustion of ethyl alcohol is -1367 kJ mol^-1; (ii) Heat of formation of carbon dioxide is -393.7 kJ mol^-1; (iii) Heat of formation of water is -285.8 kJ mol^-1.

Step 1 - Combustion equation: C2H5OH(l) + 3O2(g) -> 2CO2(g) + 3H2O(l), delta H = -1367 kJ mol^-1.

Step 2 - By Hess's law: delta H(combustion) = [2 x delta Hf(CO2) + 3 x delta Hf(H2O)] - delta Hf(C2H5OH).

Step 3 - So delta Hf(C2H5OH) = [2 x (-393.7) + 3 x (-285.8)] - (-1367) = (-787.4 -

857.4) + 1367 = -277.8 kJ mol^-1.

Answer: about -278 kJ mol^-1. (Note: the book prints -278.4 kcal mol^-1; the value matches in kJ mol^-1, so the printed unit kcal is an error.)

18 If the heats of combustion of C2H4, H2 and C2H6 are -337.2, -68.3 and -372.8 kcal respectively, then calculate the heat of the following reaction: C2H4(g) + H2(g) -> C2H6(g).

Step 1 - By Hess's law: delta H(reaction) = [delta Hc(C2H4) + delta Hc(H2)] - delta Hc(C2H6).

Step 2 - Substitute: delta H = (-337.2) + (-68.3) - (-372.8) = -405.5 + 372.8 = -32.7 kcal mol^-1.

Answer: -32.7 kcal mol^-1. (FLAG: the book prints -101 kcal mol^-1, which does not follow from the printed data; -32.7 kcal mol^-1 is the value the data give, and it is also the accepted hydrogenation enthalpy of ethene.)

19 Graphite and diamond are two forms of carbon. The enthalpy of combustion of graphite at 25 degrees C is -393.51 kJ mol^-1 and that of diamond is -395.41 kJ mol^-1. What is the enthalpy change of the process Graphite -> Diamond at the same temperature?

Step 1 - Combustion of graphite: C(graphite) + O2(g) -> CO2(g), delta H = -393.51 kJ mol^-1.

Step 2 - Combustion of diamond: C(diamond) + O2(g) -> CO2(g), delta H = -395.41 kJ mol^-1.

Step 3 - Graphite -> Diamond = (graphite combustion) - (diamond combustion) = (-393.51) - (-395.41) = +1.90 kJ mol^-1.

Answer: +1.91 kJ mol^-1 as printed in the book (rounding). The process is slightly endothermic: diamond is higher in enthalpy than graphite.

20 What is the meaning of the term enthalpy of ionization? If the heat of neutralization of HCl and NaOH is -57.3 kJ mol^-1 and heat of neutralization of CH3COOH with NaOH is -55.2 kJ mol^-1, calculate the enthalpy of ionization of CH3COOH.

Step 1 - Enthalpy of ionization is the enthalpy change when one mole of a weak acid (or base) ionizes in solution, e.g. CH3COOH -> CH3COO- + H+.

Step 2 - For the weak acid, neutralization happens in two steps: ionization (delta H = x) followed by H+ + OH- -> H2O (delta H = -57.3 kJ mol^-1).

Step 3 - Total observed = x + (-57.3) = -55.2, so x = -55.2 + 57.3 = +2.1 kJ mol^-1.

Answer: 2.1 kJ mol^-1 (matches the book's printed answer). Ionization absorbs a little heat, which is why weak-acid neutralization evolves slightly less heat.

21(a) Explain what is meant by: (i) Atomization energy (ii) Lattice energy. (b) Draw a complete, fully labeled Born-Haber cycle for the formation of potassium bromide. (c) Using the information in the table below, calculate the lattice energy of potassium bromide: K(s) + 1/2 Br2(l) -> KBr(s), -392 kJ mol^-1; K(s) -> K(g), +90; K(g) -> K+(g) + e, +420; 1/2 Br2(l) -> Br(g), +112; Br(g) + e -> Br-(g), -342 (all kJ mol^-1).

(a) (i) Atomization energy: the enthalpy change when one mole of a substance is

converted into gaseous atoms, e.g. Na(s) -> Na(g), +108 kJ mol^-1. (ii) Lattice energy: the enthalpy change when one mole of an ionic crystal is formed from its gaseous ions, e.g. K+(g) + Br-(g) -> KBr(s) (negative, energy released). (b) Labelled drawing instruction: draw the Born-Haber cycle for KBr as a stepwise energy diagram, labelled at each step: K(s) -> K(g) (sublimation/atomization, +90 kJ mol^-1); K(g) -> K+(g) + e- (ionization energy, +420 kJ mol^-1); 1/2 Br2(l) -> Br(g) (atomization, +112 kJ mol^-1); Br(g) + e- -> Br-(g) (electron affinity, -342 kJ mol^-1); K+(g) + Br-(g) -> KBr(s) (lattice energy, to be found); and the direct step K(s) + 1/2 Br2(l) -> KBr(s) (heat of formation, -392 kJ mol^-1). (c) By Hess's law: -392 = 90 + 420 + 112 - 342 + (lattice energy), so lattice energy = -392 - 280 = -672 kJ mol^-1.

Chapter 8 - Chemical Equilibrium

Exercise 8 - Multiple Choice Questions

1(i) For which system does the equilibrium constant, Kc, have units of (concentration)^-1?

A. N2 + 3H2 ⇌ 2NH3

B. H2 + I2 ⇌ 2HI

✓ C. 2NO2 ⇌ N2O4

D. 2HF ⇌ H2 + F2

1(ii) Which statement about the following equilibrium is correct? 2SO2(g) + O2(g) ⇌ 2SO3(g), delta H = -188.3 kJ mol^-1

✓ A. The value of Kp falls with a rise in temperature

B. The value of Kp falls with increasing pressure

C. Adding V2O5 catalyst increases the equilibrium yield of sulphur trioxide

D. The value of Kp is equal to Kc

1(iii) The pH of 10^-3 mol dm^-3 of an aqueous solution of H2SO4 is

A. 3.0

✓ B. 2.7

C. 2.0

D. 1.5

1(iv) The solubility product of AgCl is 2.0 x 10^-10 mol^2 dm^-6. The maximum concentration of Ag+ ions in the solution is

A. 2.0 x 10^-10 mol dm^-3

✓ B. 1.41 x 10^-5 mol dm^-3

C. 1.0 x 10^-10 mol dm^-3

D. 4.0 x 10^-20 mol dm^-3

1(v) An excess of aqueous silver nitrate is added to aqueous barium chloride and precipitate is removed by filtration. What are the main ions in the filtrate?

A. Ag+ and NO3- only

✓ B. Ag+ and Ba2+ and NO3-

C. Ba2+ and NO3- only

D. Ba2+ and NO3- and Cl-

Exercise 8 - Fill in the Blanks

2(i) Law of mass action states that the rate at which a reaction proceeds, is directly proportional to the product of the active masses of the reactants.

2(ii) In an exothermic reversible reaction, a decrease in temperature will shift the equilibrium towards the forward direction.

2(iii) The equilibrium constant for the reaction 2O3 ⇌ 3O2 is 10^55 at 25 degrees C, it tells that ozone is unstable at room temperature.

2(iv) In a gas phase reaction, if the number of moles of reactants are equal to the

number of moles of the products, Kc of the reaction is equal to the Kp.

2(v) Buffer solution is prepared by mixing together a weak base and its salt with a strong acid or a weak acid and its salt with a strong base.

Exercise 8 - True or False

3(i) When a reversible reaction attains equilibrium both reactants and products are present in a reaction mixture.

True

3(ii) The Kc of the reaction A + B ⇌ C + D is given by Kc = [C][D]/[A][B], therefore it is assumed that [A] = [B] = [C] = [D].

False

3(iii) A catalyst is a compound which increases the speed of the reaction and consequently increases the yield of the product.

False

3(iv) Ionic product Kw of pure water at 25 degrees C is 10^-14 mol^2 dm^-6 and is represented by an expression Kw = [H+][OH-] = 10^-14 mol^2 dm^-6.

True

3(v) AgCl is a sparingly soluble ionic solid in water. Its solution produces excess of Ag+ and Cl- ions.

False

Exercise 8 - Short Questions

3(ii) reason - Kc = [C][D]/[A][B], therefore it is assumed that [A] = [B] = [C] = [D].

False. The Kc expression is a ratio of concentrations; it holds whatever the individual concentrations are. It never assumes [A] = [B] = [C] = [D]; in general the four concentrations are all different.

3(iii) reason - A catalyst increases the speed of the reaction and consequently

increases the yield of the product.

False. A catalyst increases the speed of both the forward and the reverse reactions equally, so equilibrium is reached faster, but the equilibrium position and the yield are unchanged.

3(v) reason - AgCl solution produces excess of Ag+ and Cl- ions.

False. AgCl is sparingly soluble, so only a very small, equal amount of Ag+ and Cl- ions goes into solution (about 1.4 x 10^-5 mol dm^-3). There is no 'excess' of ions; most of the solid stays undissolved.

4(a) Explain the terms 'reversible reaction' and 'state of equilibrium'.

A reversible reaction is one that can proceed in both directions: reactants form products and products re-form reactants, e.g. N2 + 3H2 ⇌ 2NH3. The state of equilibrium is reached when the forward and reverse reactions proceed at the same rate, so the concentrations of reactants and products stop changing, although both reactions continue.

4(b) Define and explain the Law of mass action and derive the expression for the equilibrium constant (Kc).

Law of mass action: the rate at which a reaction proceeds is directly proportional to the product of the active masses (molar concentrations) of the reactants. For aA + bB ⇌ cC + dD, forward rate = kf [A]^a [B]^b and reverse rate = kr [C]^c [D]^d. At equilibrium the two rates are equal, so kf/kr = [C]^c [D]^d / [A]^a [B]^b = Kc, the equilibrium constant.

4(c) Write Kc for: (i) Sn2+ + 2Fe3+ ⇌ Sn4+ + 2Fe2+ (ii) Ag+ + Fe2+ ⇌ Fe3+ + Ag(s) (iii) N2 + O2 ⇌ 2NO (iv) 4NH3 + 5O2 ⇌ 4NO + 6H2O (v) PCl5 ⇌ PCl3 + Cl2.

(i) Sn2+(aq) + 2Fe3+(aq) ⇌ Sn4+(aq) + 2Fe2+(aq): Kc = [Sn4+][Fe2+]^2 / ([Sn2+][Fe3+]^2). (ii) Ag+(aq) + Fe2+(aq) ⇌ Fe3+(aq) + Ag(s): Kc = [Fe3+] / ([Ag+][Fe2+]) (pure solid Ag is omitted). (iii) N2(g) + O2(g) ⇌ 2NO(g): Kc = [NO]^2 / ([N2][O2]). (iv) 4NH3(g) + 5O2(g) ⇌ 4NO(g) + 6H2O(g): Kc = [NO]^4 [H2O]^6 /

([NH3]^4 [O2]^5). (v) PCl5(g) ⇌ PCl3(g) + Cl2(g): Kc = [PCl3][Cl2] / [PCl5].

5(a) Reversible reactions attain the position of equilibrium which is dynamic in nature and not static. Explain it.

At equilibrium the forward and reverse reactions continue at equal rates, so molecules keep reacting in both directions while the overall concentrations stay constant. It is therefore dynamic (reactions still running), not static (reactions stopped). A static state would mean no reaction at all.

5(b) Why do the rates of forward reactions slow down when a reversible reaction approaches the equilibrium stage?

As the reaction approaches equilibrium, reactants are used up, so their concentrations fall and the forward rate (which depends on reactant concentrations) falls. At the same time products build up, so the reverse rate rises. When the two rates become equal, equilibrium is reached.

6 When a graph is plotted between time on x-axis and concentrations of reactants and products on y-axis for a reversible reaction, the curves become parallel to time axis at a certain stage. (a) At what stage do the curves become parallel? (b) Before the curves become parallel, the steepness of curves falls? Give reasons. (c) The rate of decrease of concentrations of reactants and rate of increase of concentrations of products may or may not be equal, for various types of reactions, before the equilibrium time. Explain it.

(a) The curves become parallel at the equilibrium stage, when the concentrations stop changing. (b) Before equilibrium, the steepness falls because the net rate of change of concentration falls: as reactants are consumed the forward rate decreases, and as products accumulate the reverse rate increases, so the net change per unit time gets smaller until it reaches zero. (c) The rates of decrease of reactants and increase of products need not be equal before equilibrium, because they are linked by the stoichiometry: e.g. in N2 + 3H2 ⇌ 2NH3, hydrogen is

consumed three times as fast as nitrogen. Only the rates divided by their stoichiometric coefficients are equal.

7(a) Write down the relationship of different types of equilibrium constants, i.e. Kc and Kp, for the general reaction aA + bB ⇌ cC + dD. (b) Decide the comparative magnitudes of Kc and Kp for: (i) Ammonia synthesis (ii) Dissociation of PCl5.

(a) For aA + bB ⇌ cC + dD, Kp = Kc (RT)^(delta n), where delta n = (c + d) - (a + b), R is the gas constant and T the absolute temperature. (b) Ammonia synthesis, N2 + 3H2 ⇌ 2NH3: delta n = 2 - 4 = -2, so Kp = Kc (RT)^-2, and Kp is smaller than Kc. Dissociation of PCl5, PCl5 ⇌ PCl3 + Cl2: delta n = 2 - 1 = +1, so Kp = Kc (RT), and Kp is larger than Kc.

8(a) Write down Kc for the following reversible reactions, supposing the volume of reaction mixture in all cases is V dm^3 at equilibrium stage: (i) CH3COOH + CH3CH2OH ⇌ CH3COOC2H5 + H2O (ii) H2 + I2 ⇌ 2HI (iii) 2HI ⇌ H2 + I2 (iv) PCl5 ⇌ PCl3 + Cl2 (v) N2 + 3H2 ⇌ 2NH3.

With volume V dm^3 and equilibrium amounts in moles, concentrations are moles/V. (i) CH3COOH + C2H5OH ⇌ CH3COOC2H5 + H2O: Kc = [ester][water]/([acid][alcohol]); the V terms cancel (delta n = 0), so Kc = (moles of ester x moles of water)/(moles of acid x moles of alcohol). (ii) H2 + I2 ⇌ 2HI: Kc = [HI]^2/([H2][I2]), V cancels. (iii) 2HI ⇌ H2 + I2: Kc = [H2][I2]/[HI]^2, V cancels. (iv) PCl5 ⇌ PCl3 + Cl2: Kc = [PCl3][Cl2]/[PCl5] = (moles PCl3 x moles Cl2)/(moles PCl5 x V). (v) N2 + 3H2 ⇌ 2NH3: Kc = [NH3]^2/([N2][H2]^3) = (moles NH3)^2 x V^2 / (moles N2 x (moles H2)^3).

8(b) How do you explain that some of the reactions mentioned above are affected by change of volume at equilibrium stage?

Reactions with delta n = 0 (cases (i), (ii), (iii)) are not affected by a change of volume, because changing the volume changes all concentrations by the same factor and the ratio cancels. Reactions with delta n not zero (cases (iv) and (v)) are

affected: increasing the volume (lowering the pressure) shifts equilibrium towards the side with more moles of gas, and decreasing the volume shifts it towards fewer moles.

9 Explain the following two applications of equilibrium constant. Give examples: (i) Direction of reaction (ii) Extent of reaction.

(i) Direction of reaction: comparing the reaction quotient Qc with Kc tells which way the reaction will go: if Qc < Kc the reaction goes forward; if Qc > Kc it goes backward; if Qc = Kc it is at equilibrium. Example: for the HI equilibrium, adding HI makes Qc < Kc, so more HI dissociates. (ii) Extent of reaction: the size of Kc tells how far the reaction goes: a very large Kc (e.g. 10^55 for ozone conversion) means the reaction goes almost to completion; a very small Kc means it hardly proceeds; a Kc near 1 means appreciable amounts of reactants and products coexist at equilibrium.

10 Explain with reasons: (a) The change of volume disturbs the equilibrium position for some gaseous phase reactions but not the equilibrium constant. (b) The change of temperature disturbs both the equilibrium position and the equilibrium constant of a reaction. (c) The solubility of glucose in water is increased by increasing the temperature.

(a) Changing the volume changes the concentrations (and partial pressures), so by Le Chatelier's principle the equilibrium position shifts to counteract the change, but only for reactions with delta n not zero. The equilibrium constant itself does not change, because Kc depends only on temperature. (b) Changing the temperature changes the value of Kc itself (for an exothermic reaction Kc decreases with rising temperature; for an endothermic reaction it increases), and therefore also shifts the equilibrium position: temperature is the only factor that changes both. (c) Dissolving glucose in water is endothermic (heat is absorbed to break the solid lattice), so raising the temperature supplies that heat and shifts the dissolution equilibrium

forward: solubility increases with temperature.

11(a) What is ionic product of water? How does this value vary with temperature? Is it true that its value increases 75 times when the temperature of water is increased from 0 degrees C to 100 degrees C? (b) What is the justification for the increase of ionic product with temperature? (c) How would you prove that at 25 degrees C, 1 dm^3 of water contains 10^-7 moles of H3O+ and 10^-7 moles of OH-? (Note on (a): the 75-times figure is answered from the standard Kw values, 0.11 x 10^-14 at 0 degrees C and 51 x 10^-14 at 100 degrees C, which give about a 460-fold rise, not 75-fold.)

(a) The ionic product of water is Kw = [H+][OH-] = 1.0 x 10^-14 mol^2 dm^-6 at 25 degrees C. It increases with temperature, because the ionization of water is endothermic, so heating shifts H2O ⇌ H+ + OH- forward. It is not true that it increases 75 times from 0 to 100 degrees C as a fixed rule: Kw rises from about 0.11 x 10^-14 at 0 degrees C to about 51 x 10^-14 at 100 degrees C, roughly a 460-fold rise; the '75 times' figure in the question is not the book's value for this range. (b) Kw increases with temperature because water ionization absorbs heat (endothermic); by Le Chatelier's principle, heating favours the forward (ionization) direction, producing more H+ and OH-. (c) In pure water [H+] = [OH-], so Kw = [H+]^2 = 10^-14, giving [H+] = 10^-7 mol dm^-3. In 1 dm^3 of water there is therefore 10^-7 mol of H3O+ and 10^-7 mol of OH-.

12(a) Define pH and pOH. How are they related with pKw? (b) What happens to the acidic and basic properties of aqueous solutions when pH varies from zero to 14? (c) Is it true that the sum of pKa and pKb is always equal to 14 at all temperatures for any acid? If not, why? (c) No: pKa + pKb = pKw only for a conjugate acid-base pair, and pKw itself changes with temperature (14.00 only at 25 degrees C), so the sum equals 14 only at 25 degrees C; at other temperatures it equals the pKw at that temperature.

pH = -log [H+], the negative logarithm of the hydrogen ion concentration; pOH = -log [OH-]. Since Kw = [H+][OH-] = 10^-14 at 25 degrees C, taking negative logarithms gives pH + pOH = pKw = 14 at 25 degrees C. As pH goes from 0 to 14, the solution goes from strongly acidic (high [H+], e.g. pH 0) through neutral (pH 7, [H+] = [OH-]) to strongly basic (low [H+], high [OH-], e.g. pH 14): acidity decreases and basicity increases continuously with pH.

13(a) What is Lowry-Bronsted idea of acids and bases? Explain conjugate acid and bases.

An acid is a proton donor and a base is a proton acceptor (Lowry-Bronsted). A conjugate acid is formed when a base accepts a proton (e.g. NH3 + H+ -> NH4+), and a conjugate base is formed when an acid loses a proton (e.g. CH3COOH -> CH3COO- + H+). Every acid has a conjugate base and every base has a conjugate acid; a strong acid has a weak conjugate base and vice versa.

13(b) Acetic acid dissolves in water and gives proton to water, but when dissolved in H2SO4, it accepts protons. Discuss the role of acetic acid in both cases.

Acetic acid is amphiprotic: in water, which is a stronger base than the acetate ion, acetic acid donates a proton (CH3COOH + H2O ⇌ CH3COO- + H3O+) and acts as an acid. In concentrated sulphuric acid, which is a much stronger acid, acetic acid accepts a proton (CH3COOH + H2SO4 ⇌ CH3COOH2+ + HSO4-) and acts as a base. Whether it donates or accepts depends on the relative strength of the other acid-base system present.

14 In the equilibrium PCl5(g) ⇌ PCl3(g) + Cl2(g), delta H = 90 kJ mol^-1, what is the effect on (a) the position of equilibrium (b) equilibrium constant, if (i) temperature is increased (ii) volume of the container is decreased (iii) catalyst is added (iv) chlorine is added? Explain your answer.

PCl5(g) ⇌ PCl3(g) + Cl2(g), delta H = +90 kJ mol^-1 (endothermic dissociation). (i) Temperature increased: equilibrium shifts forward (endothermic direction), so the

position moves right and Kc increases. (ii) Volume decreased (pressure increased): equilibrium shifts towards fewer moles of gas, i.e. left, towards PCl5; Kc is unchanged (temperature unchanged). (iii) Catalyst added: no effect on position or on Kc; equilibrium is reached faster. (iv) Chlorine added: equilibrium shifts left, towards PCl5, to consume the added Cl2; Kc is unchanged. Summary: position: (i) right, (ii) left, (iii) no change, (iv) left; Kc: (i) increases, (ii)-(iv) unchanged.

15 Synthesis of ammonia by Haber's process is an exothermic reaction: N2(g) + 3H2(g) ⇌ 2NH3(g), delta H = -92.46 kJ. (a) What should be the possible effect of change of temperature at equilibrium stage? (b) How does the change of pressure or volume shift the equilibrium position of this reaction? (c) What is the role of the catalyst in this reaction? (d) What happens to equilibrium position if NH3 is removed from the reaction vessel from time to time?

(a) Temperature: the reaction is exothermic, so raising the temperature shifts equilibrium left (towards N2 and H2) and lowers the yield of ammonia; lowering the temperature raises the yield but slows the reaction. (b) Pressure/volume: the reaction goes from 4 moles of gas to 2, so increasing the pressure (decreasing the volume) shifts equilibrium right and increases the ammonia yield; decreasing the pressure shifts it left. (c) Catalyst: iron catalyst speeds up both directions equally, so it does not change the yield, but it lets equilibrium be reached in a practical time. (d) Removing NH3: the equilibrium shifts right to replace the ammonia removed, so more ammonia is formed; removing product continuously drives the reaction forward.

16 Sulphuric acid is produced by the burning of SO2 to SO3 through an exothermic reversible process. (a) Write the balanced reversible reaction. (b) What is the effect of pressure change on this reaction? (c) Reaction is exothermic but still the temperature of 400-500 degrees C is required to increase the yield of SO3. Give reasons.

(a) Balanced reversible reaction: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$. (b) Pressure: the reaction goes from 3 moles of gas to 2, so increasing the pressure shifts equilibrium right, increasing the SO_3 yield; decreasing the pressure shifts it left. (c) Although the reaction is exothermic and low temperature would favour SO_3, at low temperature the reaction is too slow to be useful. A temperature of 400-500 degrees C is a compromise: high enough for a practical rate (with the V_2O_5 catalyst) while still giving a good yield. Lower temperature would give higher equilibrium yield but an uneconomically slow reaction.

17(a) What are buffer solutions? Why do we need them in daily life? (b) How does the mixture of sodium acetate and acetic acid give us the acidic buffer? (c) Explain that a mixture of NH_4OH and NH_4Cl gives us the basic buffer. (d) How do you justify that the greater quantity of CH_3COONa in acetic acid decreases the dissociating power of acetic acid and so the pH increases? (e) Explain the term buffer capacity.

(a) Buffer solutions resist changes in pH when small amounts of acid or base are added. We need them in daily life because many processes need a steady pH: blood stays near pH 7.4; medicines, foods, and laboratory reactions are kept at a fixed pH. (b) Acidic buffer: acetic acid (weak acid) with sodium acetate (its salt) gives an acidic buffer: added H^+ is removed by acetate ions ($CH_3COO^- + H^+ \rightarrow CH_3COOH$) and added OH^- is removed by the acid ($CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$), so the pH stays near the pKa of acetic acid (4.74). (c) Basic buffer: ammonium hydroxide (weak base) with ammonium chloride (its salt) gives a basic buffer: added H^+ is removed by NH_4OH ($NH_4OH + H^+ \rightarrow NH_4^+ + H_2O$) and added OH^- is removed by NH_4^+ ($NH_4^+ + OH^- \rightarrow NH_4OH + H_2O$), so the pH stays near 9.25. (d) Adding more sodium acetate (common ion CH_3COO^-) suppresses the dissociation of acetic acid by the common ion effect, so less H^+ is produced and the pH rises. (e) Buffer capacity is the amount of acid or base a buffer can absorb without a large change in pH; it is greatest when the buffer components are concentrated and

present in roughly equal amounts.

18(a) What is the solubility product? Derive the solubility product expression for sparingly soluble compounds AgCl, Ag2CrO4 and PbCl2. (b) How do you determine the solubility product of a substance when its solubility is provided in grams/100 g of water? (c) How do you calculate the solubility of a substance from the value of solubility product?

(a) Solubility product, Ksp, is the product of the ion concentrations in a saturated solution of a sparingly soluble salt, each raised to its stoichiometric power. AgCl(s) ⇌ Ag+(aq) + Cl-(aq): Ksp = [Ag+][Cl-]. Ag2CrO4(s) ⇌ 2Ag+(aq) + CrO4^2-(aq): Ksp = [Ag+]^2 [CrO4^2-]. PbCl2(s) ⇌ Pb2+(aq) + 2Cl-(aq): Ksp = [Pb2+][Cl-]^2. (b) If solubility is given in grams per 100 g of water, convert to moles per dm^3: moles = (grams per 100 g water x 10) / molar mass, giving grams per dm^3 (taking 100 g water as about 100 cm^3, i.e. 1 dm^3 holds 10 x 100 g), then divide by the molar mass to get mol dm^-3. (c) From Ksp, find the molar solubility S from the Ksp expression: for a 1:1 salt Ksp = S^2, so S = sqrt(Ksp); for Ag2CrO4, Ksp = 4S^3, so S = (Ksp/4)^(1/3); for PbCl2, Ksp = 4S^3 likewise.

Exercise 8 - Numerical Questions

19 Kc value for the following reaction is 0.016 at 520 degrees C: 2HI(g) ⇌ H2(g) + I2(g). Equilibrium mixture contains [HI] = 0.08 M, [H2] = 0.01 M, [I2] = 0.01 M. To this mixture more HI is added so that its new concentration is 0.096 M. What will be the concentration of [HI], [H2] and [I2] when equilibrium is re-established?

Step 1 - Check the first equilibrium: Kc = [H2][I2]/[HI]^2 = (0.01 x 0.01)/(0.08)^2 = 0.0156, about 0.016, as given.

Step 2 - After adding HI, [HI] = 0.096 M, [H2] = [I2] = 0.01 M. Let x mol dm^-3 of HI dissociate to re-establish equilibrium: 2HI -> H2 + I2, so [HI] = 0.096 - 2x, [H2] =

[I2] = 0.01 + x.

Step 3 - Kc = (0.01 + x)^2 / (0.096 - 2x)^2 = 0.016. Taking square roots: (0.01 + x)/(0.096 - 2x) = 0.1265.

Step 4 - Solve: 0.01 + x = 0.01214 - 0.253x, so 1.253x = 0.00214, x = 0.00171 mol dm^-3.

Step 5 - New concentrations: [HI] = 0.096 - 2(0.00171) = 0.0926 mol dm^-3; [H2] = [I2] = 0.01 + 0.00171 = 0.01171 mol dm^-3.

Answer: [HI] = 0.0926 mol dm^-3, [H2] = 0.01168 mol dm^-3, [I2] = 0.01168 mol dm^-3 (matches the book's printed answers).

20 The equilibrium constant for the reaction between acetic acid and ethyl alcohol is 4.0. A mixture of 3 moles of acetic acid and one mole of C2H5OH is allowed to come to equilibrium. Calculate the amount of ethyl acetate at equilibrium stage in number of moles and grams. Also calculate the masses of reactants left behind.

Step 1 - Reaction: CH3COOH + C2H5OH ⇌ CH3COOC2H5 + H2O. Let x moles react: at equilibrium, acid = 3 - x, alcohol = 1 - x, ester = x, water = x moles.

Step 2 - Kc = x^2 / ((3 - x)(1 - x)) = 4.0, so x^2 = 4(3 - 4x + x^2) = 12 - 16x + 4x^2.

Step 3 - Rearranged: 3x^2 - 16x + 12 = 0. Solving: x = (16 - sqrt(256 - 144))/6 = (16 - 10.58)/6 = 0.903 mol (the other root, 4.43, is impossible since only 1 mol of alcohol is present).

Step 4 - Ethyl acetate: 0.903 mol x 88 g mol^-1 = 79.5 g.

Step 5 - Reactants left: acetic acid = 3 - 0.903 = 2.097 mol x 60 g mol^-1 = 126 g; ethyl alcohol = 1 - 0.903 = 0.097 mol x 46 g mol^-1 = 4.5 g.

Answer: ethyl acetate 0.90 mol, 79.5 g; acetic acid left 126 g; ethyl alcohol left 4.6 g (matches the book's printed answers).

21 Study the equilibrium H2O(g) + CO(g) ⇌ H2(g) + CO2(g). (a) Write an expression of Kp. (b) When 1.00 mole of steam and 1.00 mole of carbon monoxide are allowed to reach equilibrium, 33.3% of the equilibrium mixture is hydrogen. Calculate the value of Kp. State the units of Kp.

Step 1 - Kp = (pH2 x pCO2) / (pH2O x pCO).

Step 2 - Let x moles react: at equilibrium, H2O = 1 - x, CO = 1 - x, H2 = x, CO2 = x. Total moles = 2 at all stages (2 moles -> 2 moles).

Step 3 - Hydrogen is 33.3% of the mixture: x/2 = 0.333, so x = 0.667 mol. Equilibrium moles: H2O = CO = 0.333, H2 = CO2 = 0.667.

Step 4 - Mole fractions are proportional to partial pressures, so Kp = (0.667 x 0.667)/(0.333 x 0.333) = 4.0.

Answer: Kp = 4, and Kp has no units, because the number of moles of gas is the same on both sides (delta n = 0), so all pressure units cancel (matches the book's printed answer).

22 Calculate the pH of: (a) 10^-4 mol dm^-3 of HCl (b) 10^-4 mol dm^-3 of Ba(OH)2 (c) 1.0 mol dm^-3 of H2X, which is only 50% dissociated (d) 1.0 mol dm^-3 of NH4OH which is 1% dissociated.

Step 1 - (a) HCl is a strong acid: [H+] = 10^-4 mol dm^-3, so pH = -log(10^-4) = 4.

Step 2 - (b) Ba(OH)2 gives two OH- per formula unit: [OH-] = 2 x 10^-4 mol dm^-3. pOH = -log(2 x 10^-4) = 3.70, so pH = 14 - 3.70 = 10.3.

Step 3 - (c) H2X 50% dissociated (first dissociation counted): [H+] = 0.50 x 1.0 = 0.50 mol dm^-3, so pH = -log(0.50) = 0.30, about zero as printed.

Step 4 - (d) NH4OH 1% dissociated: [OH-] = 0.01 x 1.0 = 0.01 mol dm^-3, so pOH = 2 and pH = 14 - 2 = 12.

Answer: (a) 4 (b) 10.3 (c) zero (d) 12 (matches the book's printed answers).

23(a) Benzoic acid, C6H5COOH, is a weak mono-basic acid (Ka = 6.4 x 10^-5 mol dm^-3). What is the pH of a solution containing 7.2 g of sodium benzoate in one dm^3 of 0.02 mol dm^-3 benzoic acid?

Step 1 - Molar mass of sodium benzoate C6H5COONa = 144 g mol^-1, so 7.2 g = 7.2/144 = 0.05 mol in 1 dm^3, i.e. [salt] = 0.05 mol dm^-3. [acid] = 0.02 mol dm^-3.

Step 2 - pKa = -log(6.4 x 10^-5) = 4.19.

Step 3 - Henderson equation: pH = pKa + log([salt]/[acid]) = 4.19 + log(0.05/0.02) = 4.19 + log(2.5) = 4.19 + 0.40 = 4.59.

Answer: pH = 4.59 (matches the book's printed answer).

23(b) A buffer solution has been prepared by mixing 0.2 M CH3COONa and 0.5 M CH3COOH in 1 dm^3 of solution. Calculate the pH of solution. pKa of acid = 4.74 at 25 degrees C. How will the values of pH change by adding 0.1 mole of NaOH and 0.1 mole of HCl separately?

Step 1 - Initial pH = pKa + log([salt]/[acid]) = 4.74 + log(0.2/0.5) = 4.74 - 0.40 = 4.34.

Step 2 - Adding 0.1 mol NaOH: it converts 0.1 mol of acid into salt, so [salt] = 0.3 M, [acid] = 0.4 M. pH = 4.74 + log(0.3/0.4) = 4.74 - 0.12 = 4.62.

Step 3 - Adding 0.1 mol HCl (separately, to a fresh buffer): it converts 0.1 mol of salt into acid, so [salt] = 0.1 M, [acid] = 0.6 M. pH = 4.74 + log(0.1/0.6) = 4.74 - 0.78 = 3.96.

Answer: initial pH 4.34; after NaOH 4.62; after HCl 3.96 (matches the book's printed answers).

24 The solubility of CaF2 in water at 25 degrees C is found to be 2.05 x 10^-4 mol dm^-3. What is the value of Ksp at this temperature?

Step 1 - CaF2(s) ⇌ Ca2+(aq) + 2F-(aq). If solubility is S, then [Ca2+] = S and [F-] = 2S.

Step 2 - Ksp = [Ca2+][F-]^2 = S x (2S)^2 = 4S^3.

Step 3 - Ksp = 4 x (2.05 x 10^-4)^3 = 4 x 8.615 x 10^-12 = 3.446 x 10^-11.

Answer: 3.446 x 10^-11 (matches the book's printed answer).

25 The solubility product of Ag2CrO4 is 2.6 x 10^-2 at 25 degrees C. Calculate the solubility of the compound.

Step 1 - Ag2CrO4(s) ⇌ 2Ag+(aq) + CrO4^2-(aq). If solubility is S, then [Ag+] = 2S and [CrO4^2-] = S.

Step 2 - Ksp = [Ag+]^2 [CrO4^2-] = (2S)^2 x S = 4S^3, so S = (Ksp/4)^(1/3).

Step 3 - S = (2.6 x 10^-2 / 4)^(1/3) = (6.5 x 10^-3)^(1/3) = 0.1866 mol dm^-3.

Answer: 0.1866 mol dm^-3 (matches the book's printed answer, which follows the printed Ksp value; note that the printed Ksp of 10^-2 is far larger than the usual literature value of about 10^-12 for Ag2CrO4, so the printed figure itself appears to be a misprint, but the calculation follows the book's own data).

Chapter 9 - Solutions

Exercise 9 - Multiple Choice Questions

1(i) Molarity of pure water is

A. 1

B. 18

✓ C. 55.5

D. 6

1(ii) 18 g glucose is dissolved in 90 g of water. The relative lowering of vapour

pressure is equal to

A. 1/5

B. 5.1

✓ C. 1/51

D. 6

1(iii) A solution of glucose is 10% w/v. The volume in which 1 g mole of it is dissolved will be

A. 1 dm^3

✓ B. 1.8 dm^3

C. 200 cm^3

D. 900 cm^3

1(iv) An aqueous solution of ethanol in water may have vapour pressure

A. equal to that of water

B. equal to that of ethanol

✓ C. more than that of water

D. less than that of water

1(v) An azeotropic mixture of two liquids boils at a lower temperature than either of them when

A. it is saturated

✓ B. it shows positive deviation from Raoult's law

C. it shows negative deviation from Raoult's law

D. it is metastable

1(vi) In azeotropic mixture showing positive deviation from Raoult's law, the volume of the mixture is

✓ A. slightly more than the total volume of the components

B. slightly less than the total volume of the components

C. equal to the total volume of the components

D. none of these

1(vii) Which of the following solutions has the highest boiling point?

✓ A. 5.85% solution of sodium chloride

B. 18.0% solution of glucose

C. 6.0% solution of urea

D. All have the same boiling point

1(viii) Two solutions of NaCl and KCl are prepared separately by dissolving same amount of the solute in water. Which of the following statements is true for these solutions?

A. KCl solution will have higher boiling point than NaCl solution

✓ B. Both the solutions have different boiling points

C. KCl and NaCl solutions possess same vapour pressure

D. KCl solution possesses lower freezing point than NaCl solution

1(ix) The molal boiling point constant is the ratio of the elevation in boiling point to

A. molarity

✓ B. molality

C. mole fraction of solvent

D. mole fraction of solute

1(x) Colligative properties are the properties of

✓ A. dilute solutions which behave as nearly ideal solutions

B. concentrated solutions which behave as nearly non-ideal solutions

C. both (i) and (ii)

D. neither (i) nor (ii)

Exercise 9 - Fill in the Blanks

2(i) Number of molecules of sugar in 1 dm^3 of 1 M sugar solution is 6.02 x 10^23 .

2(ii) 100 g of a 10% aqueous solution of NaOH contains 10 g of NaOH in 90 g of water.

2(iii) When an azeotropic mixture is distilled, its composition remains constant.

2(iv) The molal freezing point constant is also known as cryoscopic constant.

2(v) The boiling point of an azeotropic solution of two liquids is lower than either of them because the solution shows positive deviation from Raoult's law.

2(vi) Among equimolal aqueous solutions of NaCl, BaCl2 and FeCl3, the maximum depression in freezing point is shown by FeCl3 solution.

2(vii) A solution of ethanol in water shows positive deviations and gives azeotropic solution with lower boiling point than other components.

2(viii) Colligative properties are used to calculate molar mass of a compound.

2(ix) The hydration energy of Br- ion is less than that of F- ion.

2(x) The aqueous solution of NH4Cl is acidic while that of Na2SO4 is neutral.

Exercise 9 - True or False

3(i) At a definite temperature the amount of a solute in a given saturated solution is fixed.

True

3(ii) Polar solvents readily dissolve non-polar covalent compounds.

False

3(iii) The solubility of a substance decreases with increase in temperature, if the heat of a solution is negative.

True

3(iv) The rate of evaporation of a liquid is inversely proportional to the intermolecular forces of attraction.

True

3(v) The molecular mass of an electrolyte determined by lowering of vapour

pressure is less than the theoretical molecular mass.

True

3(vi) Boiling point elevation is directly proportional to the molality of the solution and inversely proportional to boiling point of solvent.

False

3(vii) All solutions containing 1 g of non-volatile non-electrolyte solutes in some solvent will have the same freezing point.

False

3(viii) The freezing point of a 0.05 molal solution of a non-volatile non-electrolyte in water is -0.93 degrees C.

False

3(ix) Hydration and hydrolysis are different processes for Na2SO4.

True

3(x) The hydration energy of an ion only depends upon its charge.

False

Exercise 9 - Short Questions

3(ii) reason - Polar solvents readily dissolve non-polar covalent compounds.

False. Like dissolves like: polar solvents dissolve polar and ionic compounds, while non-polar covalent compounds dissolve in non-polar solvents. A polar solvent does not readily dissolve a non-polar compound.

3(vi) reason - Boiling point elevation is directly proportional to molality and inversely proportional to boiling point of solvent.

False. Delta Tb = Kb x m: elevation is directly proportional to molality, but Kb is a property of the solvent (Kb = R Tb^2 / (1000 x lvap)), not simply inversely proportional to the boiling point of the solvent.

3(vii) reason - All solutions containing 1 g of non-volatile non-electrolyte solutes will

have the same freezing point.

False. The freezing point depression depends on the number of moles of particles (molality), which for 1 g of solute depends on its molar mass: 1 g of urea (60 g mol^-1) and 1 g of glucose (180 g mol^-1) give different molalities and hence different freezing points.

3(viii) reason - Freezing point of a 0.05 molal solution is -0.93 degrees C.

False. Delta Tf = Kf x m = 1.86 x 0.05 = 0.093 degrees C, so the freezing point is -0.093 degrees C, not -0.93 degrees C: the printed value is ten times too large.

3(x) reason - Hydration energy of an ion only depends upon its charge.

False. Hydration energy depends on the charge AND the size of the ion (charge density): a smaller ion of the same charge has a larger hydration energy, e.g. F- is more strongly hydrated than Br-.

4 Define and explain the following with one example in each case: (a) A homogeneous phase (b) A concentrated solution (c) A solution of solid in a solid (d) A consolute temperature (e) A non-ideal solution (f) Zeotropic solutions (g) Heat of hydration (h) Water of crystallization (i) Azeotropic solution (j) Conjugate solution. (Q4 spans book pp. 281-282.)

(a) Homogeneous phase: a part of a system uniform in composition and properties throughout, e.g. a salt solution (one liquid phase). (b) Concentrated solution: a solution containing a relatively large amount of solute, e.g. 40% NaOH solution. (c) A solution of solid in a solid: a solid solution, e.g. brass (zinc dissolved in copper). (d) Consolute temperature: the temperature above (or below) which two partially miscible liquids become completely miscible, e.g. the phenol-water system becomes fully miscible above about 66 degrees C. (e) A non-ideal solution: a solution that does not obey Raoult's law, e.g. ethanol-water (positive deviation). (f) Zeotropic solutions: liquid mixtures whose composition changes on distillation, i.e. non-azeotropic mixtures, e.g. an ethanol-water mixture away from its azeotropic

composition. (g) Heat of hydration: the heat released when one mole of gaseous ions is surrounded by water molecules, e.g. Na+(g) + aq -> Na+(aq) is strongly exothermic. (h) Water of crystallization: water molecules built into a crystal lattice in a definite proportion, e.g. CuSO4.5H2O; Na2CO3.10H2O. (i) Azeotropic solution: a liquid mixture that distils without change of composition at a constant temperature, e.g. 95.6% ethanol-water, boiling at 78.2 degrees C. (j) Conjugate solution: one of a pair of partially miscible liquid layers in equilibrium with each other, e.g. the two layers formed by phenol and water below the consolute temperature.

5(a) What are the concentration units of solutions? Compare molar and molal solutions.

Concentration units: percentage composition (w/w, w/v, v/v), molarity (moles per dm^3 of solution), molality (moles per kg of solvent), mole fraction, parts per million. Molar vs molal: molarity uses the volume of the solution, which changes with temperature, so molarity changes with temperature; molality uses the mass of the solvent, which does not change with temperature, so molality is temperature-independent. Molar solutions are easier to prepare by volume; molal solutions are preferred for precise work and for colligative properties.

5(b) One has one molal solution of NaCl and one molal solution of glucose. (i) Which solution has greater number of particles of solute? (ii) Which solution has greater amount of the solvent? (iii) How do we convert these concentrations into weight by weight percentage?

(i) The NaCl solution has the greater number of solute particles: 1 molal NaCl gives about 2 moles of particles per kg of water (Na+ and Cl-), while 1 molal glucose gives 1 mole of particles. (ii) Both have the same amount of solvent: 1 kg of water per molal solution. (iii) To convert to weight-by-weight percentage: mass of solute = molality x molar mass (per kg of solvent); percentage = 100 x mass of solute / (1000 + mass of solute). For NaCl: 58.5/(1058.5) x 100 = 5.53% w/w; for glucose:

180/1180 x 100 = 15.25% w/w.

6 Explain with reasons: (i) Molality is independent of temperature but molarity depends on temperature. (ii) Sum of mole fractions is always unity. (iii) 100 g of 98% H2SO4 has a volume of 54.34 cm^3 (density 1.84 g cm^-3). (iv) Relative lowering of vapour pressure is independent of temperature. (v) Colligative properties are obeyed when the solute is non-electrolyte and solutions are dilute. (vi) Total volume of solution by mixing 100 cm^3 water with 100 cm^3 alcohol may not be 200 cm^3. (vii) One molal urea solution is dilute compared with one molar urea solution, but the number of solute particles is the same. (viii) Non-ideal solutions do not obey Raoult's law.

(i) Molality depends only on masses (moles of solute per kg of solvent), and masses do not change with temperature, so molality is temperature-independent. Molarity depends on the volume of the solution, which expands or contracts with temperature, so molarity changes with temperature. (ii) Mole fractions are parts of a whole: X1 + X2 + ... = 1 always, because the sum of the moles of all components divided by the total moles is unity by definition. (iii) 100 g of 98% H2SO4 contains 98 g of acid; volume = 98/1.84 = 53.26 cm^3, and the book's figure of 54.34 cm^3 corresponds to 100 g of the acid solution (100/1.84 = 54.35 cm^3): the printed statement is about the volume of 100 g of the 98% acid solution itself. (iv) Relative lowering of vapour pressure = mole fraction of solute, which depends on composition, not on temperature (both the solution and pure solvent vapour pressures change with temperature in the same proportion for an ideal solution). (v) Colligative properties depend only on the number of particles: electrolytes dissociate into more particles and concentrated solutions deviate from ideal behaviour, so the simple laws hold only for dilute solutions of non-electrolytes. (vi) When water and alcohol are mixed, hydrogen bonding between unlike molecules packs the molecules more closely (and some volume is lost to intermolecular

association), so the total volume is less than 200 cm^3; volumes are not additive for non-ideal mixtures. (vii) One molal urea in water contains 1 mole of urea per 1000 g of water, while one molar urea contains 1 mole per dm^3 of solution (about 1000 cm^3 of solution, which contains less than 1000 g of water); both contain the same number of solute particles (1 mole) but the molal solution is more dilute in terms of solvent proportion because its solvent mass is larger. (viii) Non-ideal solutions show deviations from Raoult's law because the attractions between unlike molecules differ from those between like molecules, so the vapour pressure is higher (positive deviation) or lower (negative deviation) than Raoult's law predicts.

7 What are non-ideal solutions? Discuss their types and give three examples of each.

Non-ideal solutions do not obey Raoult's law: their vapour pressure is higher or lower than predicted. Types: (i) Positive deviation: solute-solvent attractions are weaker than solvent-solvent and solute-solute attractions, so molecules escape more easily and vapour pressure is higher; examples: ethanol-water; acetone-carbon disulphide; benzene-methanol. (ii) Negative deviation: solute-solvent attractions are stronger than like attractions, so vapour pressure is lower; examples: acetone-chloroform; nitric acid-water; phenol-aniline. Positive-deviation mixtures can form minimum-boiling azeotropes; negative-deviation mixtures form maximum-boiling azeotropes.

8(a) Explain fractional distillation. Justify the two curves when composition is plotted against boiling point of solutions. (b) The solutions showing positive and negative deviations cannot be fractionally distilled at their specific compositions. Explain it.

(a) Fractional distillation separates liquids of different boiling points by repeated evaporation and condensation in a fractionating column. On a boiling-point vs composition diagram, the liquid curve and the vapour curve are separate: at any temperature the vapour in equilibrium with a liquid is richer in the more volatile

component, so repeated condensation enriches the distillate in the volatile component. (b) Solutions showing positive or negative deviations have vapour-pressure curves that pass through a maximum or minimum at the azeotropic composition, where the liquid and vapour curves meet: at that composition the mixture distils unchanged, so it cannot be separated by fractional distillation.

9(a) What are azeotropic mixtures? Explain them with the help of graphs. (b) Explain the effect of temperature on phenol-water system.

(a) Azeotropic mixtures are liquid mixtures that boil at a constant temperature and distil without change of composition, because at the azeotropic composition the liquid and vapour have the same composition. On a boiling-point vs composition graph, the liquid and vapour curves touch at the azeotrope: a minimum-boiling azeotrope for positive deviation (e.g. 95.6% ethanol-water, 78.2 degrees C) and a maximum-boiling azeotrope for negative deviation (e.g. 68% nitric acid-water, 121 degrees C). (b) Effect of temperature on phenol-water: below the consolute temperature (about 66 degrees C) phenol and water are only partially miscible and form two conjugate layers; as the temperature rises the mutual solubility increases and the two layers approach each other in composition; above the consolute temperature they merge into a single homogeneous phase.

10(a) What are colligative properties? Why are they called so? (b) What is the physical significance of Kb and Kf values of solvents?

(a) Colligative properties are properties of a solution that depend only on the number of solute particles, not on their nature: lowering of vapour pressure, elevation of boiling point, depression of freezing point, and osmotic pressure. They are called 'colligative' (from the Latin for 'collected together') because they depend on the total number of particles collected in the solution. (b) Kb (ebullioscopic constant) is the elevation of boiling point produced by a one-molal solution; Kf (cryoscopic constant) is the depression of freezing point produced by a one-molal

solution. Physically, Kb and Kf are characteristic of the solvent only (they depend on its boiling/freezing point and latent heat), and they let us calculate molar masses from measured delta Tb or delta Tf.

11 How do you explain that the lowering of vapour pressure is a colligative property? How do we measure the molar mass of a non-volatile, non-electrolyte solute in a volatile solvent?

Lowering of vapour pressure is colligative because it depends only on the mole fraction of solute particles (Raoult's law: delta p/p0 = X_solute), not on what the solute is. To measure the molar mass of a non-volatile, non-electrolyte solute: dissolve a known mass w2 of solute in a known mass w1 of solvent, measure the vapour-pressure lowering delta p, and use M2 = (w2 x p0 x M1)/(delta p x w1), where p0 is the vapour pressure of the pure solvent and M1 its molar mass.

12 How do you justify that: (a) boiling points of solvents increase due to the presence of solutes (b) freezing points are depressed due to the presence of solutes (c) the boiling point of one molal urea solution is 100.52 degrees C but the boiling point of two molal urea solution is less than 101.04 degrees C (d) Beckmann's thermometer is used to note the depression in freezing point.

(a) A non-volatile solute lowers the vapour pressure of the solvent, so a higher temperature is needed for the solution's vapour pressure to reach atmospheric pressure: the boiling point rises. (b) The solute lowers the vapour pressure of the liquid, so the solid solvent and the solution are in equilibrium only at a lower temperature: the freezing point is depressed. (c) Delta Tb = Kb x m: for urea in water Kb = 0.52 K kg mol^-1, so a one-molal solution boils at 100 + 0.52 = 100.52 degrees C, and a two-molal solution at 100 + 1.04 = 101.04 degrees C. The printed statement that the two-molal boiling point is 'less than 101.04 degrees C' is the ideal-solution value approached from below in practice, since real solutions deviate slightly from ideality at higher concentration. (d) Beckmann's thermometer

measures small temperature differences very precisely (it reads differences of 0.01 degrees C over a 5-6 degree range), which is what a freezing-point depression measurement needs, since depressions are often only a fraction of a degree.

12(e)(f) (from the top of book p. 283) (e) In summer the antifreeze solutions protect the liquid of the radiator from boiling over. (f) NaCl and KNO3 are used to lower the melting point of ice.

(e) In summer, antifreeze solutions protect radiator liquid from boiling over because the dissolved solute (e.g. ethylene glycol) raises the boiling point of the coolant. (f) NaCl and KNO3 are used to lower the melting point of ice because they depress the freezing point of water, so ice melts even below 0 degrees C.

13 What is Raoult's law? Give its three statements. How does this law help us to understand the ideality of a solution?

Raoult's law has three statements: (i) the relative lowering of vapour pressure of a solvent by a non-volatile solute equals the mole fraction of the solute; (ii) the vapour pressure of a volatile component in a solution equals its mole fraction times its pure vapour pressure (p = X x p0); (iii) for a mixture of volatile liquids, the total vapour pressure is the sum of the partial pressures. A solution is ideal if it obeys Raoult's law at all compositions and temperatures, which happens when solute-solvent attractions equal solvent-solvent and solute-solute attractions, so there is no heat or volume change on mixing.

14 Give graphical explanation for elevation of boiling point of a solution. Describe one method to determine the boiling point elevation of a solution.

Labelled graph instruction: plot temperature on the y-axis and time (or heat supplied) on the x-axis for pure solvent and for solution: the solution's vapour-pressure curve lies below the solvent's, so it reaches atmospheric pressure at a higher temperature; the horizontal gap between the two boiling points is delta Tb. Method (Landsberger or Cottrell method): a known mass of solute is dissolved in

a known mass of solvent in a boiling-point apparatus; the boiling point of the pure solvent and of the solution are measured with a Beckmann thermometer, and the difference gives delta Tb, from which the molar mass follows by M2 = (Kb x w2 x 1000)/(delta Tb x w1).

15 Freezing points of solutions are depressed when non-volatile solutes are present in volatile solvents. Justify it. Plot a graph to elaborate your answer. Also give one method to record the depression of freezing point of a solution.

A non-volatile solute lowers the vapour pressure of the solvent, so the vapour-pressure curve of the solution lies below that of the pure solvent. The freezing point is where the liquid's vapour-pressure curve meets the solid's curve; since the solution curve is lower, it meets the solid curve at a lower temperature. Labelled graph instruction: plot vapour pressure vs temperature, showing the solid curve, the pure-solvent liquid curve, and the lower solution curve meeting the solid curve at a lower temperature (delta Tf below the pure freezing point). Method (Beckmann method): the freezing points of the pure solvent and of the solution are measured with a Beckmann thermometer in a freezing-point apparatus, and the difference gives delta Tf.

16 Discuss the energetics of solution. Justify the heats of solutions as exothermic and endothermic properties.

Energetics of solution: dissolving involves breaking solute-solute and solvent-solvent attractions (endothermic) and forming solute-solvent attractions (exothermic, hydration). If the heat released on hydration exceeds the heat needed to break the attractions, the heat of solution is negative (exothermic), e.g. NaOH, H2SO4 in water. If the breaking terms dominate, the heat of solution is positive (endothermic), e.g. NH4Cl, KNO3 in water. The sign of the heat of solution therefore decides whether solubility rises or falls with temperature.

Exercise 9 - Numerical Questions

17(a) Calculate the molarity of glucose solution when 9 g of it are dissolved in 250 cm^3 of solution.

Step 1 - Molar mass of glucose C6H12O6 = 180 g mol^-1. Moles of glucose = 9/180 = 0.05 mol.

Step 2 - Volume of solution = 250 cm^3 = 0.25 dm^3.

Step 3 - Molarity = 0.05/0.25 = 0.2 mol dm^-3.

Answer: 0.2 mol dm^-3 (matches the book's printed answer).

17(b) Calculate the mass of urea in 100 g of H2O in 0.3 molal solution.

Step 1 - Molality = moles of solute per kg of solvent. Mass of water = 100 g = 0.1 kg.

Step 2 - Moles of urea = 0.3 mol kg^-1 x 0.1 kg = 0.03 mol.

Step 3 - Molar mass of urea = 60 g mol^-1, so mass = 0.03 x 60 = 1.8 g.

Answer: 1.8 g (matches the book's printed answer).

17(c) Calculate the concentration of a solution in terms of molality kg^-1, which is obtained by mixing 250 g of 20% solution of NaCl with 200 g of 40% solution of NaCl.

Step 1 - NaCl from first solution = 20% of 250 g = 50 g; water = 200 g. NaCl from second solution = 40% of 200 g = 80 g; water = 120 g.

Step 2 - Total NaCl = 50 + 80 = 130 g. Moles of NaCl = 130/58.5 = 2.222 mol.

Step 3 - Total water = 200 + 120 = 320 g = 0.32 kg.

Step 4 - Molality = 2.222/0.32 = 6.94 mol kg^-1.

Answer: 6.94 m (matches the book's printed answer).

18(a) An aqueous solution of sucrose has been labeled as 1 molal. Find the mole fraction of the solute and the solvent.

Step 1 - 1 molal means 1 mol of sucrose per 1000 g of water.

Step 2 - Moles of water = 1000/18 = 55.56 mol.

Step 3 - Mole fraction of sucrose = 1/(1 + 55.56) = 1/56.56 = 0.0177.

Step 4 - Mole fraction of water = 55.56/56.56 = 0.9823.

Answer: solute 0.0176, solvent 0.9823 (matches the book's printed answers, rounding).

18(b) You are provided with 80% H2SO4 w/w having density 1.8 g cm^-3. How much volume of this H2SO4 sample is required to obtain one dm^3 of 20% w/w H2SO4, which has a density of 1.25 g cm^-3?

Step 1 - Mass of 1 dm^3 of 20% acid = 1000 cm^3 x 1.25 g cm^-3 = 1250 g.

Step 2 - Mass of pure H2SO4 needed = 20% of 1250 g = 250 g.

Step 3 - Mass of 80% stock needed = 250/0.80 = 312.5 g.

Step 4 - Volume of stock = 312.5/1.8 = 173.6 cm^3.

Answer: 173.5 cm^3 (matches the book's printed answer, rounding).

19 250 cm^3 of 0.2 molar K2SO4 solution is mixed with 250 cm^3 of 0.2 molar KCl solution. Calculate the molar concentration of K+ ions in the solution.

Step 1 - Moles of K+ from K2SO4 = 2 x 0.2 mol dm^-3 x 0.25 dm^3 = 0.10 mol.

Step 2 - Moles of K+ from KCl = 1 x 0.2 mol dm^-3 x 0.25 dm^3 = 0.05 mol.

Step 3 - Total K+ = 0.10 + 0.05 = 0.15 mol in a total volume of 0.50 dm^3.

Step 4 - [K+] = 0.15/0.50 = 0.3 mol dm^-3.

Answer: 0.3 molar (matches the book's printed answer).

20 5 g of NaCl are dissolved in 1000 g of water. The density of resulting solution is 0.997 g/cm^3. Calculate molality, molarity and mole fraction of this solution. Assume that the volume of the solution is equal to that of solvent.

Step 1 - Moles of NaCl = 5/58.5 = 0.08547 mol. Moles of water = 1000/18 = 55.56 mol.

Step 2 - Molality = 0.08547 mol / 1 kg = 0.0855 mol kg^-1, about 0.0854 m as printed.

Step 3 - Volume of solution = volume of solvent = 1000 cm^3 = 1 dm^3 (as instructed), so molarity = 0.08547/1 = 0.0855 mol dm^-3, about 0.08542 M as printed.

Step 4 - Mole fraction of NaCl = 0.08547/(0.08547 + 55.56) = 0.00154. Mole fraction of water = 1 - 0.00154 = 0.99846, about 0.9984.

Answer: M = 0.08542, m = 0.0854, mole fraction of NaCl = 0.00154, mole fraction of H2O = 0.9984 (matches the book's printed answers).

21 4.675 g of a compound with empirical formula C3H5O were dissolved in 212.5 g of pure benzene. The freezing point of solution was found 1.02 degrees C less than that of pure benzene. The molal freezing point constant of benzene is 5.1 degrees C. Calculate (i) the relative molar mass and (ii) the molecular formula of the compound.

Step 1 - Use delta Tf = Kf x (w2 x 1000)/(M2 x w1), so M2 = Kf x w2 x 1000/(delta Tf x w1).

Step 2 - M2 = 5.1 x 4.675 x 1000/(1.02 x 212.5) = 23842.5/216.75 = 110 g mol^-1.

Step 3 - Empirical formula mass of C3H5O = 3 x 12 + 5 x 1 + 16 = 57 g mol^-1. Ratio = 110/57 = 1.93, about 2.

Step 4 - Molecular formula = 2 x C3H5O = C6H10O2.

Answer: (i) 110 g mol^-1 (ii) C6H10O2 (matches the book's printed answers).

22 The boiling point of a solution containing 0.2 g of a substance A in 20.0 g of ether (molar mass = 74) is 0.17 K higher than that of pure ether. Calculate the molar mass of A. Molal boiling point constant of ether is 2.16 K.

Step 1 - Use delta Tb = Kb x (w2 x 1000)/(M2 x w1), so M2 = Kb x w2 x 1000/(delta Tb x w1).

Step 2 - M2 = 2.16 x 0.2 x 1000/(0.17 x 20.0) = 432/3.4 = 127 g mol^-1.

Answer: 127 g mol^-1 (matches the book's printed answer).

23 3 g of a non-volatile, non-electrolyte solute X are dissolved in 50 g of ether (molar mass = 74) at 293 K. The vapour pressure of ether falls from 442 torr to 426 torr under these conditions. Calculate the molar mass of solute X.

Step 1 - Lowering of vapour pressure = 442 - 426 = 16 torr. Relative lowering = 16/442 = 0.0362.

Step 2 - By Raoult's law in the approximate form used in the book: relative lowering = (w2/M2)/(w1/M1) = (3/M2)/(50/74).

Step 3 - So 3/M2 = 0.0362 x (50/74) = 0.02446, giving M2 = 3/0.02446 = 122.6 g mol^-1.

Answer: 122.6 g mol^-1 (matches the book's printed answer; the exact Raoult form, with the solute moles in the denominator, gives about 118 g mol^-1).

Chapter 10 - Electrochemistry

Exercise 10 - Multiple Choice Questions

1(i) The cathodic reaction in the electrolysis of dil. H_2SO_4 with Pt electrodes is

✓ A. Reduction

B. Oxidation

C. Both oxidation and reduction

D. Neither oxidation nor reduction

1(ii) Which of the following statements is not correct about galvanic cell?

A. Anode is negatively charged

✓ B. Reduction occurs at anode

C. Cathode is positively charged

D. Reduction occurs at cathode

1(iii) Stronger the oxidizing agent, greater is the

A. oxidation potential

✓ B. reduction potential

C. redox potential

D. E.M.F of cell

1(iv) If the salt bridge is not used between two half cells, then the voltage

A. Decreases rapidly

B. Decreases slowly

C. Does not change

✓ D. Drops to zero

1(v) If a strip of Cu metal is placed in a solution of $FeSO_4$

A. Cu will be deposited

B. Fe is precipitated out

C. Cu and Fe both dissolve

✓ D. No reaction takes place

Exercise 10 - Fill in the Blanks

2(i) The oxidation number of O-atom is +2 in OF_2 and is -1 in H_2O_2.

2(ii) Conductivity of metallic conductors is due to the flow of electrons while that of electrolytes is due to flow of ions.

2(iii) Reaction taking place at the anode is termed as oxidation and at the cathode is called as reduction.

2(iv) Electrode potential is set up when a metal is dipped in its own ions.

2(v) Cu metal is deposited on the Cu-cathode when electrolysis is performed for $CuSO_4$ solution with Cu-cathodes.

2(vi) The reduction potential of Zn is -0.76 volts and its oxidation potential is +0.76 volts.

2(vii) In a fuel cell, hydrogen and oxygen react together in the presence of a catalyst.

Exercise 10 - True or False

3(i) In electrolytic conduction, electrons flow through the electrolyte.

False

3(ii) In the process of electrolysis, the electrons in the external circuit flow from cathode to anode.

False

3(iii) Sugar is a non-electrolyte in solid form and when dissolved in water will allow the passage of an electric current.

False

3(iv) A metal will only allow the passage of an electric current when it is in cold state.

False

3(v) The electrolytic products of aqueous copper (II) chloride solution are copper and chlorine.

True

3(vi) Zinc can displace iron from its solution.

True

3(vii) S.H.E. acts as cathode when connected with Cu-electrode.

False

3(viii) A voltaic cell produces electrical energy at the expense of chemical energy.

True

3(ix) Lead storage battery is not a reversible cell.

False

3(x) Cr changes its oxidation number when $K_2Cr_2O_7$ is reacted with HCl.

True

Exercise 10 - Short Questions

3(i) reason - In electrolytic conduction, electrons flow through the electrolyte.

False. In an electrolyte the current is carried by ions moving towards the electrodes, not by electrons. Electrons flow only in the external metallic circuit.

3(ii) reason - In electrolysis, electrons in the external circuit flow from cathode to anode.

False. In the external circuit electrons flow from the anode (where oxidation releases electrons) to the cathode (where reduction consumes them), i.e. from anode to cathode, not from cathode to anode.

3(iii) reason - Sugar is a non-electrolyte in solid form and when dissolved will allow current.

False. Sugar dissolved in water gives neutral molecules, not ions, so the solution

does not conduct electricity. Sugar is a non-electrolyte both as a solid and in solution.

3(iv) reason - A metal will only allow current when it is in cold state.

False. Metals conduct electricity at all ordinary temperatures, hot or cold, by the flow of free electrons. Heating increases resistance but does not stop conduction.

3(vii) reason - S.H.E. acts as cathode when connected with Cu-electrode.

False. The standard hydrogen electrode (0.00 V) has a lower reduction potential than copper (+0.34 V), so when the two are connected, copper is the cathode (reduction) and the hydrogen electrode is the anode (oxidation). SHE acts as cathode only against electrodes with still lower potentials, such as zinc.

3(ix) reason - Lead storage battery is not a reversible cell.

False. The lead storage battery IS a reversible cell: it can be recharged by passing current through it in the reverse direction, restoring Pb and PbO2 from PbSO4.

4(a) Explain the term oxidation number with examples.

Oxidation number is the apparent charge an atom of an element has in a molecule or ion, as if all bonds were ionic. Examples: in H2O, H is +1 and O is -2; in NaCl, Na is +1 and Cl is -1; in H2SO4, H is +1, S is +6, O is -2.

4(b) Describe the rules used for the calculation of oxidation number of an element in molecules and ions, giving examples.

Rules: (i) elements in the free state have oxidation number zero, e.g. H2, Na, Mg. (ii) A monoatomic ion has oxidation number equal to its charge, e.g. K+ +1, Ca2+ +2, Br- -1. (iii) Hydrogen is +1 except in metal hydrides, where it is -1 (e.g. NaH). (iv) Oxygen is -2 except in peroxides (-1), superoxides (-1/2) and OF2 (+2). (v) In a neutral molecule the sum of oxidation numbers is zero. (vi) In an ion the sum equals the charge on the ion. (vii) Fluorine is always -1; alkali metals are always +1 and alkaline earth metals +2 in compounds.

4(c) Calculate the oxidation number of chromium in: (i) CrCl3 (ii) Cr2(SO4)3 (iii)

K2CrO4 (iv) K2Cr2O7 (v) CrO3 (vi) Cr2O3 (vii) Cr2O7^2-.

(i) CrCl3: 3 x (-1) = -3, so Cr = +3. (ii) Cr2(SO4)3: SO4 is -2, so 2Cr + 3(-2) = 0, Cr = +3. (iii) K2CrO4: 2(+1) + Cr + 4(-2) = 0, Cr = +6. (iv) K2Cr2O7: 2(+1) + 2Cr + 7(-2) = 0, Cr = +6. (v) CrO3: Cr + 3(-2) = 0, Cr = +6. (vi) Cr2O3: 2Cr + 3(-2) = 0, Cr = +3. (vii) Cr2O7^2-: 2Cr + 7(-2) = -2, Cr = +6. Answers: (i) +3, (ii) +3, (iii) +6, (iv) +6, (v) +6, (vi) +3, (vii) +6 (as printed).

4(d) Calculate the oxidation numbers of the underlined elements in: (i) Ca(ClO3)2 (ii) Na2CO3 (iii) Na3PO4 (iv) HNO3 (v) Cr2(SO4)3 (vi) HPO3 (vii) K2MnO4.

(i) Ca(ClO3)2: ClO3 is -1, so Cl + 3(-2) = -1, Cl = +5. (ii) Na2CO3: 2(+1) + C + 3(-2) = 0, C = +4. (iii) Na3PO4: 3(+1) + P + 4(-2) = 0, P = +5. (iv) HNO3: +1 + N + 3(-2) = 0, N = +5. (v) Cr2(SO4)3: S: 2Cr + 3S + 12(-2) = 0 with Cr +3 gives S = +6. (vi) HPO3: +1 + P + 3(-2) = 0, P = +5. (vii) K2MnO4: 2(+1) + Mn + 4(-2) = 0, Mn = +6. Answers: (i) +5, (ii) +4, (iii) +5, (iv) +5, (v) +6, (vi) +5, (vii) +6 (as printed; note the compound in (vii) is potassium manganate, K2MnO4, not permanganate).

5(a) Describe the general rules for balancing a redox equation by oxidation number method.

General rules (oxidation number method): (i) write the skeleton equation; (ii) find the oxidation numbers of all atoms and identify which are oxidized and which are reduced; (iii) balance the total increase in oxidation number against the total decrease, by multiplying the formulae by suitable numbers; (iv) balance the remaining atoms (other than O and H) by inspection; (v) balance oxygen with H2O and then hydrogen with H+ (in acidic medium), or with OH- and H2O in basic medium; (vi) check that atoms and charges balance.

5(b) Balance the following equations by oxidation number method: (i) Cu + HNO3 -> Cu(NO3)2 + NO2 + H2O (ii) Zn + HNO3 -> Zn(NO3)2 + NO + H2O (iii) Br2 + NaOH -> NaBr + NaBrO3 + H2O (iv) MnO2 + HCl -> MnCl2 + H2O + Cl2 (v) FeSO4 + K2Cr2O7 + H2SO4 -> Fe2(SO4)3 + Cr2(SO4)3 + K2SO4 + H2O (vi) HNO3 + HI ->

NO + H2O + I2 (vii) Cu + H2SO4 -> CuSO4 + SO2 + H2O (viii) HI + H2SO4 -> I2 + SO2 + H2O (ix) NaCl + H2SO4 + MnO2 -> Na2SO4 + MnSO4 + H2O + Cl2.

(i) Cu + 4HNO3 -> Cu(NO3)2 + 2NO2 + 2H2O. (ii) 3Zn + 8HNO3 -> 3Zn(NO3)2 + 2NO + 4H2O. (iii) 3Br2 + 6NaOH -> 5NaBr + NaBrO3 + 3H2O. (iv) MnO2 + 4HCl -> MnCl2 + 2H2O + Cl2. (v) 6FeSO4 + K2Cr2O7 + 7H2SO4 -> 3Fe2(SO4)3 + Cr2(SO4)3 + K2SO4 + 7H2O. (vi) 2HNO3 + 6HI -> 2NO + 4H2O + 3I2. (vii) Cu + 2H2SO4 -> CuSO4 + SO2 + 2H2O. (viii) 2HI + H2SO4 -> I2 + SO2 + 2H2O. (ix) 2NaCl + 2H2SO4 + MnO2 -> Na2SO4 + MnSO4 + 2H2O + Cl2.

6(a) Describe the general rules for balancing a redox equation by ion-electron method.

General rules (ion-electron method): (i) split the equation into oxidation and reduction half-reactions; (ii) balance all atoms except O and H; (iii) balance O with H2O; (iv) balance H with H+ (acidic) or with OH- and H2O (basic); (v) balance charge with electrons; (vi) multiply the half-reactions so the electrons cancel, then add and simplify.

6(b) Balance the following ionic equations by ion-electron method: (i) Fe3+ + Sn2+ -> Fe2+ + Sn4+ (ii) MnO4- + C2O4^2- -> Mn2+ + CO2 (iii) Cr2O7^2- + Cl- -> 2Cr3+ + 3Cl2 (iv) Cu + NO3- -> Cu2+ + 2NO2 (v) Cr2O7^2- + Fe2+ -> Cr3+ + Fe3+ (acidic) (vi) S2O3^2- + OCl- -> Cl- + S4O6^2- (acidic) (vii) IO3- + AsO3^3- -> I- + AsO4^3- (acidic) (viii) Cr3+ + BiO3- -> Cr2O7^2- + Bi3+ (acidic) (ix) H3AsO3 + Cr2O7^2- -> 3H3AsO4 + 2Cr3+ (acidic) (x) CN- + MnO4- -> CNO- + MnO2 (basic).

(i) 2Fe3+ + Sn2+ -> 2Fe2+ + Sn4+. (ii) 2MnO4- + 5C2O4^2- + 16H+ -> 2Mn2+ + 10CO2 + 8H2O. (iii) Cr2O7^2- + 6Cl- + 14H+ -> 2Cr3+ + 3Cl2 + 7H2O. (iv) Cu + 2NO3- + 4H+ -> Cu2+ + 2NO2 + 2H2O. (v) Cr2O7^2- + 6Fe2+ + 14H+ -> 2Cr3+ + 6Fe3+ + 7H2O. (vi) 2S2O3^2- + OCl- + 2H+ -> Cl- + S4O6^2- + H2O (acidic media). (vii) IO3- + 3AsO3^3- -> I- + 3AsO4^3- (acidic media). (viii) 2Cr3+ + 3BiO3- + 4H+ -> Cr2O7^2- + 3Bi3+ + 2H2O (acidic media). (ix) 3H3AsO3 +

Cr2O7^2- + 8H+ -> 3H3AsO4 + 2Cr3+ + 4H2O (acidic media). (x) 3CN- + 2MnO4- + H2O -> 3CNO- + 2MnO2 + 2OH- (basic media).

7 Describe the electrolysis of molten sodium chloride, and a concentrated solution of sodium chloride.

Molten NaCl: NaCl is melted and electrolysed with inert electrodes. At the cathode, Na+ + e- -> Na (molten sodium collects). At the anode, 2Cl- -> Cl2 + 2e- (chlorine gas). Concentrated aqueous NaCl (brine): at the cathode, water is reduced in preference to Na+ (Na+ is hard to reduce), giving H2 and OH-: 2H2O + 2e- -> H2 + 2OH-. At the anode, Cl- is oxidized to Cl2 (concentrated Cl- is discharged in preference to water). The solution becomes NaOH solution, with H2 at the cathode and Cl2 at the anode.

8 What is the difference between single electrode potential and standard electrode potential? How can it be measured? Give its importance.

Single electrode potential is the potential developed when a metal is dipped in a solution of its own ions (a single half-cell), e.g. Zn in ZnSO4. Standard electrode potential is that potential measured under standard conditions (1 mol dm^-3, 1 atm, 25 degrees C) against the standard hydrogen electrode, whose potential is taken as 0.00 V. It is measured by coupling the half-cell to the standard hydrogen electrode through a salt bridge and reading the cell voltage. Importance: standard potentials predict whether a redox reaction is spontaneous, which metal displaces which, and the voltage of a cell built from two half-cells.

9 Outline the important applications of electrolysis. Write the electrochemical reactions involved therein. Discuss the electrolysis of CuSO4 using Cu-electrodes and AgNO3 solution using Ag electrode.

Applications of electrolysis: extraction of reactive metals (Na, Al), electroplating, purification of metals (electrolytic refining), and manufacture of chemicals (NaOH, Cl2, H2). Electrolysis of CuSO4 with Cu electrodes: at the anode, Cu -> Cu2+ + 2e-

(the anode dissolves); at the cathode, Cu2+ + 2e- -> Cu (pure copper deposits). The CuSO4 concentration stays constant; impure copper anode is refined. Electrolysis of AgNO3 with Ag electrodes: at the anode, Ag -> Ag+ + e-; at the cathode, Ag+ + e- -> Ag. Silver transfers from anode to cathode, giving pure silver at the cathode.

10 Describe the construction and working of standard hydrogen electrode.

Labelled drawing instruction: draw a platinum foil coated with platinum black, dipping in 1 mol dm^-3 HCl, with hydrogen gas at 1 atm bubbled over it, connected by a salt bridge to the other half-cell. Construction: a platinized platinum electrode in contact with hydrogen gas at 1 atmosphere and H+ ions at 1 mol dm^-3 (usually 1 M HCl) at 25 degrees C. Working: the equilibrium H2 ⇌ 2H+ + 2e- is set up at the platinum surface; the electrode can act as anode (H2 -> 2H+ + 2e-) or cathode (2H+ + 2e- -> H2) depending on the other half-cell. Its potential is arbitrarily fixed at 0.00 V, and all other electrode potentials are measured against it.

11 Is the reaction Fe3+ + Ag -> Fe2+ + Ag+ spontaneous? If not, write spontaneous reaction involving these species.

As written, Fe3+ + Ag -> Fe2+ + Ag+ is NOT spontaneous: E(cell) = E(cathode) - E(anode) = 0.77 - 0.80 = -0.03 V, and a negative cell voltage means a non-spontaneous reaction. The spontaneous reaction is the reverse: Fe2+ + Ag+ -> Fe3+ + Ag, with E(cell) = 0.80 - 0.77 = +0.03 V.

12 Explain the difference between: (a) Ionization and electrolysis (b) Electrolytic cell and voltaic cell (c) Conduction through metals and molten electrolytes.

(a) Ionization is the formation of ions when a substance dissolves (e.g. NaCl -> Na+ + Cl- in water) without any external current; electrolysis is a chemical change brought about by passing an electric current through an electrolyte. (b) An electrolytic cell uses electrical energy to drive a non-spontaneous reaction (anode positive, cathode negative); a voltaic (galvanic) cell produces electrical energy from

a spontaneous reaction (anode negative, cathode positive). (c) Conduction through metals is by free electrons, with no chemical change and no movement of matter; conduction through molten electrolytes is by ions moving to the electrodes, with chemical change (discharge) at the electrodes.

13 Describe a galvanic cell explaining the functions of electrodes and the salt bridge.

A galvanic cell has two half-cells: e.g. Zn in ZnSO4 (anode, negative) and Cu in CuSO4 (cathode, positive), joined by a salt bridge and an external wire. At the anode, oxidation occurs: Zn -> Zn2+ + 2e-. The electrons travel through the external wire to the cathode, where reduction occurs: Cu2+ + 2e- -> Cu. The salt bridge (a tube of KCl or NH4NO3 in agar) completes the circuit and maintains electrical neutrality by letting ions migrate: anions move towards the anode compartment and cations towards the cathode compartment, preventing charge build-up that would stop the cell. Without it the voltage drops to zero.

14 Write comprehensive notes on: (a) Spontaneity of oxidation reduction reactions (b) Electrolytic conduction (c) Alkaline, silver oxide and nickel-cadmium batteries, fuel cell (d) Lead accumulator, its desirable and undesirable features.

(a) Spontaneity of oxidation-reduction reactions: a redox reaction is spontaneous if its cell voltage is positive, E(cell) = E(cathode) - E(anode) > 0, which corresponds to a negative Gibbs energy change, delta G = -nFE. The species with the higher reduction potential is reduced; the other is oxidized. (b) Electrolytic conduction: conduction in electrolytes is by the movement of ions; cations move to the cathode and anions to the anode, and are discharged there. It depends on the concentration and mobility of ions and on temperature, and it is always accompanied by chemical change at the electrodes. (c) Alkaline battery: Zn anode, MnO2 cathode, KOH electrolyte, 1.5 V, non-rechargeable, longer life than the dry cell. Silver oxide battery: Zn anode, Ag2O cathode, alkaline electrolyte, about 1.5 V, used in watches

and calculators. Nickel-cadmium battery: Cd anode, NiO2 cathode, alkaline electrolyte, 1.4 V, rechargeable, sealed, used in power tools and portable computers. Fuel cell: hydrogen and oxygen react (with a platinum catalyst) in KOH electrolyte to give water and electricity, about 75% efficient, used in space vehicles, where the water produced is also used as drinking water. (d) Lead accumulator, its desirable and undesirable features: desirable: rechargeable, gives a large current, steady voltage (12 V from six cells), reliable. Undesirable: heavy, contains corrosive sulphuric acid and toxic lead, loses charge on standing, and is damaged by overcharging or complete discharge.

15 Will the reaction be spontaneous for the following set of half reactions? What will be the value of E(cell)? (i) Cr3+(aq) + 3e- -> Cr(s), E = -0.74 V (ii) MnO2(s) + 4H+(aq) + 2e- -> Mn2+(aq) + 2H2O(l), E = +1.28 V.

For the spontaneous cell, MnO2 is reduced (cathode, +1.28 V) and Cr is oxidized (anode, -0.74 V as a reduction potential). E(cell) = E(cathode) - E(anode) = 1.28 - (-0.74) = +2.02 V. Since E(cell) is positive, the reaction is spontaneous. Balanced overall reaction: 2Cr(s) + 3MnO2(s) + 12H+(aq) -> 2Cr3+(aq) + 3Mn2+(aq) + 6H2O(l).

16 Explain with reasons: (a) A porous plate or a salt bridge is not required in lead storage cell. (b) The standard oxidation potential of Zn is 0.76 V and its reduction potential is -0.76 V. (c) Na and K can displace hydrogen from acids but Pt, Pd and Cu can not. (d) The equilibrium is set up between metal atoms of electrode and ions of metal in a cell. (e) A salt bridge maintains the electrical neutrality in the cell. (f) Lead accumulator is a chargeable battery. (g) Impure Cu can be purified by electrolytic process. (h) SHE acts as anode when connected with Cu electrode but as cathode with Zn electrode.

(a) In a lead storage cell the electrodes are close together in the same electrolyte (dilute H2SO4), and the solid products (PbSO4) stick to the plates, so no porous

plate or salt bridge is needed to separate the half-cells. (b) Oxidation potential and reduction potential of the same couple are equal in magnitude and opposite in sign: for Zn2+/Zn the reduction potential is -0.76 V (Zn2+ + 2e- -> Zn) and the oxidation potential is +0.76 V (Zn -> Zn2+ + 2e-). (c) Na and K are above hydrogen in the electrochemical series (negative reduction potentials), so they can reduce H+ to H2 from acids; Pt, Pd and Cu are below hydrogen (positive reduction potentials), so they cannot. (d) At a metal electrode, metal atoms leave the electrode as ions and ions from the solution deposit as metal atoms; when the two rates become equal, an equilibrium is set up and a potential difference (the electrode potential) appears between the metal and the solution. (e) The salt bridge maintains electrical neutrality: as cations are produced in the anode compartment and consumed in the cathode compartment, ions from the bridge migrate to balance the charge, so the solutions stay neutral and current keeps flowing. (f) The lead accumulator is rechargeable because its discharge products are solids (PbSO4 on both plates) that stay in place and are converted back to Pb and PbO2 when current is passed in reverse. (g) Impure copper is made the anode in a CuSO4 electrolytic cell: copper dissolves from the anode and pure copper deposits at the cathode, while impurities fall as anode mud or stay in solution. (h) SHE (0.00 V) is below Cu (+0.34 V), so against Cu it is the anode (oxidation); it is above Zn (-0.76 V), so against Zn it is the cathode (reduction).

Chapter 11 - Reaction Kinetics

Exercise 11 - Multiple Choice Questions

1(i) In zero order reaction, the rate is independent of

A. temperature of reaction

✓ B. concentration of reactants

C. concentration of products

D. none of these

1(ii) If the rate equation of a reaction 2A + B -> products is, rate = k[A]^2 [B], and A is present in large excess, then order of reaction is

✓ A. 1

B. 2

C. 3

D. none of these

1(iii) The rate of reaction

A. increases as the reaction proceeds

✓ B. decreases as the reaction proceeds

C. remains the same as the reaction proceeds

D. may decrease or increase as the reaction proceeds

1(iv) With increase of 10 degrees C temperature the rate of reaction doubles. This increase in rate of reaction is due to

A. decrease in activation energy of reaction

B. decrease in the number of collisions between reactant molecules

C. increase in activation energy of reactants

✓ D. increase in number of effective collisions

1(v) The unit of the rate constant is the same as that of the rate of reaction in

A. first order reaction

B. second order reaction

✓ C. zero order reaction

D. third order reaction

Exercise 11 - Fill in the Blanks

2(i) The rate of an endothermic reaction increases with the increase in temperature.

2(ii) All radioactive disintegration nuclear reactions are of first order.

2(iii) For a fast reaction the rate constant is relatively large and half-life is short.

2(iv) The second order reaction becomes first order if one of the reactants is in large excess.

2(v) Arrhenius equation can be used to find out activation energy of a reaction.

Exercise 11 - True or False

3(i) The half life of a first order reaction increases with temperature.

False

3(ii) The reactions having zero activation energies are instantaneous.

True

3(iii) A catalyst makes a reaction more exothermic.

False

3(iv) There is difference between rate law and the law of mass action.

True

3(v) The order of reaction is strictly determined by the stoichiometry of the balanced equation.

False

Exercise 11 - Short and Numerical Questions

3(i) reason - The half life of a first order reaction increases with temperature.

False. For a first order reaction the half life is t1/2 = 0.693/k. The rate constant k increases with temperature (Arrhenius equation), so the half life DECREASES with temperature; it does not increase.

3(iii) reason - A catalyst makes a reaction more exothermic.

False. A catalyst lowers the activation energy and so speeds up both forward and reverse reactions equally; it does not change the enthalpy change (delta H) of the reaction, so it cannot make a reaction more exothermic.

3(v) reason - The order of reaction is strictly determined by the stoichiometry of the balanced equation.

False. The order of a reaction is found from experiment (from the rate law), and it may differ from the stoichiometric coefficients. For example, the reaction 2NO + 2H2 -> 2H2O + N2 is third order overall, not fourth, and many reactions have fractional or zero orders.

4 What is chemical kinetics? How do you compare chemical kinetics with chemical equilibrium and thermodynamics?

Chemical kinetics is the study of the rates of chemical reactions and the factors that affect them, together with the mechanism (the step-by-step path) of reactions. Chemical equilibrium tells us how far a reaction goes and the composition at equilibrium, but not how fast it gets there. Thermodynamics tells us whether a reaction is energetically possible (delta G negative) and how much heat is involved, but not its rate: a thermodynamically feasible reaction (e.g. diamond turning into graphite) may be immeasurably slow. Kinetics, equilibrium and thermodynamics are therefore complementary: thermodynamics says 'can it happen', equilibrium says 'how far', kinetics says 'how fast'.

5 The rate of a chemical reaction with respect to products is written with positive sign, but with respect to reactants is written with a negative sign. Explain it with reference to the hypothetical reaction aA + bB -> cC + dD.

For aA + bB -> cC + dD, the rate with respect to a product is positive because the product concentration increases with time, while the rate with respect to a reactant is negative because the reactant concentration decreases with time. To get a single positive rate for the reaction, each concentration change is divided by its

stoichiometric coefficient: rate = -(1/a) d[A]/dt = -(1/b) d[B]/dt = +(1/c) d[C]/dt = +(1/d) d[D]/dt. The minus signs on reactants simply cancel their negative concentration changes.

6 What are instantaneous and average rates? Is it true that the instantaneous rate of a reaction at the beginning of the reaction is greater than average rate and becomes far less than the average rate near the completion of reaction?

Average rate is the change in concentration over a stated time interval, divided by that interval. Instantaneous rate is the rate at one particular instant, given by the slope of the concentration-time curve at that point (tangent). Yes, the statement is true for an ordinary reaction: at the start, concentrations are highest, so the instantaneous rate is highest and greater than the average rate over the whole reaction; near completion, concentrations are low, so the instantaneous rate is much smaller than the average rate over the whole reaction. The average rate lies between the initial and final instantaneous rates.

7 Differentiate between: (i) Rate and rate constant of a reaction (ii) Homogeneous and heterogeneous catalyses (iii) Fast step and the rate determining step (iv) Enthalpy change of reaction and energy of activation of reaction.

(i) Rate is the change in concentration per unit time (mol dm^-3 s^-1) at a given moment; rate constant k is the proportionality constant in the rate law (rate = k [A]^x [B]^y), equal to the rate when all concentrations are unity, and constant at a given temperature. (ii) Homogeneous catalysis: catalyst and reactants are in the same phase, e.g. acid-catalysed hydrolysis of an ester in water. Heterogeneous catalysis: catalyst and reactants are in different phases, e.g. iron catalyst (solid) in the Haber process (gases). (iii) Fast step: a step in a mechanism that occurs quickly and does not limit the overall rate. Rate determining step: the slowest step in a mechanism, which limits the overall rate; the rate law is determined by this step. (iv) Enthalpy change of reaction (delta H) is the total heat absorbed or evolved in

the reaction (products minus reactants); energy of activation (Ea) is the extra energy reactants must gain to reach the activated complex, i.e. the height of the energy barrier, which decides the rate but not the heat change.

8 Justify the following statements (as printed, the list runs (i), (ii), (iii), (v), (iv), (vi), (vii): items (iv) and (v) are swapped in print; they are answered here in printed order).

(i) The rate of a reaction is ever changing because it depends on the concentrations of reactants, which fall as the reaction proceeds, and on temperature and other conditions, which may also change. (ii) The rate falls as reactants are consumed, but the rate constant k depends only on temperature (and the nature of the reaction), not on concentrations, so under fixed conditions k stays constant while the rate changes. (iii) In a first order reaction the same fraction reacts in each half life, not the same amount: after one hour 50% has reacted, leaving 50%; in the next hour half of the remainder (25% of the original) reacts, so the remaining 50% is NOT fully consumed in the second hour; in fact a first order reaction never quite reaches 100% completion. The statement that 'the remaining 50% needs more than one hour' is therefore true in the sense that the second half (of the original) is never completed in one further hour; strictly, each successive half of what remains takes one hour. (iv) [printed as (v)] Radioactive decay is always first order because the decay of a nucleus does not depend on the concentration of other nuclei: the rate is proportional to the number of undecayed nuclei present, which is the definition of first order kinetics. (v) [printed as (iv)] For a second order reaction, rate = k [A]^2 (or k [A][B]), so k has units (mol dm^-3 s^-1)/(mol dm^-3)^2 = dm^3 mol^-1 s^-1, while the rate itself has units mol dm^-3 s^-1: the units differ because k must cancel one concentration unit. (vi) The order is found from the experimental rate law, not from the balanced equation: the sum of stoichiometric coefficients (molecularity) may differ from the order, e.g. a reaction with four stoichiometric

coefficients can be third order, and orders can be fractional or zero. (vii) The order of a reaction is obtained from the rate expression, and the rate expression is obtained from experiment (initial-rate or concentration-time data), never from the balanced equation alone.

9 Explain that half life method for measurement of the order of a reaction can help us to measure the order of even those reactions which have a fractional order.

For a reaction of order n, the half life is proportional to 1/a^(n-1), where a is the initial concentration. Measuring the half life at two different initial concentrations gives the ratio of half lives, from which (n - 1) follows by logarithms, even if n is fractional: e.g. if doubling the concentration halves the half life, n = 2; if the half life is unchanged, n = 1; intermediate changes give fractional n. The method therefore works for fractional orders, where the simple integer-order integrated equations do not apply directly.

10 A curve is obtained when a graph is plotted between time on x-axis and concentration on y-axis. The measurement of the slopes of various points gives us the instantaneous rates of reaction. Explain with suitable examples.

On a concentration-time curve, the slope at any point is the instantaneous rate at that time: the tangent to the curve gives d[C]/dt. Example: for the decomposition of H2O2, the concentration-time curve is steep at the start (large slope, fast rate) and flattens as the reaction proceeds (small slope, slow rate). Measuring tangents at several times gives the instantaneous rates at those times, and plotting them shows how the rate falls as the reactant is consumed.

11 The rate determining step of a reaction is found out from the mechanism of that reaction. Explain it with few examples.

The rate determining step is the slowest step of the mechanism, and the experimental rate law matches the composition of the activated complex of that step. Examples: (i) the reaction 2NO + 2H2 -> 2H2O + N2 has a slow step 2NO +

H2 -> N2O + H2O, followed by a fast step N2O + H2 -> N2 + H2O; the rate law (rate = k [NO]^2 [H2]) follows from the slow step. (ii) The reaction between persulphate and iodide has a slow first step whose composition gives the observed rate law, while later fast steps do not affect it. If a proposed mechanism's slow step does not give the observed rate law, the mechanism is wrong.

12 Discuss the factors which influence the rates of chemical reactions.

Factors influencing reaction rates: (i) Nature of reactants: ionic reactions are fast, covalent bond-breaking reactions are slow. (ii) Concentration: higher concentration gives more collisions per second, hence a faster rate. (iii) Temperature: raising the temperature increases the fraction of molecules with energy above Ea, so the rate roughly doubles per 10 degrees C rise. (iv) Catalyst: lowers Ea and so increases the rate without being consumed. (v) Surface area: finely divided solids react faster because more surface is exposed. (vi) Light: photochemical reactions are speeded up by light (e.g. H2 + Cl2). (vii) Pressure: for gases, higher pressure means higher concentration and more collisions.

13 Explain the following facts about the reaction 2NO(g) + 2H2(g) -> 2H2O(g) + N2(g): (i) The changing concentrations of reactants change the rates of this reaction. (ii) Individual orders with respect to NO and H2 can be measured. (iii) The overall order can be evaluated by keeping the concentration of one of the substances constant.

(i) Changing the concentration of NO or H2 changes the rate, because the rate law contains both concentrations: rate = k [NO]^2 [H2]. (ii) The individual orders are measured by the isolation method: keep [H2] constant and vary [NO] to find the order in NO (2), then keep [NO] constant and vary [H2] to find the order in H2 (1). (iii) The overall order (2 + 1 = 3) is evaluated by adding the individual orders, each found while the other concentration is held constant, so that only one concentration changes at a time.

14 The collision frequency and the orientation of molecules are necessary conditions for determining the proper rate of reaction. Justify the statement.

Not every collision leads to reaction: molecules must collide with energy at least equal to the activation energy (otherwise they simply bounce apart) and with the correct orientation (the reacting atoms must meet). Collision frequency counts all collisions, but only the fraction with sufficient energy and proper orientation are effective. Hence the rate depends on the collision frequency multiplied by the fraction of molecules above Ea and the orientation (steric) factor: rate = P Z e^(-Ea/RT) [A][B]. Without both conditions, even a high collision frequency gives a slow rate.

15 How does Arrhenius equation help us to calculate the energy of activation of a reaction?

The Arrhenius equation is k = A e^(-Ea/RT), or log k = log A - Ea/(2.303 RT). Measuring k at two temperatures and plotting log k against 1/T gives a straight line of slope -Ea/(2.303 R); from the slope, Ea = -2.303 R x slope. Alternatively, from two values k1 and k2 at T1 and T2: log(k2/k1) = (Ea/2.303 R) x (1/T1 - 1/T2), which gives Ea directly. The equation thus converts temperature-dependence of the rate into the energy barrier.

16 Define the following terms and give examples: (i) Homogeneous catalysis (ii) Heterogeneous catalysis (iii) Activation of a catalyst (iv) Auto-catalysis (v) Catalytic poisoning (vi) Enzyme catalysis.

(i) Homogeneous catalysis: catalyst and reactants in the same phase, e.g. H+ (aqueous) catalysing the hydrolysis of an ester. (ii) Heterogeneous catalysis: catalyst and reactants in different phases, e.g. solid iron in the Haber process; solid Pt in the contact process. (iii) Activation of a catalyst (promotor): a substance that increases the activity of a catalyst without itself being a catalyst, e.g. alumina or molybdenum added to iron in the Haber process; copper and tellurium with nickel in

oil hydrogenation. (iv) Auto-catalysis: a product of the reaction acts as the catalyst, so the reaction speeds up as it proceeds, e.g. Mn2+ in the oxalic acid-KMnO4 reaction; nitrous acid in the copper-nitric acid reaction. (v) Catalytic poisoning: loss of catalytic activity due to impurities that block the catalyst surface, e.g. sulphur and arsenic compounds poison platinum catalysts; carbon monoxide poisons the iron catalyst in the Haber process. (vi) Enzyme catalysis: catalysis by protein enzymes in living cells, highly specific and efficient, e.g. urease hydrolysing urea; invertase hydrolysing sucrose; zymase fermenting glucose to ethanol.

17 Briefly describe with examples: (i) Change of physical state of a catalyst at the end of reaction (ii) A very small amount of a catalyst may prove sufficient to carry out a reaction (iii) A finely divided catalyst may prove more effective (iv) Equilibrium constant of a reversible reaction is not changed in the presence of a catalyst (v) A catalyst is specific in its action.

(i) Change of physical state: a catalyst may change its physical form during the reaction while its mass and chemical composition stay the same, e.g. granular MnO2 used in KClO3 decomposition ends as a fine powder; the shining surface of a solid catalyst may become dull. (ii) A very small amount of catalyst suffices: a catalyst is not consumed, so it can act repeatedly; e.g. 1 mg of fine platinum powder can convert 2.5 dm^3 of H2 and 1.25 dm^3 of O2 to water, and 1 g of colloidal platinum can decompose thousands of dm^3 of H2O2. (iii) A finely divided catalyst is more effective because catalysis happens at the surface: finer division gives a larger surface area for the same mass, e.g. colloidal platinum is far more active than a lump of platinum; finely divided nickel is used for hydrogenating vegetable oils. (iv) A catalyst does not change the equilibrium constant: it speeds up forward and reverse reactions equally, so equilibrium is reached sooner but the equilibrium composition and K are unchanged. (v) A catalyst is specific: a catalyst that works for one reaction may not work for another, and different catalysts can

give different products from the same reactant, e.g. formic acid gives H2O + CO with Al2O3 but H2 + CO2 with Cu.

18 What are enzymes? Give examples in which they act as catalyst. Mention the characteristics of enzyme catalysis.

Enzymes are complex protein molecules produced by living cells that catalyse biochemical reactions. Examples: urease (soybean) hydrolyses urea to NH3 and CO2; invertase (yeast) hydrolyses sucrose to glucose and fructose; zymase (yeast) ferments glucose to ethanol and CO2; diastase hydrolyses starch. Characteristics of enzyme catalysis: (i) enzymes are the most efficient catalysts known and greatly lower the activation energy; (ii) they are highly specific (urease acts on urea only, not even on methyl urea); (iii) they work best at an optimum temperature (activity rises to a maximum, then falls as the enzyme denatures) and an optimum pH; (iv) their activity is inhibited by poisons and enhanced by co-enzymes or activators; (v) they act through a lock-and-key fit of the substrate into active centres on the enzyme surface (Michaelis-Menten mechanism: E + S ⇌ ES -> P + E).

19 In the reaction of NO and H2, it was observed that equimolecular mixture of gases at 340.5 mm pressure was half changed in 102 seconds. In another experiment with an initial pressure of 288 mm of Hg, the reaction was half completed in 140 seconds. Calculate the order of reaction.

Step 1 - For the half-life method, (t1/2)1/(t1/2)2 = (a2/a1)^(n-1), where a is the initial pressure and n the order.

Step 2 - Substitute: 102/140 = (288/340.5)^(n-1), i.e. 0.7286 = (0.8458)^(n-1).

Step 3 - Take logarithms: log(0.7286) = (n - 1) log(0.8458), so -0.1376 = (n - 1) x (-0.0727).

Step 4 - n - 1 = 0.1376/0.0727 = 1.89, so n = 2.89, about 2.88.

Answer: order 2.88, i.e. effectively third order (matches the book's printed answer).

20 A study of chemical kinetics of a reaction A + B -> Products gave the following data at 25 degrees C: [A] = 1.00, [B] = 0.15, rate = 4.2 x 10^-6; [A] = 2.00, [B] = 0.15, rate = 8.4 x 10^-6; [A] = 1.00, [B] = 0.2, rate = 5.6 x 10^-6 (rates in mol dm^-3 s^-1). Calculate the rate law.

Step 1 - Compare experiments 1 and 2: [B] is constant (0.15) while [A] doubles (1.00 -> 2.00) and the rate doubles (4.2 -> 8.4 x 10^-6). The reaction is first order in A.

Step 2 - Compare experiments 1 and 3: [A] is constant (1.00) while [B] increases by a factor 0.2/0.15 = 1.333 and the rate increases by the same factor (5.6/4.2 = 1.333). The reaction is first order in B.

Step 3 - Rate law: rate = k [A][B]. Overall order = 1 + 1 = 2.

Step 4 - Rate constant from experiment 1: k = rate/([A][B]) = 4.2 x 10^-6/(1.00 x 0.15) = 2.8 x 10^-5 mol^-1 dm^3 s^-1.

Answer: rate = k [A][B], second order overall (matches the book's printed answer).

21 Some reactions taking place around room temperature have activation energies around 50 kJ mol^-1. (i) What is the value of the factor e^(-Ea/RT) at 25 degrees C? (ii) Calculate this factor at 35 degrees C and 45 degrees C and note the increase in this factor for every 10 degrees C rise in temperature. (iii) Prove that for every 10

degrees C rise in temperature, the factor doubles and so rate constant also doubles.

Step 1 - (i) At 25 degrees C (298 K): Ea/RT = 50000/(8.314 x 298) = 20.18, so e^(-Ea/RT) = e^-20.18 = 1.72 x 10^-9.

Step 2 - (ii) At 35 degrees C (308 K): Ea/RT = 50000/(8.314 x 308) = 19.53, so the factor = 3.31 x 10^-9. At 45 degrees C (318 K): Ea/RT = 50000/(8.314 x 318) = 18.91, so the factor = 6.12 x 10^-9.

Step 3 - Ratios: 3.31/1.72 = 1.92 and 6.12/3.31 = 1.85, i.e. the factor roughly doubles for each 10 degrees C rise.

Step 4 - (iii) Since k = A e^(-Ea/RT), doubling the exponential factor doubles k, so the rate constant (and hence the rate) roughly doubles for every 10 degrees C rise in temperature.

Answer: (i) 1.72 x 10^-9 (ii) 3.31 x 10^-9 at 35 degrees C and 6.12 x 10^-9 at 45 degrees C (iii) the factor doubles per 10 degrees C rise, so k doubles (matches the book's printed answers).

22 H2 and I2 react to produce HI. Following data for rate constant at various temperatures (K) have been collected: T = 500, 550, 600, 650, 700 K; k = 6.814 x 10^-4, 2.64 x 10^-2, 0.56, 7.31, 66.67 cm^3 mol^-1 s^-1. (i) Plot a graph between 1/T on x-axis and log k on the y-axis. (ii) Measure the slope of this straight line and calculate the energy of activation of this reaction.

Step 1 - Labelled graph instruction: plot 1/T (x-axis) against log k (y-axis) for the five points: (0.00200, -3.17), (0.00182, -1.58), (0.00167, -0.25), (0.00154, 0.86), (0.00143, 1.82). The points lie on a straight line with negative slope, as the

Arrhenius equation predicts.

Step 2 - From the plotted line, the slope = -Ea/(2.303 R). The slope of the best-fit line through the five points is about -8326 K.

Step 3 - Ea = 2.303 x R x 8326 = 2.303 x 8.314 x 8326 = 159400 J mol^-1, about 160 kJ mol^-1.

Answer: slope about 8326 (magnitude); Ea about 160.6 kJ mol^-1 (matches the book's printed answers).